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Data-driven camera calibration for light field camera systems

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Abstract

This thesis presents the development and evaluation of a data-driven calibration software for multi-camera light field systems. The applications of this type of camera are gaining more importance, such as in robotics, autonomous driving, and 3D reconstruction. Therefore, an accurate calibration is essential for the use of light field cameras. The standard method to calibrate a camera is by using a target (e.g. a checkerboard), but for multi-camera systems the complexity increases. Additionally, the need for constant human intervention or the inaccessibility of cameras motivates this project to overcome these limitations. The proposed software uses a method presented in the LiFCal paper that uses geometric properties of depth maps and scenes to periodically calibrate a camera without the need for a target. In addition, an initial calibration using a checkerboard is performed to create a reference for deciding whether the calibration improves or not. The decision is also supported by a health monitoring system that determines the quality of a calibration based on the reprojection error and categorizes it into a score between 0 and 100. This approach allows a target-free recalibration system to ensure the accuracy of a light field camera over a long-term period, avoiding decalibration over time.

The software was evaluated on synthetically generated image data created in Blender. The system was represented by a cross-shaped camera arrangement consisting of thirteen individual cameras. The initial calibration achieved high accuracy with reprojection errors around 0.03 pixels and a mean health score of 93.21 ± 4.37 . In contrast, the reprojection errors in the LiFCal method were higher and very inconsistent across different calibration runs. The errors ranged between 0.4 and 10 pixels, with a mean health score of 56.76 ± 4.09 . However, the results demonstrate that while the framework successfully integrates both calibration methods, LiFCal faces fundamental limitations when applied to multi-camera arrays, as it was originally designed for micro-lens-array cameras. In addition, this thesis identifies the specific conditions and requirements necessary for target-free recalibration in multi-camera light field systems.

Zusammenfassung

Diese Arbeit präsentiert die Entwicklung und Evaluation einer datengetriebenen Kalibriersoftware für Multikamera-Lichtfeldsysteme. Die Anwendungen dieser Art von Kamera gewinnen zunehmend an Bedeutung, beispielsweise in der Robotik, im autonomen Fahren und in der 3D-Rekonstruktion. Daher ist eine präzise Kalibrierung für den Einsatz von Lichtfeldkameras essenziell. Die Standardmethode zur Kalibrierung einer Kamera erfolgt mithilfe eines Targets (z. B. eines Schachbretts), jedoch steigt die Komplexität bei Multikamera-Systemen. Zusätzlich motivieren der Bedarf an ständiger menschlicher Intervention oder die eingeschränkte Zugänglichkeit von Kameras dieses Projekt, diese Einschränkungen zu überwinden. Die vorgeschlagene Software verwendet eine im LiFCal-Paper vorgestellte Methode, die geometrische Eigenschaften von Tiefenkarten und Szenen nutzt, um eine Kamera periodisch ohne die Notwendigkeit eines Targets zu kalibrieren. Zusätzlich wird eine initiale Kalibrierung mithilfe eines Schachbretts durchgeführt, um eine Referenz zu schaffen, anhand derer entschieden werden kann, ob sich die Kalibrierung verbessert oder nicht. Die Entscheidung wird zusätzlich durch ein Health-Monitoring-System unterstützt, das die Qualität einer Kalibrierung anhand des Reprojektionsfehlers bestimmt und diese in einen Score zwischen 0 und 100 einordnet. Dieser Ansatz ermöglicht ein targetfreies Rekalibrierungssystem, das die Genauigkeit einer Lichtfeldkamera über einen langfristigen Zeitraum sicherstellt und eine Dekalibrierung im Laufe der Zeit vermeidet.

Die Software wurde anhand synthetisch erzeugter Bilddaten evaluiert, die in Blender erstellt wurden. Das System wurde durch eine kreuzförmige Kameraanordnung bestehend aus dreizehn einzelnen Kameras repräsentiert. Die initiale Kalibrierung erreichte eine hohe Genauigkeit mit Reprojektionsfehlern von etwa 0.03 Pixeln und einem mittleren Health Score von $93,21 \pm 4,37$. Im Gegensatz dazu waren die Reprojektionsfehler der LiFCal-Methode höher und über verschiedene Kalibrierläufe hinweg sehr inkonsistent. Die Fehler lagen zwischen 0.4 und 10 Pixeln, mit einem mittleren Health Score von 56.76 ± 4.09 . Die Ergebnisse zeigen jedoch, dass das Framework zwar beide Kalibrierungsmethoden erfolgreich integriert, LiFCal jedoch grundlegenden Einschränkungen unterliegt, wenn es auf Multikamera-Arrays angewendet wird, da es ursprünglich für Mikro-Linsen-Array-Kameras entwickelt wurde. Zusätzlich identifiziert diese Arbeit die spezifischen Bedingungen und Anforderungen, die für eine targetfreie Rekalibrierung in Multikamera-Lichtfeldsystemen notwendig sind.

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1. Introduction

In recent years, light field cameras have gained significance in many applications, such as robotics, augmented reality, and 3D reconstruction. In these application areas, the accuracy of the camera system plays a crucial role in overall performance. Camera calibration is therefore an essential step, as it enables an accurate geometric description of the imaging process and ensures reliable results.

There are various types of light field camera systems; however, this thesis focuses on multi-camera light field systems. These systems consist of multiple individual cameras arranged in a specific geometric configuration in order to capture different viewpoints of a scene. Compared to single-camera setups, the calibration of such systems is considerably more complex. The standard approach relies on target-based calibration using physical patterns such as checkerboards. In contrast, a more recent target-free calibration approach has been proposed by Fleith et al.[8]. The motivation of this thesis is to develop a calibration software that combines both approaches within a data-driven recalibration framework, enabling automatic calibration of a multi-camera array without human intervention.

Despite its potential, the proposed software approach also introduces limitations and challenges that must be carefully examined. For this reason, this thesis investigates the feasibility and performance of the target-free calibration method in the context of multi-camera light field systems. The research questions guiding this work are presented in the following section.

1.1. Research Questions

In order to systematically investigate the feasibility and performance of the target-free calibration method LiFCal for multi-camera light field systems, the following research questions are formulated. These questions aim to explore the applicability, accuracy, and conditions under which LiFCal can effectively replace traditional target-based calibration methods.

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

The first research question examines whether the LiFCal approach is suitable for the intended multi-camera light field system. The second research question investigates whether LiFCal can achieve a calibration quality comparable to a classical target-based approach and thus potentially replace it in the context of a multi-camera system. The third research question focuses on identifying the conditions and requirements necessary to enable a reliable and practical target-free recalibration within an automated software framework.

1.2. Structure of the Thesis

To provide the necessary background and context, the thesis begins with the theoretical foundations in Chapter 2. This chapter introduces the basic knowledge of camera models, lens distortion, camera calibration, and the different types of light field cameras. Chapter 3 describes the methods and materials used to implement the calibration framework. This includes the main idea of the project, the use case description, and the software architecture. The results of the implementation and evaluation are presented in Chapter 4. It contains the visualization of the results and their accuracy. Finally, Chapter 5 discusses the findings, answers the research questions, and provides an outlook on future work. A final conclusion completes the thesis in the last chapter.

2. Theoretical Background

This chapter introduced the theoretical foundations required for this thesis. It covered the pinhole camera model, including intrinsic and extrinsic parameters as well as lens distortion, as the basis for geometric camera modeling. Building on this, common camera calibration methods and error metrics were presented, including target-based calibration and bundle adjustment. The chapter also introduced light field camera systems, their main system types, and the state of the art in their calibration. Together, these concepts provide the necessary background for the calibration approach and experiments presented in the following chapters.

2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [17].

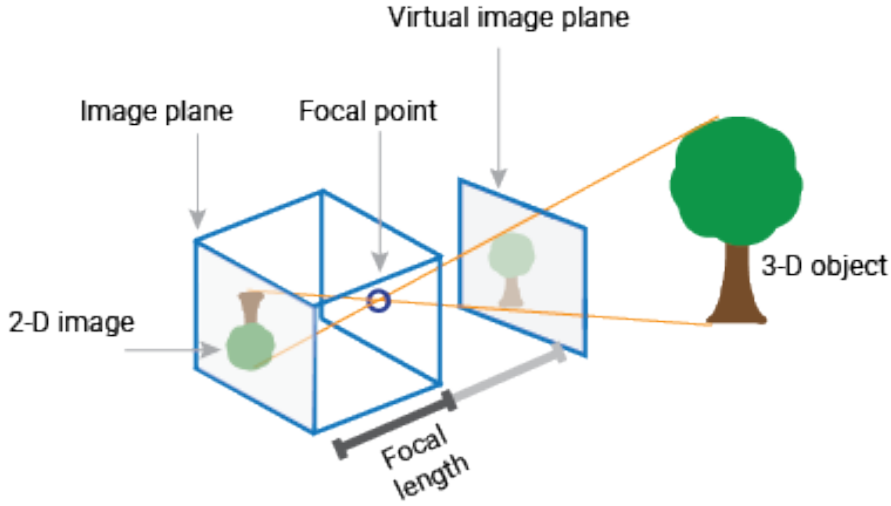


Figure 1: Illustration of the pinhole camera model. Source: [19].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [17]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [19].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where (f_x, f_y) are the focal lengths in pixel units along the x and y axes, s is the skew coefficient between the axes, and (c_x, c_y) is the principal point (optical center) in pixel coordinates. K is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

where R is a 3×3 rotation matrix representing the orientation of the camera with respect to the world coordinate system, and t is a 3×1 translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [19].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [7]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [23].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

2.2.1. Radial Distortion

Radial distortion is caused by the spherical shape of camera lenses, which leads to different refraction of light from the image center to the outer regions. This causes straight lines in the image to appear curved, resulting in barrel or pincushion distortion [6].

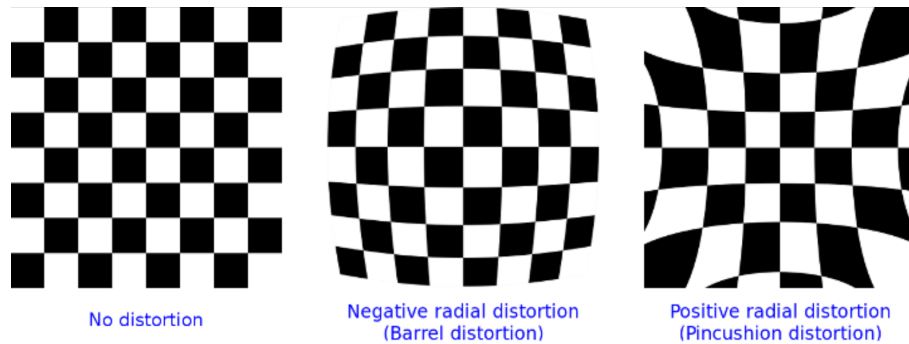


Figure 2: Comparison of different types of radial distortion. Source: [21].

The formal definition of radial distortion is given as follows [19]:

$$x_{distorted} = x \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (1)$$

$$y_{distorted} = y \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (2)$$

where x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (3)$$

The coefficients k_1 , k_2 , and k_3 in (1) and (2) control the strength of radial lens distortion. In practice, the first coefficient k_1 has the dominant influence on the distortion type: negative values of k_1 typically result in barrel distortion, while positive values lead to pincushion distortion.

2.2.2. Tangential Distortion

Tangential distortion occurs when the image sensor and the camera lens are not perfectly aligned or parallel to each other [19]. As a result, image points are displaced tangentially with respect to the optical axis. Figure 3 illustrates the geometric cause of tangential distortion and its effect on the captured image.

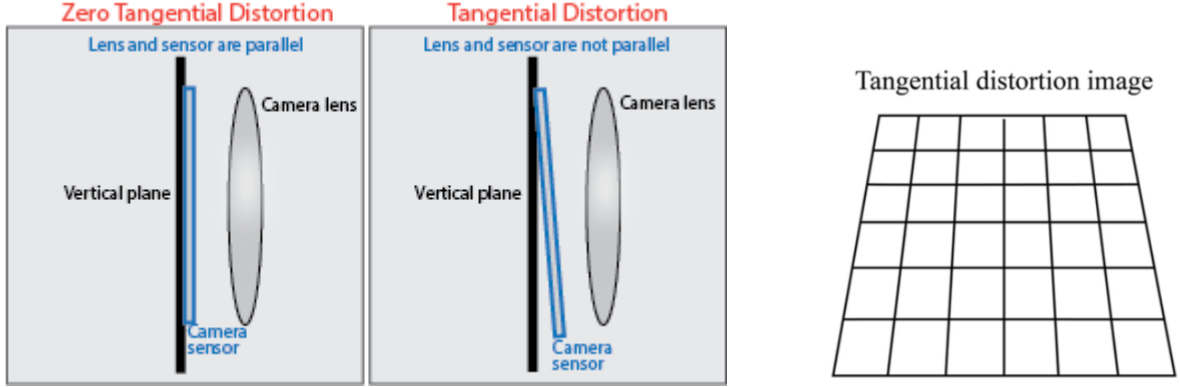


Figure 3: Visualization of tangential distortion. Source: [19], [18].

The formal definition of tangential distortion is given by the following equations [19]:

$$x_{distorted} = x + \left(2p_1 xy + p_2 (r^2 + 2x^2) \right) \quad (4)$$

$$y_{distorted} = y + \left(p_1 (r^2 + 2y^2) + 2p_2 xy \right) \quad (5)$$

Here, x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (6)$$

The coefficients p_1 and p_2 describe how strong the tangential lens distortion is. This type of distortion is caused by mechanical misalignment of the lens elements relative to the image sensor, resulting in a displacement of image points. Although tangential distortion is typically less pronounced than radial distortion, it can still negatively affect the camera calibration accuracy if not properly modeled.

2.3. Camera Calibration

The goal of camera calibration, also known as camera resectioning, is to estimate the parameters of a camera model. These parameters include the intrinsic parameters, the extrinsic parameters, and the lens distortion coefficients [19]. Knowing these parameters makes it possible to correct distortions in the recorded images and to perform metric measurements in the scene, for example to estimate object sizes or distances from image data.

Camera calibration is required for many tasks in 3D computer vision and photometry. Applications such as depth estimation, 3D reconstruction, or pose estimation rely on a precise mapping from 3D world coordinates to 2D image coordinates [25]. If the calibration is inaccurate, errors can propagate through the processing pipeline and reduce the quality of the final results.

In practice, cameras are often calibrated using targets with known geometric properties. Planar targets, such as checkerboard patterns, are widely used because they are easy to handle and lead to reliable results [32]. Several images of the target are captured from different viewpoints and orientations. Based on these images, the camera parameters can be estimated with relatively small calibration errors. For this reason, target-based calibration methods are commonly used in many computer vision applications.

2.3.1. Target-Based Camera Calibration

There are various approaches for target-based camera calibration. One of the most common methods uses a checkerboard pattern, as mentioned before. The checkerboard consists of a regular grid of black and white squares with known dimensions, as shown in Figure 4.

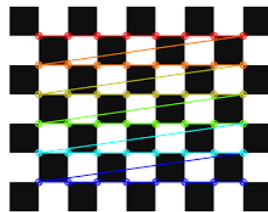


Figure 4: Example of a checkerboard and detected corner points. Source: [14].

During the calibration process, control points are extracted from the images. In the case of a checkerboard, these control points correspond to the corner points of the squares [25]. The positions of these corners are known in the coordinate system of the calibration target. By capturing multiple images of the checkerboard from different angles and distances, a set of 2D image points can be obtained that correspond to the known 3D points of the target. Based on these point correspondences, an optimization problem can be formulated to estimate the intrinsic and extrinsic camera parameters as well as the lens distortion coefficients [32].

This calibration approach is implemented in widely used computer vision libraries such as OpenCV [20]. It can be applied to single-camera setups and multi-camera systems. Besides checkerboards, other types of calibration targets exist, for example circular dot patterns [25]. In some cases, these patterns can achieve higher accuracy, especially when the image quality is low. However, they also require physical targets and a manual calibration setup.

The main limitations of target-based calibration methods are the required manual placement of the target and the need to capture images with different poses. In addition, the calibration accuracy depends on factors such as the quality of the corner detection, the number of images used, and the diversity of viewpoints during data acquisition [32]. Despite these limitations, target-based calibration remains a standard method in many computer vision applications due to its simplicity and reliable performance.

2.3.2. Calibration Parameters and Error Metrics

To evaluate the quality of a camera calibration, different error metrics are used. These metrics describe how well the estimated camera parameters fit the observed image data. In practice, this is done by comparing detected image points with points that are projected using the calibrated camera model. The most common metric is the reprojection error [19].

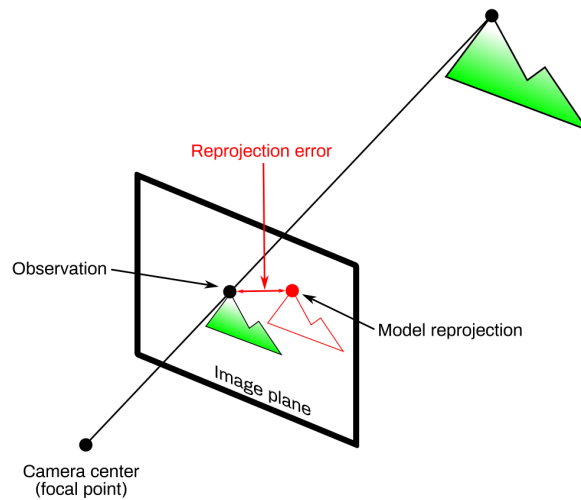


Figure 5: Visualization of reprojection error. Source: [2].

After a calibration, known 3D points are projected back onto the image plane. The reprojection error measures the distance between the observed image points and the

corresponding projected points. It is usually computed as the average Euclidean distance and is given in pixel units [2]. It can be defined as

$$e = \frac{1}{N} \sum_{i=0}^{N-1} \|\mathbf{p}_i - \mathbf{q}_i\|_2. \quad (7)$$

Here, p_i represents the observed feature points in the image, while q_i denotes the projected points obtained from the calibrated camera model. In general, a smaller reprojection error indicates a better calibration. What is considered a good value depends on the application, but for accurate calibrations the reprojection error is often between 0.1 and 0.5 pixels [12].

Another commonly used measure is the root mean square (RMS) error [11]. It summarizes the reprojection error over all calibration points and is computed as:

$$\epsilon_{\text{res}} = \left(\frac{1}{2n} \sum_{i=1}^n \|\mathbf{p}'_i - \mathbf{q}'_i\|_2^2 \right)^{1/2} \quad (8)$$

The RMS error provides a single value that describes the overall calibration accuracy and is often used to compare different calibration results.

2.3.3. Bundle Adjustment

Bundle adjustment is a method for improving a 3D reconstruction by refining both the positions of 3D points and the camera parameters at the same time [27]. The name comes from the “bundles” of light rays from the 3D points to the cameras, which are adjusted together to make the reconstruction more accurate. The parameters that are optimized together usually include the 3D coordinates of the points, the extrinsic parameters, and sometimes the intrinsic parameters [4].

This joint optimization improves calibration accuracy because it ensures that all cameras and points are consistent with each other. As a result, projection errors are reduced and small misalignments are corrected that would remain if cameras or points were adjusted separately [27]. This idea is illustrated in Figure 6.

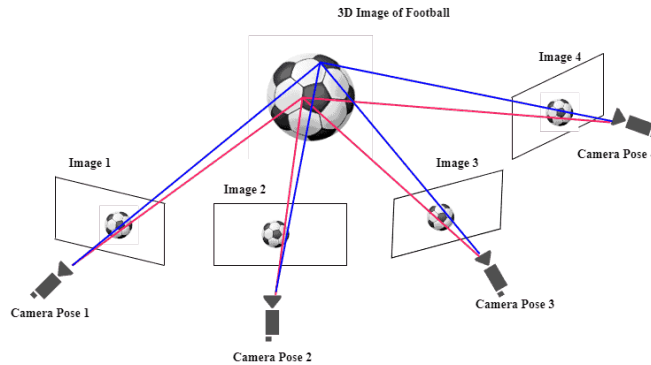


Figure 6: Visual representation of bundle adjustment. Source: [15].

Mathematically, bundle adjustment is formulated as an optimization problem that minimizes the reprojection error. This optimization is usually solved using nonlinear least squares methods and takes advantage of the fact that the problem structure is largely sparse [4].

2.4. Light Field Camera Systems

A standard camera captures a 2D projection of the 3D world. Light field cameras, on the other hand, have the ability to capture more information. They record not only the intensity of light rays but also their direction [10]. This property allows the creation of accurate depth maps and refocus the image after it has been taken [8]. Therefore light field cameras are growing in importance for applications in computer vision, robotics, augmented reality, and other applications.

The four-dimensional nature of light fields means that their processing currently needs relatively high computational resources [10]. However, the evolution of hardware and algorithms is making light field cameras more practical for real-world applications.

There are mainly two different types of light field camera systems: multi-camera arrays and micro-lens-array-based cameras. Each type has its own advantages and disadvantages, which will be discussed in the following sections.

2.4.1. Multi-Camera Light Field Systems

Multi-camera arrays are one way to capture light fields. This type of system is structured by arranging multiple individual cameras in a geometric configuration [29]. Each camera captures a different viewpoint of the scene, enabling the system to record a 4D light field. In practice, there are different approaches to arranging the cameras. For example, in [26], an array of 128 cameras in a square grid is used, while in other works cross-shaped structures with fewer cameras are employed [29], as shown in Figure 7.

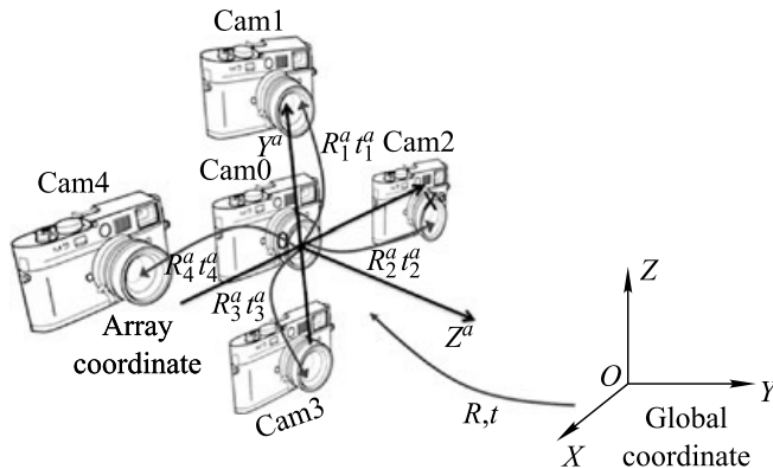


Figure 7: Representation of a multi-camera system. Source: [29].

Besides the advantages of a multi-camera setup, such as improved depth accuracy and refocusing capabilities, there are also challenges. The calibration of such systems is more

complex and requires higher computational effort due to the larger number of cameras involved [29].

It is also essential to keep the cameras properly aligned and synchronized. Even small errors in alignment or timing can cause artifacts or undesired results in the final image. Therefore, multi-camera light field systems require a very precise setup and calibration to work as intended.

2.4.2. Micro-Lens-Array-Based Light Field Cameras

The second type of light field camera is based on a micro-lens array (MLA). In this case instead of having different cameras, the system uses only one sensor combined with an array of small lenses placed between the main lens and the image sensor [8]. The MLA allows the camera to capture also the 4D light field information.

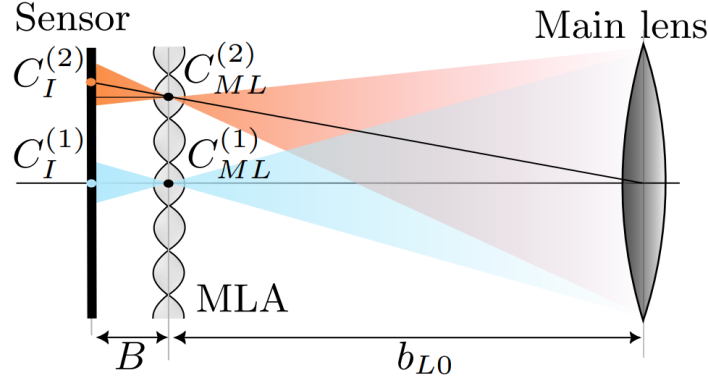


Figure 8: Structure of a micro-lens-array-based light field camera. Source: [8].

As illustrated in Figure 8, each lens in the MLA focuses light from different directions onto different regions of the sensor. This way, the camera is also able to capture the same information as a multi-camera array, but with a more compact design. Because of the smaller size and lower cost, MLA-based light field cameras are mostly used in consumer applications [16]. However, microlens-array cameras have to compromise between spatial and angular resolution, while multi-camera systems capture full-resolution images with sharper results and more accurate depth information.

For MLA-based light field cameras, the pattern-based calibration method provides satisfactory results in terms of reprojection error. In practice, the overall calibration quality strongly depends on the calibration target, particularly its flatness [1].

2.5. State of the Art in Light Field Camera Calibration

Accurate calibration is a central requirement for light field camera systems, as even small geometric errors directly affect depth estimation and view consistency. In camera arrays, where many individual views must agree, calibration errors accumulate and lead to visible artifacts in depth maps and reconstructed images.

Early work on light field calibration mainly relies on pattern-based approaches. Dima et al. systematically evaluate several multi-camera calibration methods using ground truth

data and show that calibration accuracy strongly depends on the array geometry and the physical setup of the calibration target [5]. In particular, small deviations such as bending or uneven surfaces of the calibration board already lead to increased reprojection errors across the array. Yang et al. demonstrate that using a simple checkerboard target can still yield stable calibration results for camera arrays while reducing setup complexity [30]. However, the accuracy of these methods remains closely tied to target quality and careful placement.

To overcome the limitations of target-based calibration, several self-calibration approaches have been proposed. Zhang et al. introduce a fixed relative pose prior to stabilize self-calibration in camera arrays [31]. By constraining relative camera poses, their method reduces pose drift in situations where image features are sparse or unevenly distributed. Zhou et al. address low-texture sports scenes and use temporal fusion across multiple frames to estimate calibration parameters [33]. These approaches reduce the need for calibration targets, but still depend on sufficient scene structure and a reasonable initial alignment.

Some methods focus on specific system layouts or downstream tasks. Ge et al. study a ring-shaped multi-camera system and use a transparent calibration target to support panoramic measurements [9]. Xie et al. show that accurate camera array calibration is essential for reliable depth map fusion, as calibration errors propagate directly into depth artifacts, especially near object boundaries [28].

The approach most relevant to this thesis is LiFCal by Fleith et al. [8]. Instead of using a physical calibration target, LiFCal moves the MLA-based camera through the scene and estimates calibration parameters via bundle adjustment driven by depth maps from multiple viewpoints. This shifts the calibration problem from controlled target acquisition to scene structure and camera motion, making the method particularly suitable for scenarios where calibration targets are impractical or unavailable.

Overall, existing work shows that calibration accuracy in light field systems remains sensitive to setup quality, motion, and scene structure. Larger camera arrays amplify small errors, and target-free methods require reliable depth estimates and sufficient camera motion to properly constrain the geometry.

3. Methods and Materials

This chapter introduces the methods and materials used in this thesis, covering both the design of the calibration framework and the technical setup used for its implementation and evaluation.

3.1. Methods

In this first section, the design and implementation of the calibration software are described. It describes the overall workflow, the use case scenario, and the specific requirements that guided the development process. The UML class diagram and state diagram illustrate the software architecture and behavior.

3.1.1. Overall approach

The goal of this thesis is to develop and evaluate an automatic data-driven calibration framework for multi-camera light field systems based on the target-free calibration method proposed by Fleith et al. [8]. As discussed in the state of the art, classical target-based calibration approaches require manual intervention and careful placement of calibration targets, which limits their applicability in long-term deployments or scenarios with restricted human access.

The proposed approach uses a reliable target-based calibration as an initial reference and ground truth. This initial calibration is performed using a checkerboard pattern and serves as a baseline for evaluating subsequent target-free recalibration steps. After this initial setup, the system uses a small set of images from the scene and applies LiFCal to automatically recalibrate the multi-camera system. To assess whether the target-free recalibration is acceptable, a health monitoring mechanism is introduced. This mechanism compares the calibration results obtained from LiFCal with the baseline calibration using defined decision metrics. If the recalibration achieves a calibration quality comparable to or better than the initial calibration, it is accepted and replaces the previous baseline. Otherwise, the system retains the existing calibration parameters.

This workflow is designed to enable periodic self-recalibration without manual intervention. A potential application scenario is a robotic system deployed in a remote or inaccessible environment, where mechanical shifts or environmental changes may degrade calibration quality over time. By regularly capturing image sets and applying the proposed recalibration process, the system can maintain accurate camera parameters throughout its operation.

To clearly describe the interaction between the individual components and calibration phases, a use case description is introduced in the following section.

3.1.2. Use Case Description

The following use case describes the intended workflow of the proposed automatic calibration framework for a cross-shaped multi-camera light field system consisting of a central camera and twelve surrounding cameras, arranged in four cardinal directions (North, East, South, and West). The workflow is structured into three consecutive phases: an

initial reference calibration, periodic target-free recalibration, and a health-based decision step.

Phase 1 — Initial Reference Calibration (once)

- **Target-based calibration (OpenCV):** A one-time checkerboard-based calibration is performed to estimate the intrinsic and extrinsic parameters for all cameras of the cross-shaped array.
- **Optional refinement:** An optional global refinement using bundle adjustment is applied to improve consistency across all camera parameters.
- **Baseline creation:** A global health score is computed for the initial calibration and stored as `baseline.json`. This baseline represents the known-good reference calibration for subsequent recalibration decisions.
- **Logging:** The complete initial calibration is stored as a timestamped JSON file and a summary entry is appended to `calibration.log`.

Phase 2 — Automated Recalibration Attempts (periodic scheduler)

- **Trigger:** An automated recalibration cycle is executed at fixed time intervals (e.g. every five minutes).
- **Conditional data preparation:** If LiFCal-compatible data (focus and depth directories) already exists and is complete, it is reused. Otherwise, the system generates the required inputs by running `run_imageset_creation.py`.
- **LiFCal recalibration run:** The script `run_LiFCal_calibration.py` executes LiFCal-based recalibration for all cameras in the array and produces per-camera parameter files as well as a combined multi-camera result.
- **Logging and traceability:** The combined recalibration result is stored as a timestamped JSON file and all relevant execution details are appended to `calibration.log`.

Phase 3 — Health Evaluation and Baseline Update Rule

- **Health score computation:** A single global health score in the range $[0, 100]$ is computed from the combined LiFCal calibration result.
- **Comparison against baseline:** The newly computed health score is compared to the score stored in the current baseline calibration.
- **Acceptance rule:** If the new health score is strictly higher than the baseline score, the baseline calibration is replaced by the new result. Otherwise, the existing baseline remains unchanged.
- **Continuous improvement loop:** The recalibration process is repeated periodically, allowing improved calibration results to update the baseline over time without manual intervention.

This structure allows the system to maintain accurate calibration parameters over time, adapting to changes in the environment or camera setup. The health-based decision mechanism ensures that only beneficial recalibrations are accepted, preserving the integrity of the calibration throughout the system's operation.

This use case serves as the conceptual basis for the requirements analysis and system design presented in the following sections.

3.1.3. Requirements Analysis

The requirements for the proposed automatic calibration framework are divided into functional and non-functional requirements. These requirements define the necessary features and qualities of the system to achieve the intended use case.

Functional Requirements

- **FR1 – Initial Reference Calibration (Target-based):**
The system shall perform a one-time, checkerboard-based initial calibration using OpenCV to estimate intrinsic and extrinsic parameters for the cross-shaped camera array. This initial calibration shall be stored as the baseline calibration and serve as the reference state for all subsequent health evaluations and recalibration decisions.
- **FR2 – Parameter Export and Standardized Representation:**
The system shall export all calibration results (initial calibration and LiFCal-based recalibration) in a standardized, machine-readable JSON format, including intrinsic parameters, extrinsic poses, reprojection error statistics, timestamps, and metadata.
- **FR3 – Periodic Calibration Health Monitoring:**
The system shall periodically (at least once per configured scheduler interval) evaluate the calibration quality by computing a global health score.
- **FR4 – Health Score Computation (Global Scalar Metric):**
The system shall compute a single scalar health score in the range $[0, 100]$ representing the overall calibration quality of the complete camera array. For the initial calibration, the health score shall be derived from reprojection error statistics. For LiFCal-based recalibration, the health score shall be derived from LiFCal-specific reprojection and consistency metrics extracted from the combined parameter set.
- **FR5 – Target-Free Online Recalibration (LiFCal-based):**
The system shall execute a LiFCal-based target-free recalibration using automatically prepared focus images and virtual depth data whenever a recalibration cycle is triggered.
- **FR6 – Acceptance and Baseline Update Rule:**
The system shall compare the health score of a newly computed calibration against the health score of the currently stored baseline calibration. The new calibration shall be accepted and stored as the new baseline only if its health score is strictly higher than the baseline score; otherwise, the system shall retain the previous baseline calibration.

- **FR7 – Automated Periodic Recalibration Scheduling:**
The system shall support automated, periodic execution of the LiFCal-based recalibration process at a configurable time interval (minimum every 5 minutes).
- **FR8 – Logging and Traceability:**
The system shall log every calibration and recalibration run, including at least timestamps, used image sets, computed health scores, baseline update decisions, and error states.
- **FR9 – Abort and Logging on Recalibration Failure:**
The system shall abort the current recalibration cycle if a recalibration run fails. In such cases, the system shall retain the currently active baseline calibration and record a descriptive error message, including the failure reason, in the calibration log.

Non-Functional Requirements

- **NFR1 – Modularity:**
The system shall be structured into modular components (initial calibration, LiFCal recalibration, health monitoring, scheduling, and logging) to support extensibility and maintainability.
- **NFR2 – Reproducibility:**
All calibration results and health evaluations shall be reproducible using stored JSON files and logged metadata.
- **NFR3 – Robustness:**
The calibration and monitoring pipeline shall tolerate moderate variations in scene content and illumination without system failure, even if recalibration quality temporarily degrades.
- **NFR4 – Automation:**
The calibration framework shall operate fully automatically after initial setup, without requiring manual intervention during periodic monitoring and recalibration.
- **NFR5 – Transparency:**
The system shall provide access to intermediate calibration metrics and logged decision outcomes to allow inspection and analysis of health score computations and acceptance decisions.

These requirements form the basis for the system design and implementation presented in the following sections. They were derived following the guidelines and best practices of IEEE Std 830–1998 [13] and directly support the intended use case of the proposed calibration framework. The functional requirements are reflected in the architectural decisions, UML diagrams, and implementation steps, while the non-functional requirements address relevant quality aspects such as robustness, automation, and maintainability.

The requirements specification is intentionally kept compact and focused. It is not intended to represent a complete industrial requirements document, but rather a research-oriented specification that is appropriate for the scope and objectives of this bachelor thesis.

3.1.4. UML Class Diagram

Based on the requirements analysis and the defined use case, a UML class diagram was designed to structure and illustrate the main components of the project. The diagram in Figure 9 shows the key classes, their attributes, methods, and relationships. The software architecture is based on a modular design, so that individual components can be developed, tested, and maintained independently. If future extensions or modifications are needed, the structure allows the easy integration of new features without major changes to the existing codebase.

The core class is the **controller**, which manages the overall calibration workflow. It coordinates the initial calibration, periodic recalibration attempts, health monitoring, and logging. The controller interacts with several classes that contain the specific functionalities.

The initial calibration and periodic recalibration are implemented in the **InitialCalibration** and **run_LiFCal_calibration** classes, respectively. To run these classes correctly, the **run_initial_calibration**, **run_periodic_lifcal**, and **run_imageset_creation** classes are responsible for preparing the required data and executing the calibration processes. In addition, the **depthmap** class creates, based on the image set, a depth map for each image, which is used for the LiFCal calibration. For health monitoring and decision-making, the **HealthMonitoring** class computes the health scores and applies the acceptance rules based on the defined criteria. These criteria will be discussed in the next section.

The **LiFCal** class implements the target-free calibration method proposed by Fleith et al. [8]. This class encapsulates the core logic of LiFCal, including the bundle adjustment optimization. It provides methods to run the calibration process and retrieve the estimated camera parameters.

Finally, for results and traceability, the **ResultLogger**, **lifcal_to_json**, and **CalibrationState** classes handle the logging of calibration runs, health evaluations, and decisions. This ensures that all relevant information is recorded in a structured format for later analysis.

3.1.5. Health Monitoring and Decision Metrics

As mentioned in the use case description, a key component of the proposed calibration framework is the health monitoring mechanism. This mechanism evaluates the quality of each recalibration attempt and decides whether to accept or reject the new calibration based on defined metrics. To define these metrics, the following formulas are used:

$$\text{HealthScore} = 100 \cdot \left(1 - \left(0.6 \cdot E_{\text{reproj}}^2 + 0.4 \cdot E_{\text{rms}}^2\right)\right) \quad (9)$$

$$E_{\text{reproj}} = \text{clamp}\left(\frac{e_{\text{reproj}} - R_{\text{BEST}}}{R_{\text{WORST}} - R_{\text{BEST}}}, 0, 1\right) \quad (10)$$

$$E_{\text{rms}} = \text{clamp}\left(\frac{e_{\text{rms}} - S_{\text{BEST}}}{S_{\text{WORST}} - S_{\text{BEST}}}, 0, 1\right) \quad (11)$$

This health score formulation is used for the initial target-based calibration. It combines normalized reprojection error and RMS error into a single scalar score in the range $[0,100]$, where higher values indicate better calibration quality. Clamp functions ensure that the error terms are bounded between 0 and 1 based on predefined best and worst-case thresholds. The weights are chosen to give slightly higher priority to the reprojection error than to the RMS error. This choice follows common practice in the literature, where reprojection error is treated as a primary indicator of calibration correctness and overall geometric accuracy.

For the LiFCal-based recalibration, a similar health score is computed using LiFCal-specific metrics:

$$\text{HealthScore}_{\text{LiFCal}} = 100 \cdot \left[1 - \left(0.6 \cdot E_{\text{std}}^2 + 0.4 \cdot E_{\text{mae}}^2\right)\right] \quad (12)$$

$$E_{\text{std}} = \text{clamp}\left(\frac{\sqrt{x_{\text{std}}^2 + y_{\text{std}}^2} - S_{\text{BEST}}}{S_{\text{WORST}} - S_{\text{BEST}}}, 0, 1\right) \quad (13)$$

$$E_{\text{mae}} = \text{clamp}\left(\frac{\sqrt{x_{\text{mae}}^2 + y_{\text{mae}}^2} - R_{\text{BEST}}}{R_{\text{WORST}} - R_{\text{BEST}}}, 0, 1\right) \quad (14)$$

with

$$R_{\text{BEST}} = 0.03, \quad R_{\text{WORST}} = 1.0, \quad S_{\text{BEST}} = 0.0, \quad S_{\text{WORST}} = 50.0$$

This formulation is applied to LiFCal-based recalibration results. It evaluates the stability and consistency of the estimated camera parameters using standard deviation and mean absolute error metrics. The resulting health score enables a direct comparison with the baseline calibration to support automated acceptance or rejection decisions. The initial

and recalibration health scores are computed differently due to the different result metrics provided by the respective calibration methods. The specific best and worst-case values are defined also based on calibration performance which we discussed in Calibration Parameters and Error Metrics section.

3.1.6. State Diagram

In order to illustrate the workflow and interactions between the different calibration phases and components, a state diagram is presented in Figure 10. This diagram visualizes the transitions between the initial calibration, periodic recalibration attempts, and health evaluation steps.

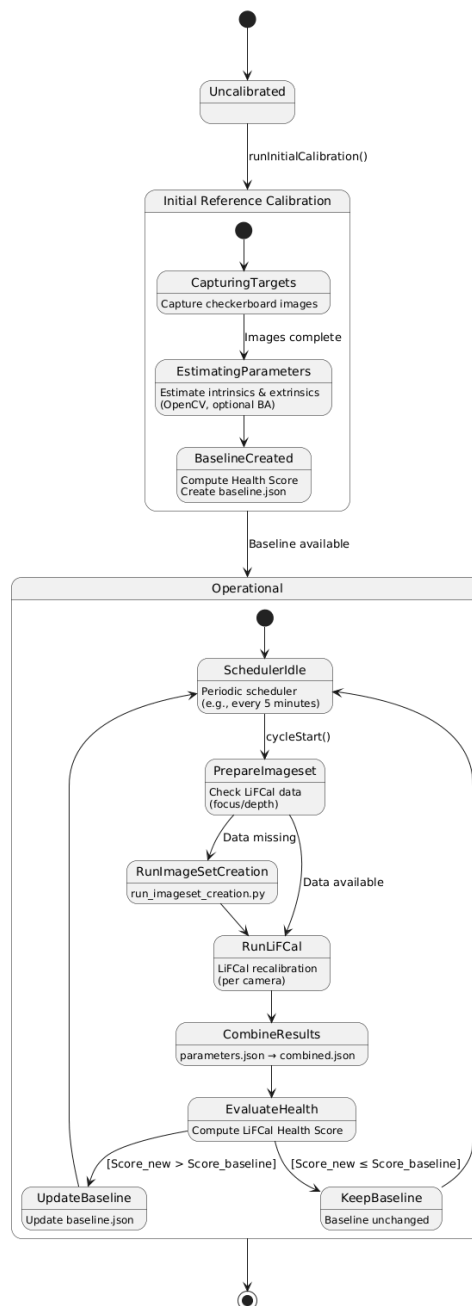


Figure 10: State Diagram

3.2. Materials

In this chapter, the materials and tools used for the implementation and testing of the calibration software are described and presented. Due to the lack of a physical multi-camera light field system, synthetic data generation with Blender was used to create realistic test scenarios. The structure of the image sets, ground truth definitions, and the runtime environment are also discussed.

3.2.1. Synthetic Data Generation with Blender

To evaluate and test the proposed system, a world scene was simulated using open-source software. This software is Blender [3], which also considers physical properties of light, materials, and camera optics to generate realistic images.

The first step was to create the multi-camera array geometry. As described in the use case, a cross-shaped arrangement with one central camera and twelve surrounding cameras was constructed. Each camera was configured with the same sensor size and a focal length of 50mm and a sensor size of 36mm to make the image sets as realistic as possible. The cameras were placed at equal distances from their neighboring cameras, forming a cross shape. Each camera was positioned on a grid to simulate the main structure of the array, as shown in Figure 11.

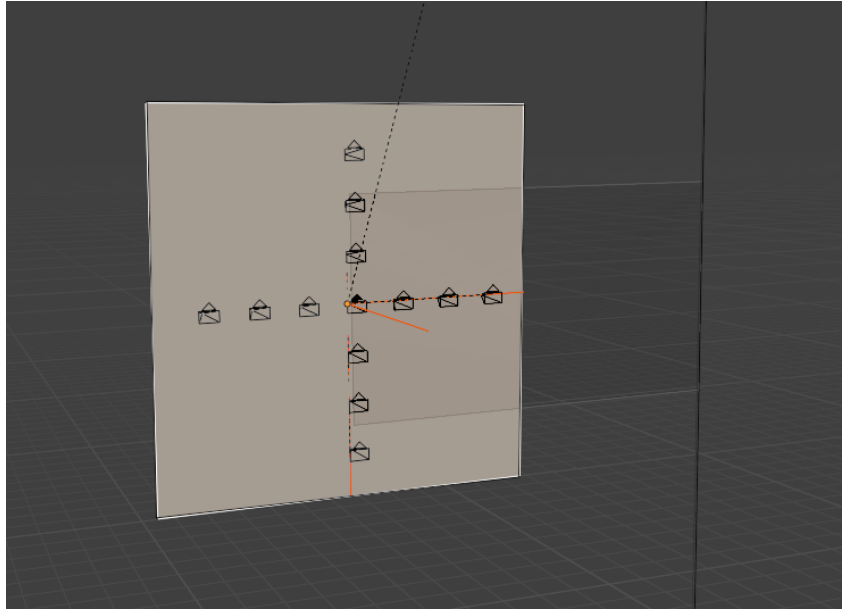


Figure 11: Blender visualization of the synthetic 13-camera light field array. It consists of one central reference camera and twelve surrounding cameras arranged in a cross-shaped configuration with equal spacing.

The first scene that needed to be created included a calibration target for the initial calibration. Therefore, a checkerboard pattern was modeled and placed in front of a scene obtained from the Blender community [24]. The checkerboard was positioned in different locations and orientations to simulate a realistic calibration scenario. Due to the multi-camera setup, rendering the images took several minutes per pose, because each camera view had to be rendered individually. After rendering, the images were organized into folders according to the naming conventions required by the calibration

scripts. The scene created for the initial calibration was modeled as a simple room, as shown in Figure 12.

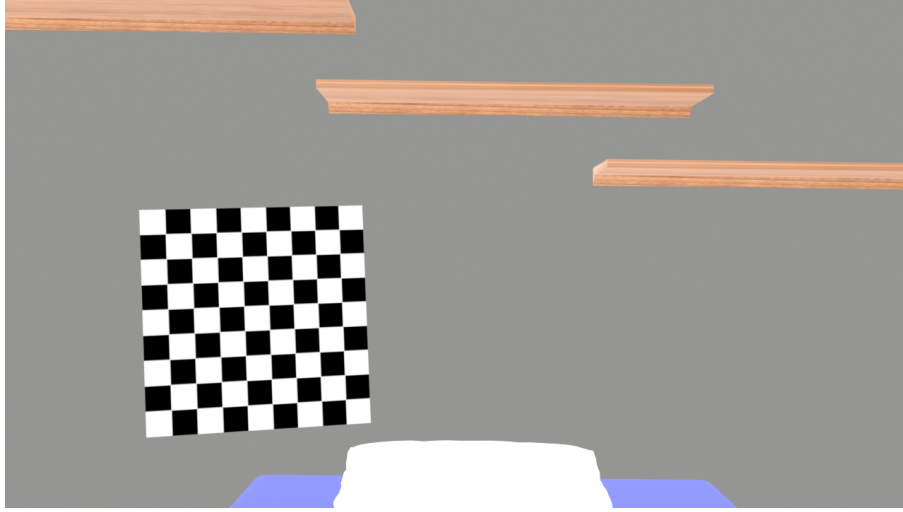


Figure 12: Blender scene used for the initial target-based calibration. The checkerboard pattern, placed at varying positions and orientations in front of the camera array, serves as the calibration target for estimating intrinsic and extrinsic parameters.

For the LiFCal-based recalibration, a different scene was created without a calibration target. Since LiFCal relies on feature detection and depth maps, the scene had to be more complex and include a variety of objects and textures. Therefore, another scene from the Blender community was used [22], which represents a modern living room with furniture and decorations. The same camera setup was used, and the scene was rendered from different viewpoints to create the required image sets. An example of the scene used for LiFCal recalibration is shown in Figure 13.

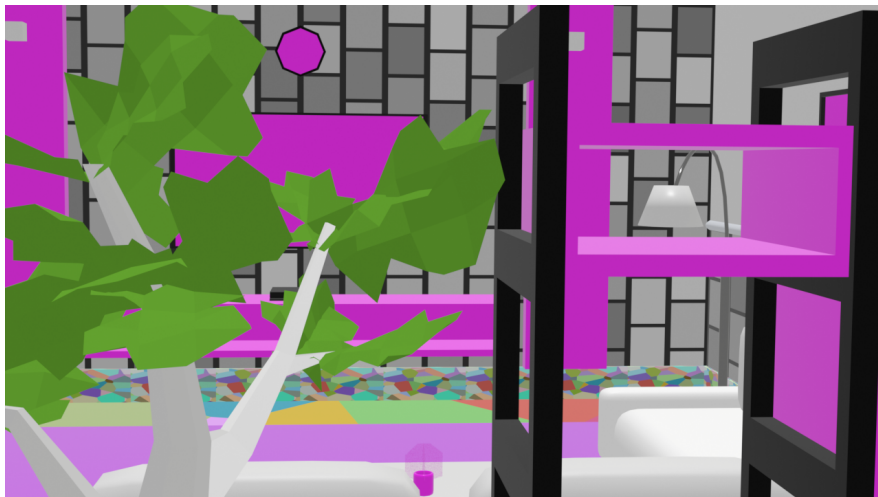


Figure 13: Blender scene used for LiFCal-based recalibration without a calibration target. The scene contains multiple objects with diverse colors, textures, and lighting conditions to provide rich visual features and depth variation, enabling robust feature detection.

The motivation behind using synthetic data was to maintain full control over the scene configuration and to simulate different calibration scenarios without the need for physical hardware. Another major advantage is the availability of ground truth camera parameters, which is essential for comparing the estimated calibration results with the actual values. This enables a more objective evaluation of calibration accuracy and the effectiveness of the implemented software.

3.2.2. Image Sets and Ground Truth Definition

To evaluate the calibration methods effectively, multiple image sets were created with different numbers of poses and camera views. In addition, for each image set, the multi-camera system was deliberately decalibrated to simulate real-world scenarios in which calibration quality may degrade over time due to mechanical or environmental influences.

The naming conventions for the image sets followed a structured and consistent format to ensure maintainability and usability. Each image set was organized into folders according to the calibration phase (initial calibration or LiFCal-based recalibration). File names encoded both the pose number and the camera identifier, for example `pose_001_Up1.png`.

The ground truth camera parameters were directly extracted from Blender. Since camera positions, orientations, and intrinsic parameters were explicitly defined within the 3D scene, the exact intrinsic and extrinsic parameters were known internally. The ground truth data was stored in JSON format, following the same structure as the calibration results, to enable direct and systematic comparison during evaluation. For each camera, both intrinsic parameters and extrinsic poses were included.

4. Results and Evaluation

In this chapter, the results, plots, and tables obtained from the implementation and from the comparison with the available ground truth data are presented. In addition, the runtime environment and toolchain used for the evaluation are described to ensure reproducibility of the experimental setup. The evaluation focuses on the accuracy of the initial target-based calibration as well as on the performance of the LiFCal-based recalibration. The results are reported in terms of intrinsic and extrinsic parameter accuracy, reprojection errors, and the evolution of health scores.

4.1. Runtime Environment and Toolchain

Most parts of the implementation are based on established open-source libraries and tools. The primary programming language used is Python, chosen for its extensive ecosystem supporting computer vision, numerical computation, and scientific prototyping. Core libraries include OpenCV for calibration and image processing, NumPy for numerical operations, and Matplotlib for visualization. The LiFCal implementation provided by Fleith et al. [8] was integrated into the workflow to perform target-free recalibration.

The development and evaluation were conducted on a system running Ubuntu 20.04.6 LTS. A complete list of software dependencies and library versions is documented in the project repository README files to support reproducibility. It should be noted that the LiFCal implementation relies on GPU acceleration and requires a compatible NVIDIA GPU to execute the optimization efficiently. Consequently, LiFCal-based recalibration cannot be performed on systems without suitable GPU support.

4.2. Results of Initial Calibration

The results of the initial calibration are categorized into intrinsic accuracy, extrinsic accuracy, and reprojection and RMS error. Each category is discussed in the following subsections. To visualize the results effectively, the plots are taken from one specific image set in which the camera system was deliberately decalibrated by a certain degree to simulate real-world conditions. Additional plots for all recalibration runs are provided in subsection A.1.

4.2.1. Intrinsic Accuracy

The accuracy of the intrinsic parameters is evaluated by comparing the estimated focal lengths and principal points with the ground truth values obtained from Blender. The deviations in f_x , f_y , c_x , and c_y are shown in Figure 14 and Figure 15.

Figure 14 shows the deviations of f_x and f_y from the ground truth for each camera in pixels. The blue curve represents the deviation of f_x , while the orange curve represents the deviation of f_y . The estimated f_x values remain close to zero across all cameras, whereas the f_y values exhibit a nearly constant offset of approximately 275 pixels. This systematic offset originates from the Blender camera model, where the sensor fit can only be configured as either horizontal or vertical. As a consequence, one focal length component (f_x or f_y) is scaled accordingly, resulting in a constant deviation. In Figure 15, the normalized relative error of f_x is shown to improve the visualization of the deviations,

with the maximum error reaching approximately 0.1%. A normalized representation for f_y is not included due to the systematic scaling offset described above.

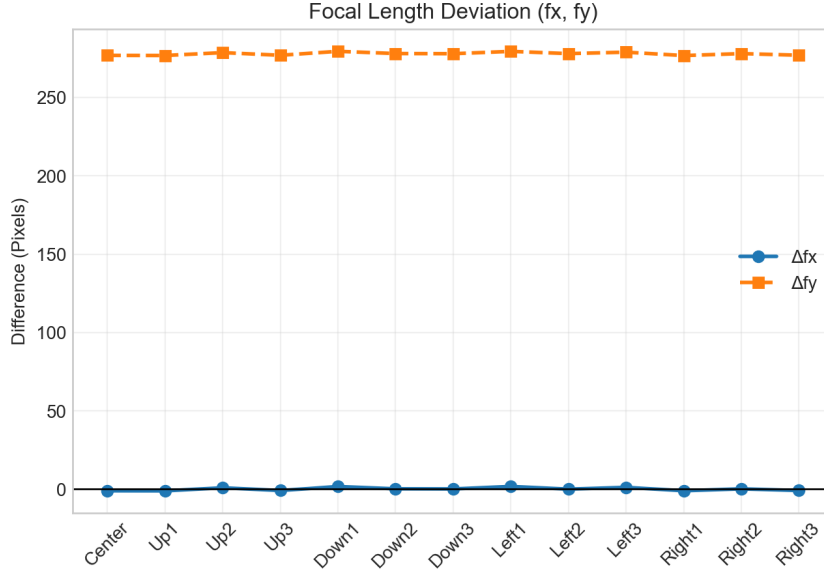


Figure 14: Absolute deviation of the estimated focal lengths from ground truth for all cameras in the initial calibration. The blue curve represents the deviation of f_x , while the orange curve represents the deviation of f_y , both measured in pixels. A horizontal zero line indicates perfect agreement with the ground-truth parameters.

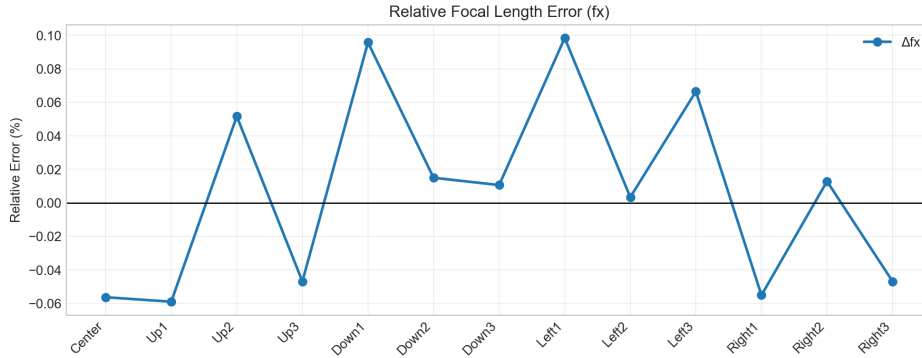


Figure 15: Relative focal length error of f_x for all cameras in the initial calibration, computed as $\frac{f_{\text{est}} - f_{\text{gt}}}{f_{\text{gt}}} \cdot 100\%$. The plot shows the normalized percentage deviation with respect to the ground-truth focal length, allowing a scale-independent interpretation of the estimation accuracy.

The ground-truth intrinsic parameters were computed from the Blender camera configuration using a focal length of 50mm, a sensor size of 36mm and an image resolution of 1280×720 pixels. This results in ground-truth values of $f_x = 1777.78$ and $f_y = 1500.0$, with principal point coordinates $c_x = 640$ and $c_y = 360$. In comparison, the calibration process estimates values of approximately $f_x \approx 1776.78$ and $f_y \approx 1776.86$, and principal point coordinates of approximately $c_x \approx 639.12$ and $c_y \approx 358.92$ for the central camera.

While the estimated f_x and principal point coordinates are close to the ground-truth values, the estimated f_y differs significantly from the ground-truth value of 1500.0, which shows the nearly constant offset observed in the deviation plots.

The deviations of c_x and c_y are illustrated in Figure 16. The principal point coordinates show small deviations from the ground-truth values across all cameras. The deviations remain within a narrow pixel range and are distributed around zero, indicating only minor differences between the estimated and reference values.

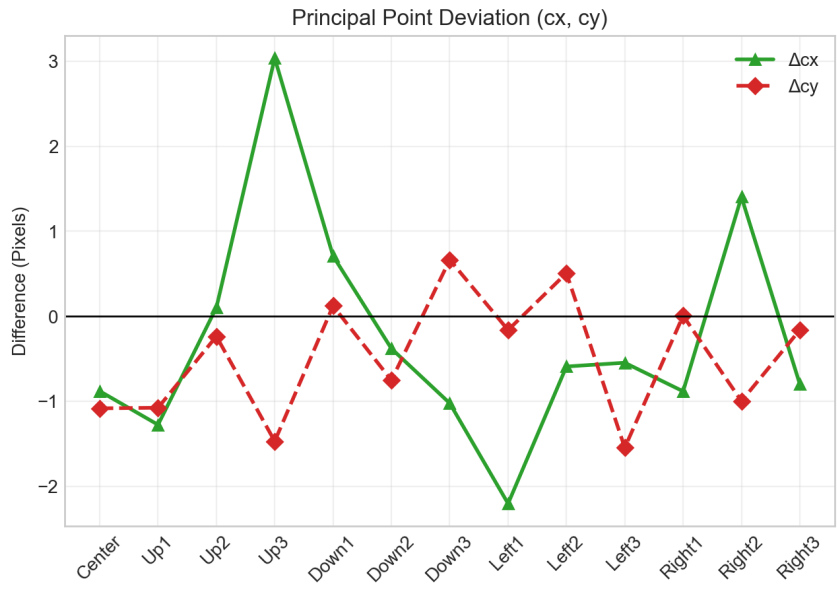


Figure 16: Absolute deviation of the estimated principal point coordinates from ground truth for all cameras in the initial calibration. The green curve represents the deviation of c_x , while the red dashed curve represents the deviation of c_y , both measured in pixels. The horizontal zero line indicates perfect agreement with the ground-truth values.

4.2.2. Extrinsic Accuracy

For the evaluation of the extrinsic parameters, the estimated camera poses are compared with the corresponding ground truth poses. In this case, rotation and translation errors are evaluated using two separate plots to clearly visualize the results, as shown in Figure 17 and Figure 18.

In the first plot (Figure 17), the 3D positions of the multi-camera array from the ground truth are shown, while the second plot (Figure 18) illustrates the estimated camera positions after the initial calibration. The mean translation error is shown in the lower part of the figure. The overall structure size is larger than that of a typical multi-camera array due to the synthetic setup in Blender. In both plots the main structure of the cross-shaped camera array is clearly visible.

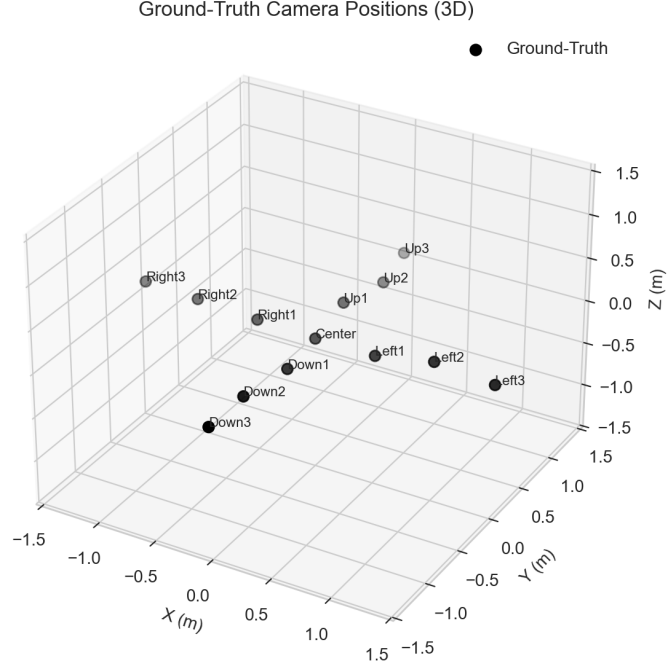


Figure 17: Ground-truth 3D positions of all cameras in the synthetic multi-camera array. The plot shows the spatial arrangement of the cameras in the (X, Y, Z) coordinate system, illustrating the cross-shaped configuration used for the simulation.

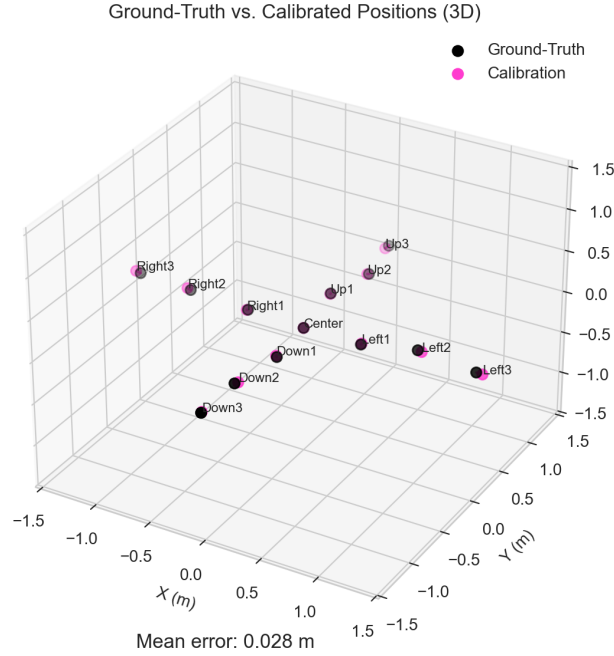


Figure 18: Comparison of ground-truth and estimated camera positions after the initial calibration. Black markers represent the ground-truth positions, while magenta markers indicate the calibrated positions. The spatial overlap illustrates the translation estimation accuracy, with a reported mean position error of 0.028m.

The translation error per camera and the rotation error per camera are further evaluated in the plots shown in Figure 19 and Figure 20.

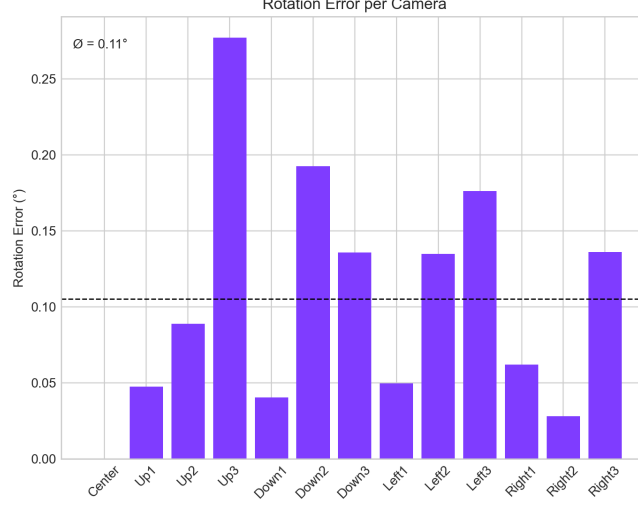


Figure 19: Rotation error per camera after the initial calibration. The bars represent the angular deviation between the estimated and ground-truth extrinsic rotations, measured in degrees. The dashed horizontal line indicates the mean rotation error across all cameras ($\emptyset = 0.11^\circ$).

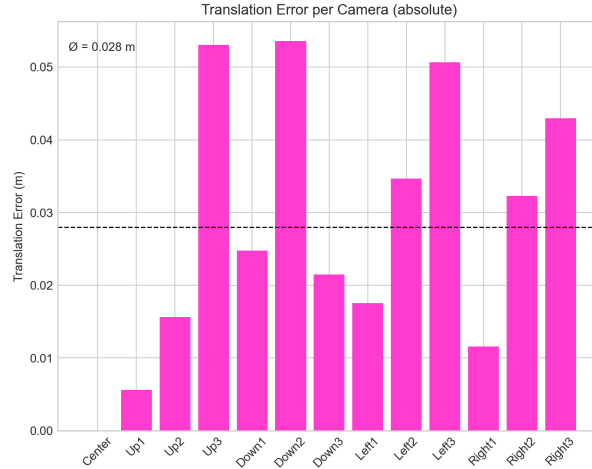


Figure 20: Absolute translation error per camera after the initial calibration. The bars represent the Euclidean distance between the estimated and ground-truth camera positions, measured in meters. The dashed horizontal line indicates the mean translation error across all cameras ($\emptyset = 0.028$ m).

The first figure presents the angular deviation for each camera, while the second figure illustrates the corresponding positional error. Since the central camera was defined as the reference frame during the calibration process, its extrinsic parameters serve as the origin of the coordinate system. Consequently, its rotation and translation errors are zero by definition.

4.2.3. Reprojection and RMS Error

As discussed in the Calibration Parameters and Error Metrics section, reprojection error and RMS error are important indicators of calibration quality. The plot shown in Figure 21 presents the reprojection error per camera of the initial calibration.

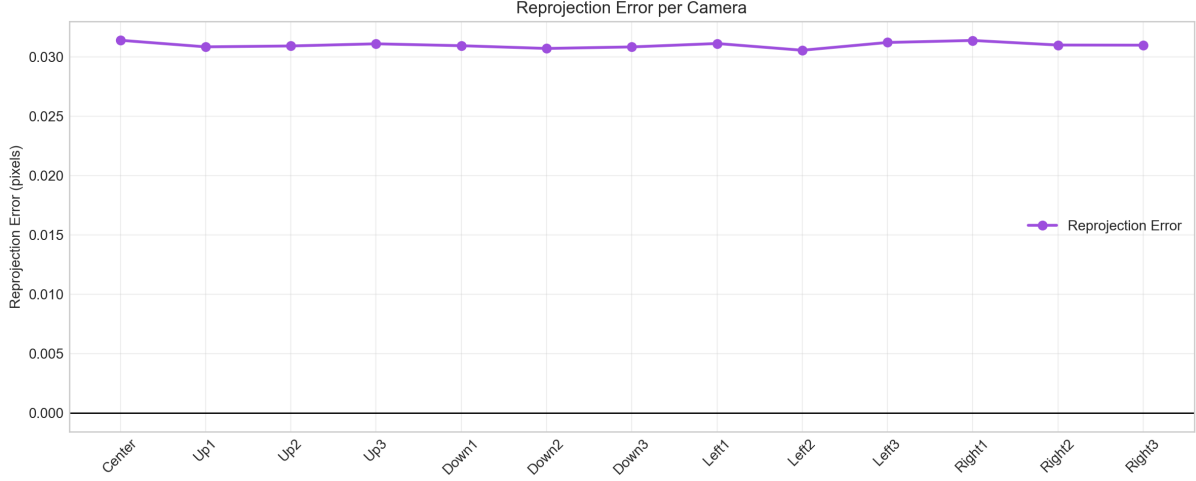


Figure 21: Reprojection error per camera of the initial calibration.

The RMS error for the multi-camera array is shown in Figure 22. These values are automatically provided by the OpenCV stereo calibration function.

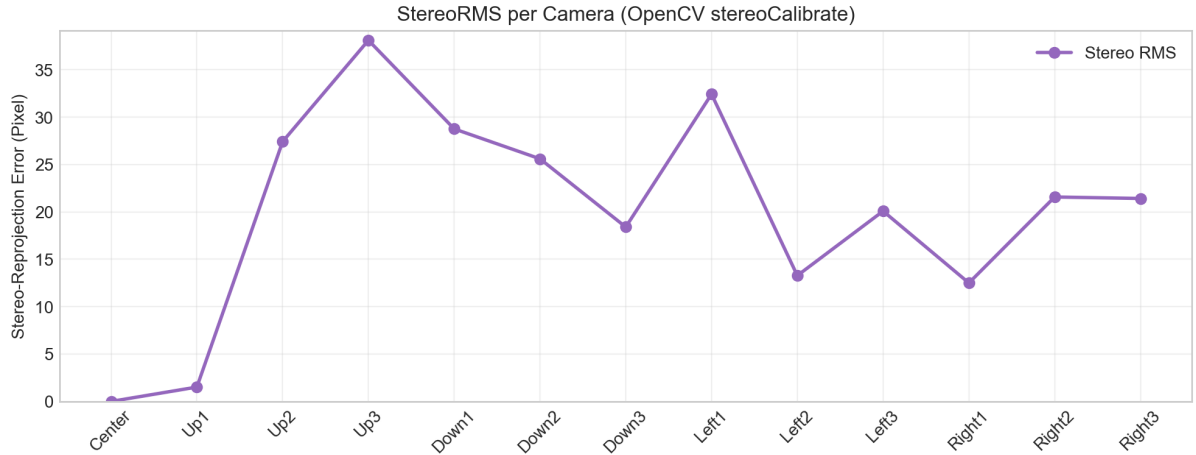


Figure 22: Stereo RMS error of the initial calibration.

Both the reprojection error and the RMS error are measured in pixels. The two plots were generated using the calibration results of the initial target-based calibration. In Figure 21 the error values are almost constant across all cameras with a reprojection error of about 0.3 pixels. In Figure 22 the RMS error values vary between 15 and 35 pixels for the different cameras.

4.2.4. Comparison of Mean Results Across Image Sets

To provide a consolidated overview of the initial calibration results, the mean values across different image sets are summarized in Table 1. The table reports the number of

poses used for each image set, whether bundle adjustment (BA) was applied, as well as the mean translation error, mean rotation error, and mean reprojection error.

Name	#Poses	BA	Mean ΔT [m]	Mean ΔR [°]	Mean Reproj. [px]
imageset1	41	yes	0.07576	0.17447	0.33859
imageset2	33	yes	0.04710	0.16089	0.03274
imageset3	41	yes	0.09500	0.18225	0.40989
imageset4	17	yes	0.05414	0.23255	0.03103
imageset5	28	yes	0.02798	0.10531	0.03100
imageset6	36	yes	0.06228	0.20084	0.03070
imageset7	20	yes	0.04716	0.21424	0.03166

Table 1: Mean calibration results for different image sets

To analyze the calibration performance under different conditions, the image sets were generated with varying numbers of poses. The first four image sets (imageset1 to imageset4) are grouped into Scenario A_Robustness. In this scenario, the same multi-camera decalibration was applied in order to evaluate how robust the calibration results are with respect to different checkerboard poses.

The remaining image sets (imageset5 to imageset7) belong to Scenario B_Generalization. In this case, different multi-camera decalibrations were applied to assess how well the calibration generalizes to different camera configurations with varying numbers of poses. A summary of both scenarios is provided in Table 2.

Scenario	#Runs	Avg. #Poses	Mean ΔT [m]	Mean ΔR [°]	Mean Reproj. [px]
A_Robustness	4	33.0	0.06800	0.18754	0.20306
B_Generalization	3	28.0	0.04581	0.17346	0.03112

Table 2: Scenario-based summary of calibration results

4.3. Results of LiFCal-Based Recalibration

In this section, the results of the LiFCal-based recalibration are presented and visualized using plots. Due to the fundamentally different nature of the LiFCal calibration results compared to the available ground truth data, the evaluation focuses exclusively on reprojection error and the resulting health score.

A direct comparison of extrinsic parameters is not meaningful in this case. LiFCal requires multiple poses of the scene, with each pose resulting in a separate parameter estimation. Consequently, extrinsic parameters vary across poses and cannot be compared directly to a single ground truth configuration. In addition, LiFCal calibrates each camera independently, leading to camera parameters defined in individual local coordinate systems rather than in a shared global reference frame. For these reasons, intrinsic and extrinsic parameters across cameras are not directly comparable in the same way as in the target-based calibration. These limitations are discussed in more detail in the Discussion of the Results section.

4.3.1. Reprojection Error after LiFCal

For the evaluation of the LiFCal-based recalibration, three independent recalibration runs using different image sets are considered. As a representative example, the reprojection error per camera for one selected run is shown in Figure 23. This visualization follows the same structure as the reprojection error plot for the initial target-based calibration shown in Figure 21. However, due to the output format of LiFCal, the reprojection error is reported separately for the x and y image directions.

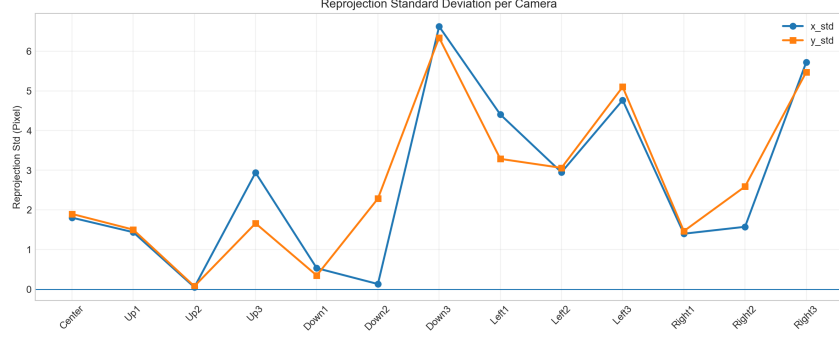


Figure 23: Reprojection error per camera for a single LiFCal-based recalibration run.

To assess the consistency of the reprojection error across multiple recalibration runs, Figure 24 shows the reprojection errors for all three runs. The upper part of the figure visualizes the x reprojection errors for each camera, while the lower part shows the corresponding y reprojection errors. This representation allows for a direct comparison of the reprojection error distributions across different runs and camera indices.

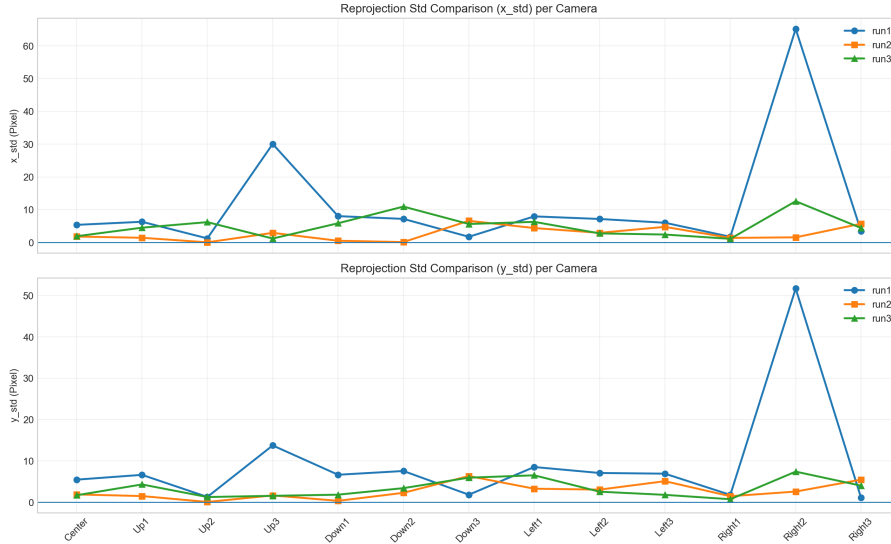


Figure 24: Reprojection error per camera across three LiFCal-based recalibration runs.

In addition, a direct comparison between the LiFCal-based recalibration and the initial target-based calibration is provided in Figure 25. The figure compares the mean reprojection errors in x and y for the best-performing LiFCal run with the reprojection error obtained from the initial calibration.

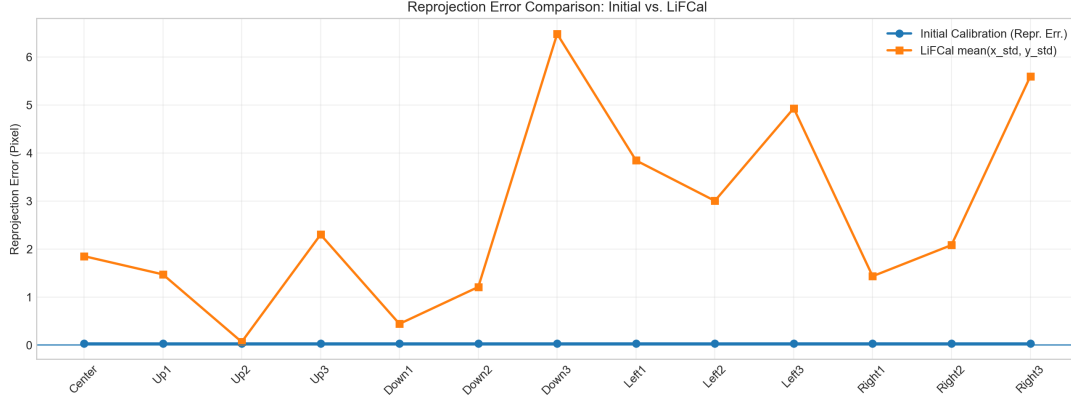


Figure 25: Comparison of mean reprojection error per camera between initial calibration and LiFCal-based recalibration. The blue curve corresponds to the one shown in Figure 21 and the orange curve corresponds to the one shown in Figure 23.

The reprojection error of the initial target-based calibration remains close to zero for all cameras, with all values around 0.03px. After LiFCal-based recalibration, the reprojection error is higher for every camera. The values show a clear separation between the two calibration stages.

4.3.2. Health Score Comparison

The health scores are calculated using the formulas presented in the Health Monitoring and Decision Metrics section. All health scores for both the LiFCal-based recalibration runs and the initial target-based calibration image sets are summarized in Table 3. Note that the score range is $[0, 100]$, with higher values indicating better calibration quality.

Name	Calibration Type	#Poses	BA	Health Score
Run 1	LiFCal	4	true	52.0538
Run 2	LiFCal	10	true	59.4736
Run 3	LiFCal	20	true	58.7394
LiFCal (Mean $\pm$ Std)				56.76 $\pm$ 4.09
Imageset1	Initial	41	true	91.6138
Imageset2	Initial	33	true	97.8905
Imageset3	Initial	41	true	87.0450
Imageset4	Initial	17	true	97.1769
Imageset5	Initial	28	true	92.9879
Imageset6	Initial	36	true	88.6221
Imageset7	Initial	17	true	97.1303
Initial (Mean $\pm$ Std)				93.21 $\pm$ 4.37

Table 3: Overview of calibration runs and corresponding health scores

The first block of the table summarizes the health scores obtained from the three LiFCal-based recalibration runs, while the second block presents the health scores from the initial target-based calibration image sets. The health scores for the LiFCal runs are significantly lower than those of the initial calibration. The highest health score among the LiFCal

runs is approximately 59.47, while the highest health score from the initial calibration is around 97.89. The lowest health score from the initial calibration is 87.05, and the lowest from the LiFCal runs is 52.05. On average, the mean health score is 56.76 ± 4.09 for the LiFCal runs and 93.21 ± 4.37 for the initial calibration.

5. Discussion of the Results

In this part of the thesis, the obtained results are analyzed and discussed in detail. The focus is on interpreting the findings from both the initial target-based calibration and the LiFCal-based recalibration. The limitations of the LiFCal method in the given setup are examined, and a comparison between the two calibration approaches is provided. The research questions outlined in the Introduction are also answered based on the results. Finally, potential extensions and improvements for future work are suggested.

5.1. Interpretation of Initial Calibration Results

The initial target-based calibration method provided very accurate results in terms of intrinsic and extrinsic parameter estimation. As shown in Figure 21, the reprojection error remained consistent across all cameras with a value of about 0.03 pixels. This indicates that the calibration process was able to accurately model the camera geometry and lens distortions. The intrinsic parameters, particularly the focal lengths and principal points, also showed small deviations from the ground truth values, as illustrated in Figure 15 and Figure 16. In this case, the f_x values were very close to the ground truth, while the f_y values exhibited a nearly constant offset of approximately 275 pixels. This systematic deviation is due to the synthetic nature of the data and the specific configuration of the Blender camera model, because the sensor width and height were automatically set. Either the width or the height could have been adjusted and the other parameter would have been calculated accordingly to maintain the aspect ratio. This led to a consistent offset in one of the focal length parameters.

The RMS error values, shown in Figure 22, were not as low as expected, ranging from 15 to 35 pixels across the cameras. However, these values are not directly comparable to the per-camera reprojection errors, as the RMS returned by OpenCV’s `stereoCalibrate` represents a joint error over both camera views and also reflects the consistency of the estimated relative camera poses. Small inaccuracies in corner detection or checkerboard poses can therefore result in higher RMS values, especially before applying bundle adjustment.

The extrinsic parameters also demonstrated good accuracy, with mean translation errors around 0.068 meters and mean rotation errors approximately 0.187 degrees across different image sets, as summarized in Table 2. Almost all image sets achieved reprojection errors below 0.04 pixels, indicating a high level of geometric consistency in the estimated camera poses. However, in `imageset1` and `imageset3`, the reprojection errors were slightly higher, around 0.34 pixels and 0.41 pixels, respectively. These two anomalies are caused by the quality of the taken images during the rendering process in Blender. In these two image sets, the checkerboard distance to the camera was larger than in the other image sets, which made it more difficult to detect the corners accurately. This led to less precise calibration results and consequently higher reprojection errors. More poses with closer checkerboard positions would likely have improved the calibration accuracy in these cases. That shows the importance of the variety and quality of the calibration images for achieving optimal results. A higher number of poses does not necessarily guarantee better calibration if the image quality is compromised.

In Table 2 it can be observed that the robustness and generalization capabilities of the initial calibration method are stable across different decalibration scenarios. Scenario A_Robustness, which tested the same decalibration with varying checkerboard poses, yielded consistent results with low mean translation and rotation errors. Scenario B_Generalization, which involved different decalibrations, also produced reliable calibration outcomes, demonstrating the method’s ability to adapt to various camera configurations. This indicates that the initial target-based calibration is effective in maintaining accuracy under different conditions, provided that the calibration images are of sufficient quality.

5.2. Limitations of LiFCal in the Given Setup

The LiFCal-based recalibration results revealed several limitations when applied to the synthetic multi-camera light field setup. The visualization in Figure 25 clearly shows that the reprojection errors after LiFCal-based recalibration are significantly higher than those obtained from the initial target-based calibration.

However, it is important to note that LiFCal is designed as a target-free calibration method that relies on feature detection and depth maps. The synthetic data generated in Blender may not fully capture the complexity and variability of real-world scenes, which can negatively affect the performance of LiFCal. In addition, the depth maps used in the LiFCal process are derived from the synthetic scenes and may not accurately represent the depth information that would be obtained from real images.

A major reason for the high reprojection errors is that LiFCal was primarily developed for micro-lens-array (MLA) light field cameras rather than for multi-camera array systems. The assumptions made by LiFCal regarding the camera model and the image formation process do not fully hold for the multi-camera array setup used in this thesis. This mismatch can lead to inaccuracies in the estimated camera parameters and consequently to higher reprojection errors. Nevertheless, LiFCal provides a systematic target-free recalibration approach, which is a significant advantage in scenarios where the use of physical calibration targets is impractical.

Furthermore, as discussed in the use case description, the main purpose of the automatic recalibration software is to maintain calibration quality over time without manual intervention. Although LiFCal does not require a calibration target, it still depends on multiple images of the scene captured from different viewpoints. This implies that the camera system must be able to move within the scene in order to generate the required input data for LiFCal. In practical applications, this requirement may not always be feasible, particularly in static environments where the camera system cannot be repositioned easily. Therefore, while LiFCal enables a target-free calibration process, it still relies on a certain degree of controlled motion or data acquisition, which can limit its applicability in fully automated and static scenarios.

With regard to the first research question (RQ1) formulated in the Introduction—“*Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?*”—the results presented in this chapter indicate that LiFCal can be applied in a target-free manner, but

its effectiveness in the given multi-camera light field setup is limited. The observed high reprojection errors suggest that LiFCal, in its current form, is not sufficient to achieve accurate calibration for this specific system without further adaptations. Nevertheless, the method provides a valuable foundation for an automated recalibration pipeline that can be improved and extended in future work.

5.3. Comparison between Initial Calibration and LiFCal

The comparison between the initial target-based calibration and the LiFCal-based recalibration highlights several key differences in terms of accuracy and limitations. As shown in Figure 24, the reprojection errors after LiFCal recalibration are inconsistent across the different runs, while the initial calibration produced stable and low reprojection errors across all cameras and image sets, as illustrated in Table 1. For the multi-camera array system used in this thesis and the synthetic data generated in Blender, the initial calibration method clearly outperformed LiFCal in terms of accuracy. The health scores summarized in Table 3 further emphasize this discrepancy, with the initial calibration achieving scores above 87, while the highest score obtained with LiFCal was only around 59. This substantial difference indicates that the initial calibration method is more effective at accurately modeling the camera system under the given conditions.

As mentioned earlier, the extrinsic parameters estimated by LiFCal cannot be directly compared to the ground truth data due to the independent calibration of each camera and the absence of a shared global reference frame. This limitation restricts the evaluation of the overall geometric consistency of the LiFCal results in the same way as for the initial calibration. Consequently, the comparison focuses primarily on reprojection errors and health scores, which provide a more indirect assessment of calibration quality.

With respect to the second research question (RQ2) posed in the Introduction—“*How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?*”—the findings show that LiFCal, in the current setup and within the design of the proposed software, does not perform well enough to replace a classical checkerboard-based calibration. The significantly higher reprojection errors and lower health scores indicate that LiFCal is currently unable to achieve the same level of accuracy as the initial calibration method. In addition, the observed outliers and inconsistencies in the LiFCal results suggest that further refinements and adaptations are necessary to improve its performance for multi-camera light field systems. Therefore, while LiFCal represents a promising approach, it cannot currently serve as a complete replacement for traditional calibration methods in this context.

5.4. Comparison to the State of the Art

The state of the art in multi-camera light field calibration is still dominated by target-based approaches that use physical calibration patterns (e.g., checkerboards) to estimate intrinsic and extrinsic parameters reliably. The initial calibration implemented in this thesis follows this classical line of work and achieved consistently low reprojection errors (around 0.03px across cameras) as well as high health scores (with a mean health score of 93.21 ± 4.37 across the evaluated image sets.). Overall, these results are in line with what is typically reported for well-controlled target-based calibration in multi-camera systems.

Target-free approaches such as LiFCal are attractive because they avoid physical targets and, in principle, allow calibration updates with less manual setup. In the setup used in this thesis, LiFCal did not reach the same level of accuracy, as the reprojection errors were clearly higher across cameras, and the resulting health scores stayed far below those of the initial calibration (with a mean health score of 56.76 ± 4.09 across the three runs). This suggests that, at least in the current form and for the considered multi-camera array scenario, LiFCal cannot replace the checkerboard-based baseline.

A practical contribution of this thesis to the state of the art is the automated calibration pipeline. It combines an initial reference calibration with periodic target-free recalibration attempts and a decision mechanism based on a health score. This directly addresses the real-world problem of calibration drift over time, where repeated manual target-based calibration can become impractical. Using Blender-based synthetic scenes also made it possible to run controlled experiments with access to ground truth parameters, which are difficult to obtain in physical multi-camera setups. This innovative approach provides a useful method for evaluating and comparing calibration techniques in a reproducible manner.

In summary, the initial target-based calibration matches established practice and produces strong results in this environment, while the LiFCal-based recalibration mainly shows potential but requires further adaptation before it can be considered a robust alternative for multi-camera light field systems.

5.5. Potential Extensions and Improvements

The current state of the implemented calibration software provides a solid foundation. However, there are several potential extensions and improvements that could be explored in future work to fulfill the goal of fully automated and reliable multi-camera light field calibration. One promising direction is to adapt and enhance the LiFCal method specifically for multi-camera array systems. This could involve modifying the underlying camera model assumptions and optimization strategies to better suit the geometry and characteristics of multi-camera setups.

An optional improvement would be to implement the software in a more performance-oriented programming language, such as C++ or Rust. This would allow faster execution of the calibration pipeline, which is particularly important for real-time applications or scenarios where frequent recalibration is necessary. Additionally, integrating parallel processing techniques could further speed up the calibration process, especially when dealing with large image sets or high-resolution images.

The evaluation of the calibration framework using a real multi-camera light field system would be a crucial next step. This would provide a more realistic assessment of the calibration methods and their performance in practical scenarios. Testing the LiFCal method on real-world data would also help to identify additional challenges or limitations that may not be apparent in synthetic environments. If the recalibration process can be successfully demonstrated on real hardware, it would be necessary to extend the existing code to capture automatically images from the multi-camera system and feed them into

the calibration pipeline without manual intervention.

In order to explicitly answer the last research question (RQ3) from the Introduction—“*Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?*”—the following conditions and requirements can be identified based on the findings of this thesis:

- **LiFCal Adaptation:** The LiFCal method needs to be specifically adapted for multi-camera array systems to account for their unique geometry and characteristics.
- **Scene Complexity:** The scenes used for target-free recalibration should contain sufficient texture and depth variation to enable robust feature detection and depth estimation.
- **Camera Movement:** The camera system must be capable of capturing multiple viewpoints of the scene to provide the necessary data for LiFCal.
- **Quality of Input Data:** High-quality images with minimal noise and distortion are essential for accurate calibration results.
- **Computational Efficiency:** The calibration software should be optimized for performance to allow timely recalibration in practical applications.
- **Evaluation with Real Data:** Testing and validating the calibration methods using real multi-camera light field systems is crucial to ensure their effectiveness in practical scenarios.
- **Health Monitoring:** Implementing a robust health monitoring mechanism to assess calibration quality and decide when recalibration is necessary.

In conclusion, the requirements outlined in the Methods and Materials section and the potential improvements discussed here provide a guideline for future research and development in the field of data-driven automatic multi-camera light field calibration. Addressing these aspects will be key to achieving reliable and efficient target-free recalibration in practical applications.

6. Conclusion and Outlook

The final chapter summarizes the key aspects of the thesis. On one hand, the main achievements and findings are recapped in the Conclusion subsection. On the other hand, in the Outlook subsection, potential future directions and improvements for the calibration software and methods are discussed.

6.1. Conclusion

In this thesis, a data-driven automatic calibration software for multi-camera light field systems was implemented and evaluated. It combines an initial target-based calibration using a checkerboard, which is used to estimate the calibration parameters as a reference, with a target-free recalibration approach based on the LiFCal method. In addition, a health score monitoring mechanism is used to decide whether a new calibration result improves the overall calibration quality. The health score ranges from 0 to 100, where higher values correspond to better calibration quality. The system is designed to perform recalibration automatically, without requiring manual intervention, in order to counteract calibration drift over time. The software was implemented in Python, and synthetic image data generated with Blender was used to evaluate the calibration results against known ground truth parameters.

The results showed that the initial target-based calibration achieved high accuracy, with reprojection errors of approximately 0.03px per camera and health scores ranging between about 87 and 98 across the evaluated image sets. In contrast, the LiFCal-based recalibration resulted in significantly higher reprojection errors for all cameras and lower health scores, which ranged between approximately 52 and 59 in the performed recalibration runs. These results indicate that, in the given multi-camera light field setup, LiFCal is currently not suitable to replace a classical checkerboard-based calibration. The main limitations of LiFCal in this context were analyzed, and possible reasons for its reduced performance were discussed.

In conclusion, the implemented software provides a useful framework for automatic multi-camera light field calibration and systematic evaluation of calibration quality. However, further adaptations and improvements are required before the target-free recalibration approach can achieve a calibration accuracy comparable to that of the initial target-based method.

6.2. Outlook

For future work, several extensions of the calibration software were identified. One possible direction is to adapt the LiFCal method more explicitly to multi-camera array systems. This would require changes to the underlying camera model and the optimization process so they match the geometry and constraints of camera arrays more closely. Such adaptations may lead to more stable results when the method is applied to multi-camera configurations.

Another potential improvement concerns the implementation itself. Rewriting the software in a faster programming language, such as C++, would reduce runtime and make re-

peated recalibration more practical. This becomes relevant in real-world scenarios where recalibration is performed frequently or where time constraints play a role.

A further step would be the evaluation of the framework on a real multi-camera light field system. Testing on real hardware would allow a more realistic assessment of calibration accuracy and robustness. It would also expose practical issues which do not appear in synthetic experiments, such as noise, mechanical inaccuracies, or synchronization effects. Overall, the project presented in this thesis provides a solid foundation for further research and development in the field of multi-camera light field calibration.

A. Appendix

A.1. Additional Results

A.1.1. Imageset1

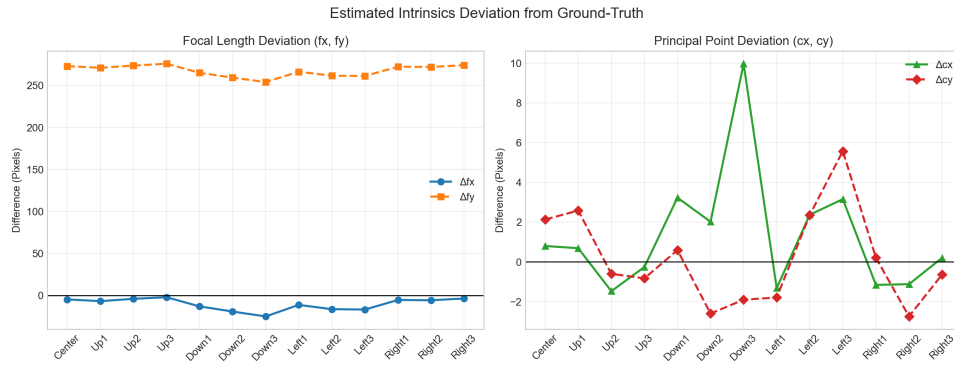


Figure 26: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset1).

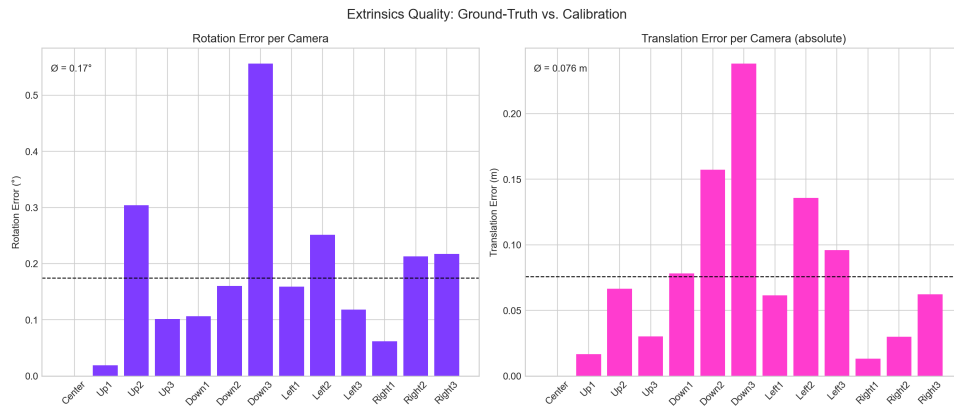


Figure 27: Extrinsic parameter errors of the initial calibration (Imageset1).

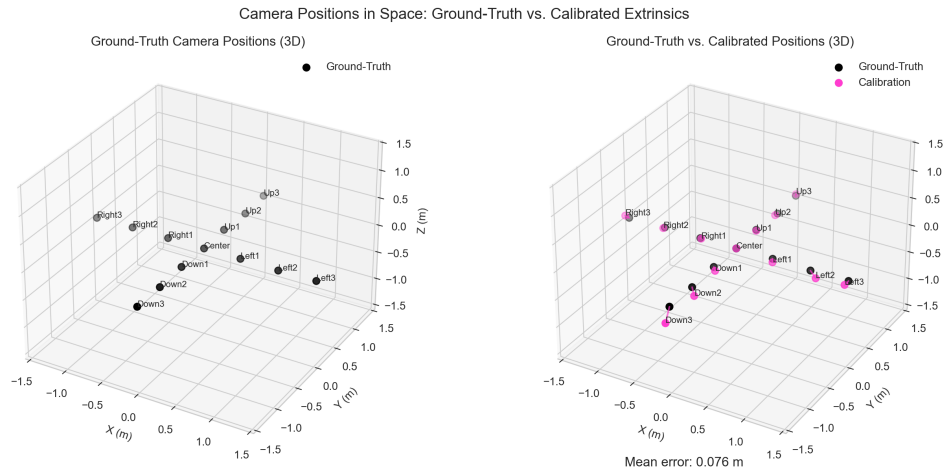


Figure 28: Translation parameter deviations from ground truth and the initial calibration (Imageset1).

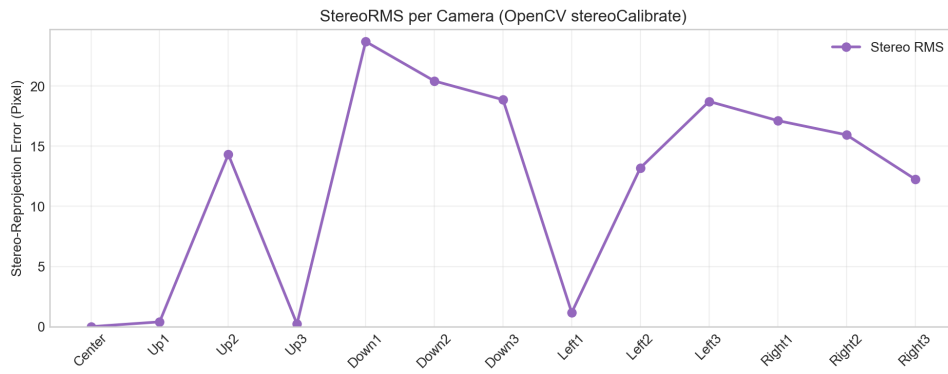


Figure 29: Stereo RMS error of the initial calibration (Imageset1).

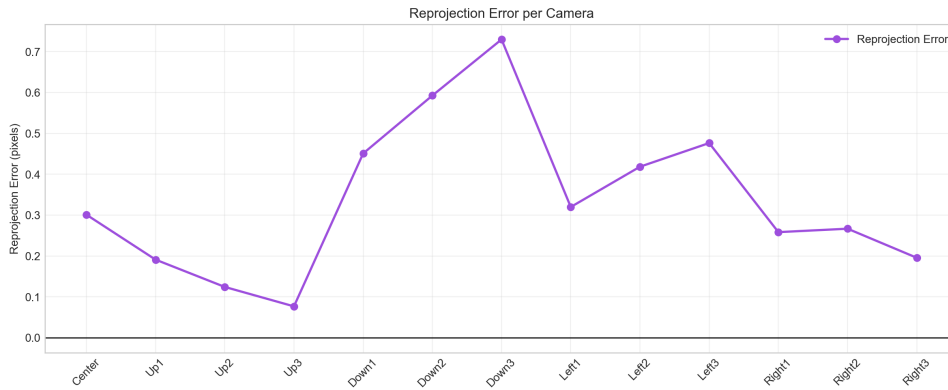


Figure 30: Reprojection error per camera of the initial calibration (Imageset1).

A.1.2. Imageset2

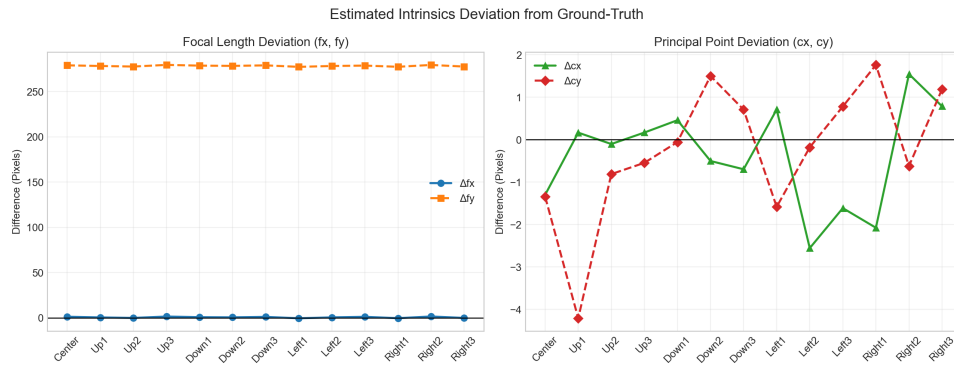


Figure 31: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset2).

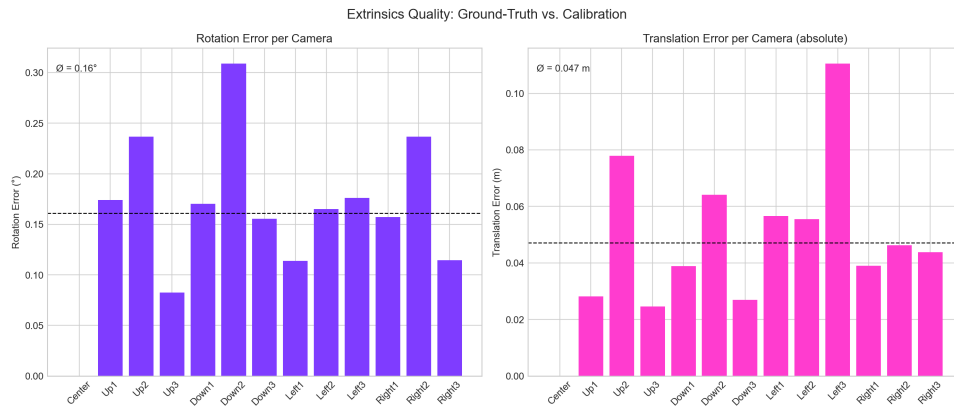


Figure 32: Extrinsic parameter errors of the initial calibration (Imageset2).

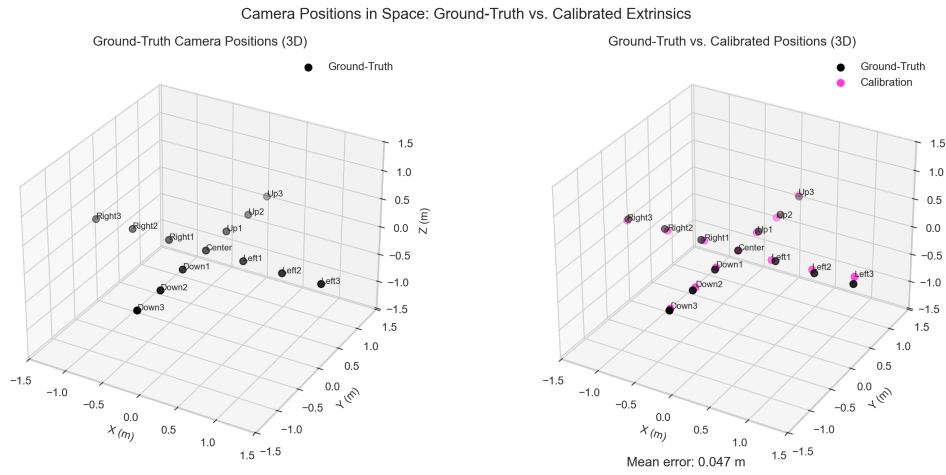


Figure 33: Translation parameter deviations from ground truth and the initial calibration (Imageset2).

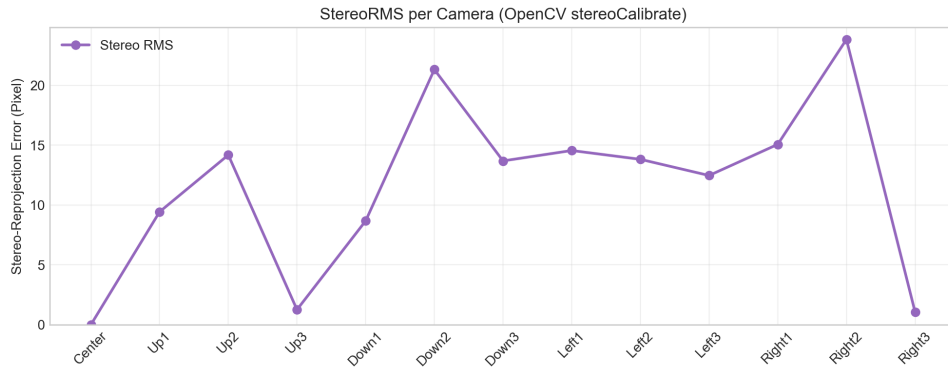


Figure 34: Stereo RMS error of the initial calibration (Imageset2).

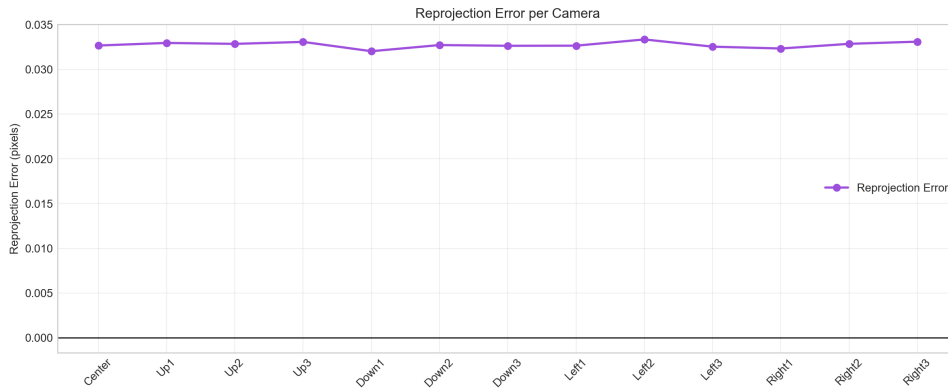


Figure 35: Reprojection error per camera of the initial calibration (Imageset2).

A.1.3. Imageset3

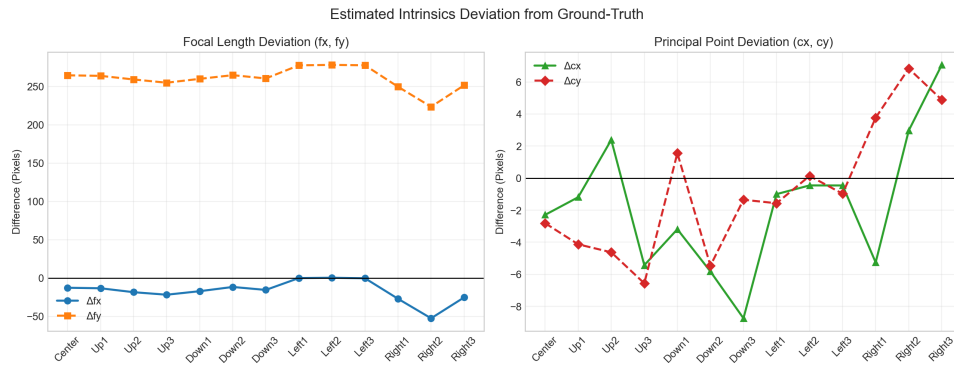


Figure 36: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset3).

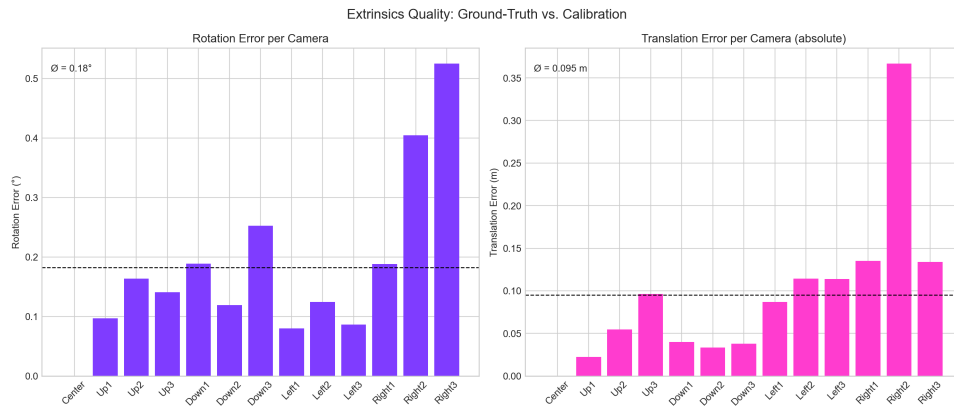


Figure 37: Extrinsic parameter errors of the initial calibration (Imageset3).

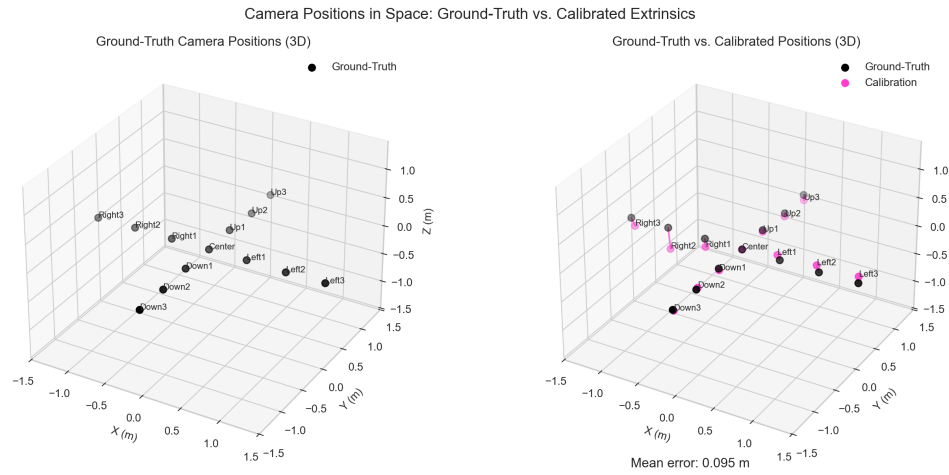


Figure 38: Translation parameter deviations from ground truth and the initial calibration (imageset3).

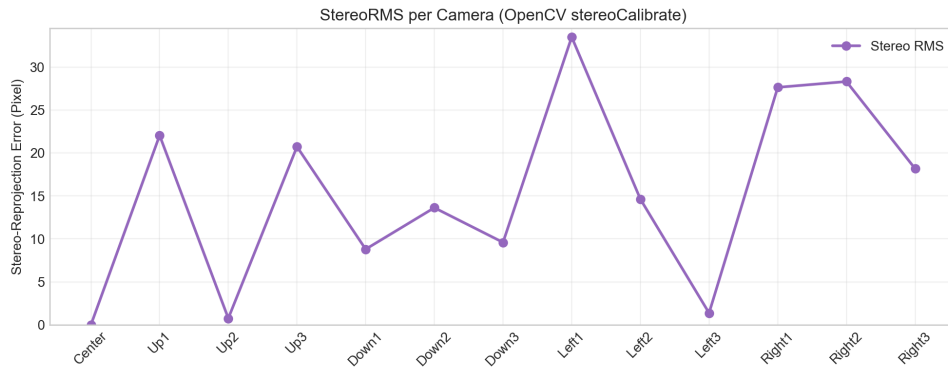


Figure 39: Stereo RMS error of the initial calibration (Imageset3).

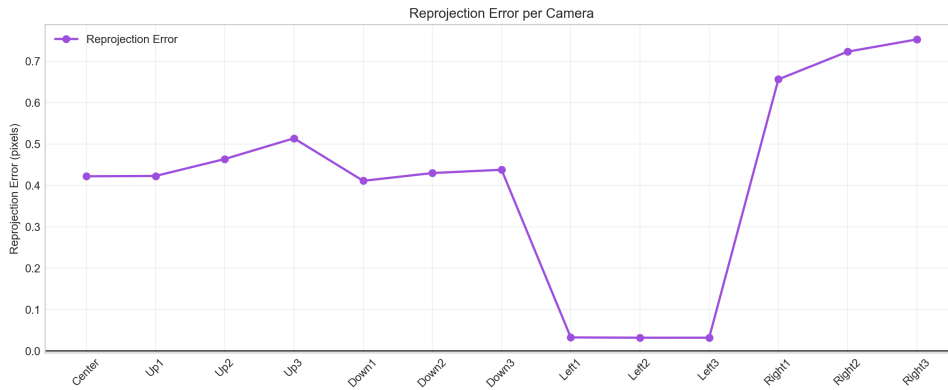


Figure 40: Reprojection error per camera of the initial calibration (Imageset3).

A.1.4. Imageset4

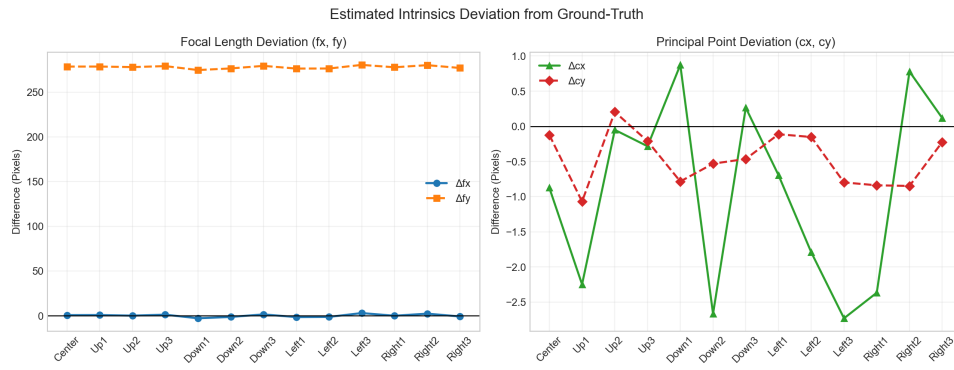


Figure 41: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset4).

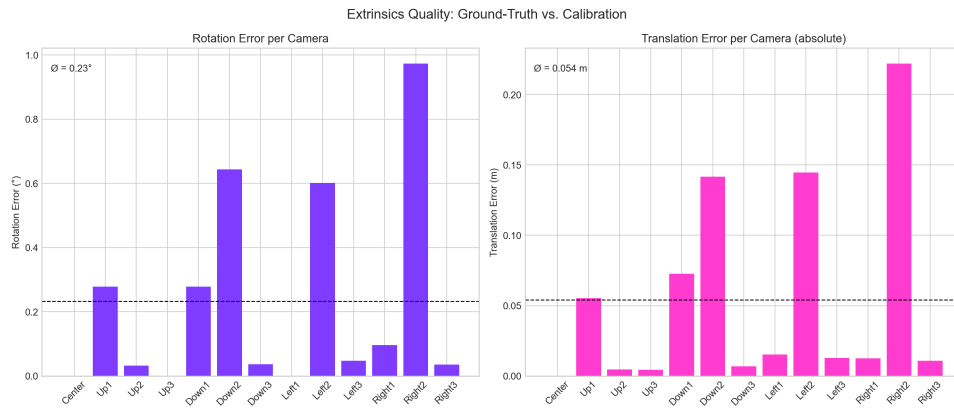


Figure 42: Extrinsic parameter errors of the initial calibration (Imageset4).

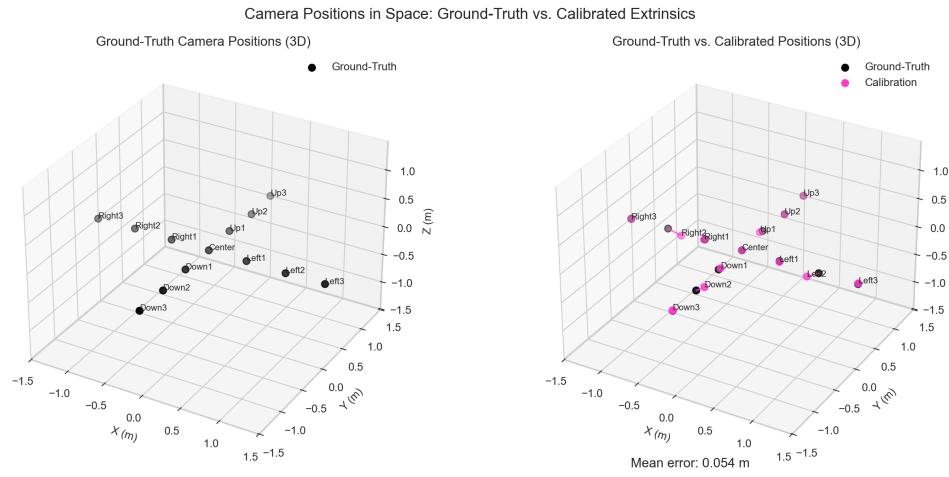


Figure 43: Translation parameter deviations from ground truth and the initial calibration (Imageset4).

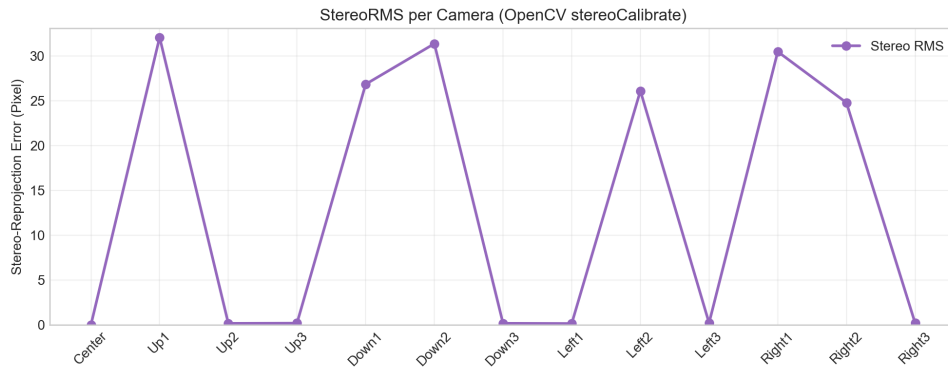


Figure 44: Stereo RMS error of the initial calibration (Imageset4).

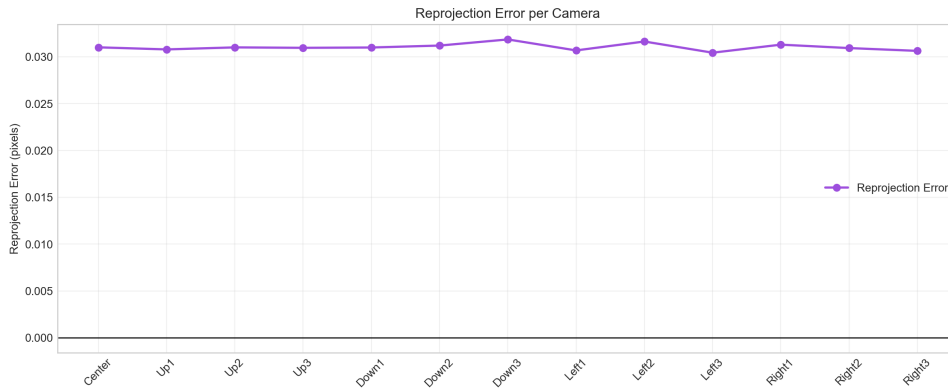


Figure 45: Reprojection error per camera of the initial calibration (Imageset4).

A.1.5. Imageset6

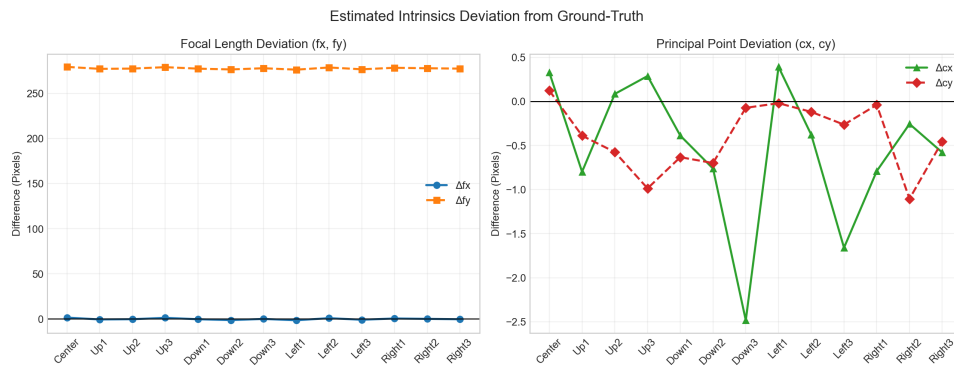


Figure 46: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset6).

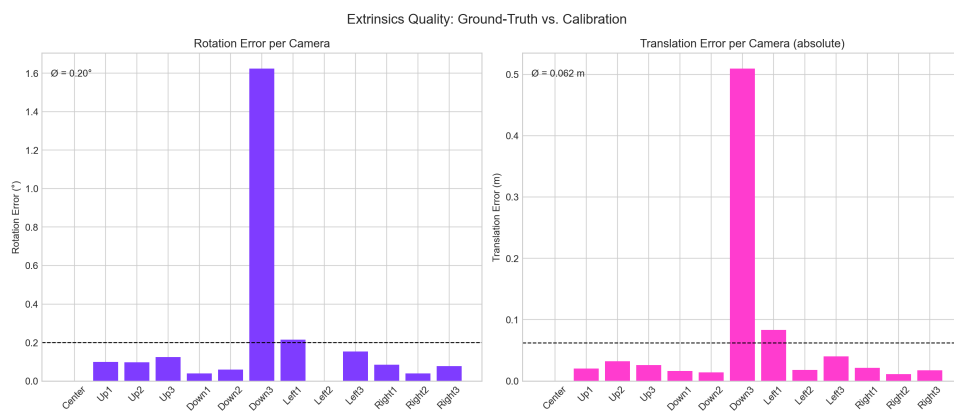


Figure 47: Extrinsic parameter errors of the initial calibration (Imageset6).

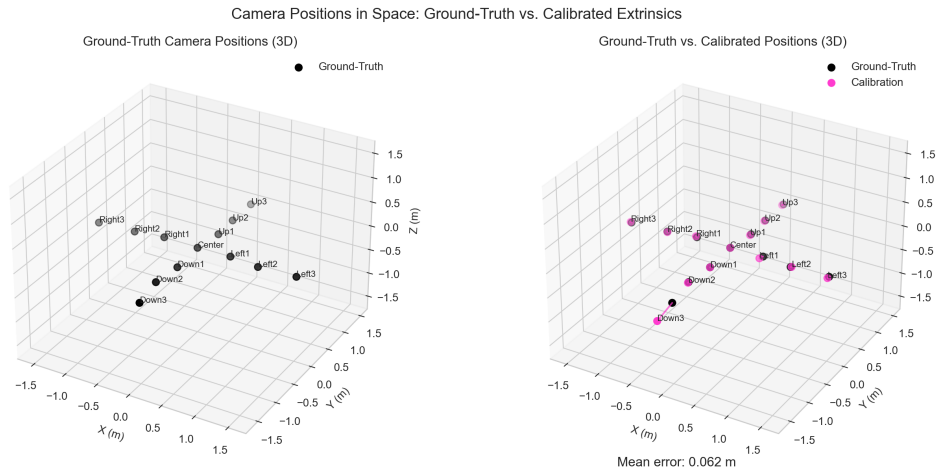


Figure 48: Translation parameter deviations from ground truth and the initial calibration (Imageset6).

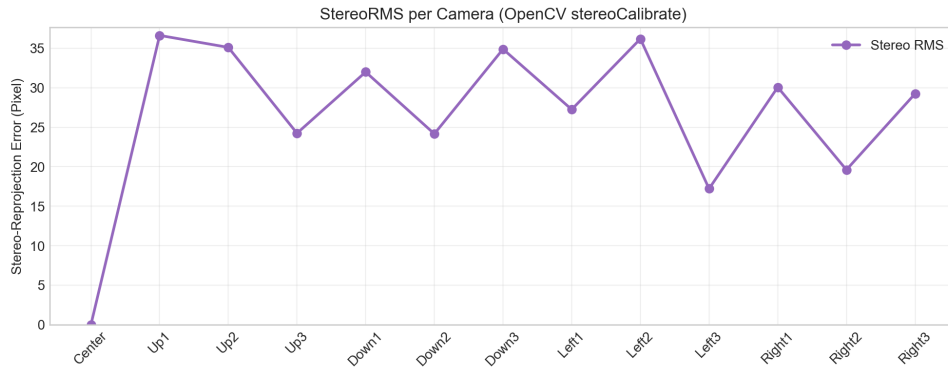


Figure 49: Stereo RMS error of the initial calibration (Imageset6).

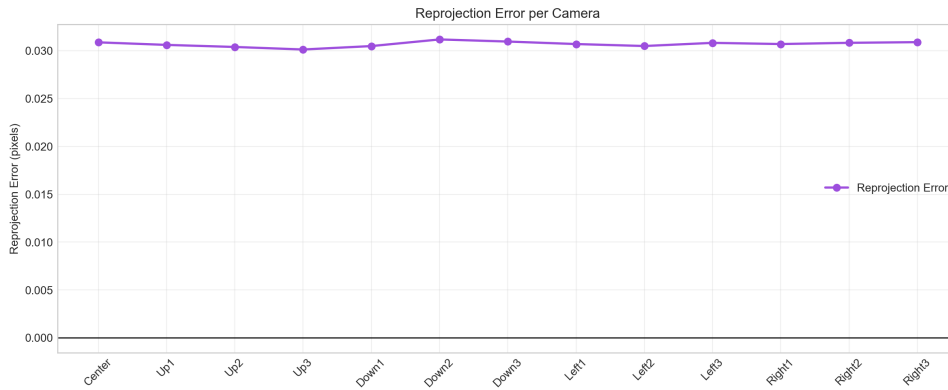


Figure 50: Reprojection error per camera of the initial calibration (Imageset6).

A.1.6. Imageset7

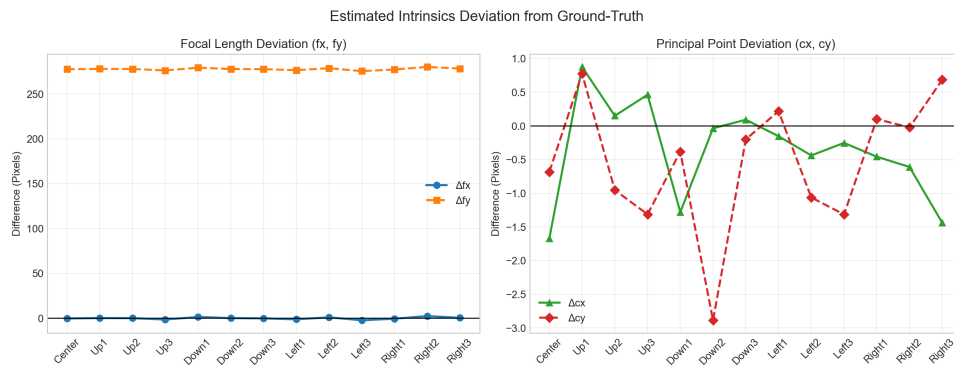


Figure 51: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset7).

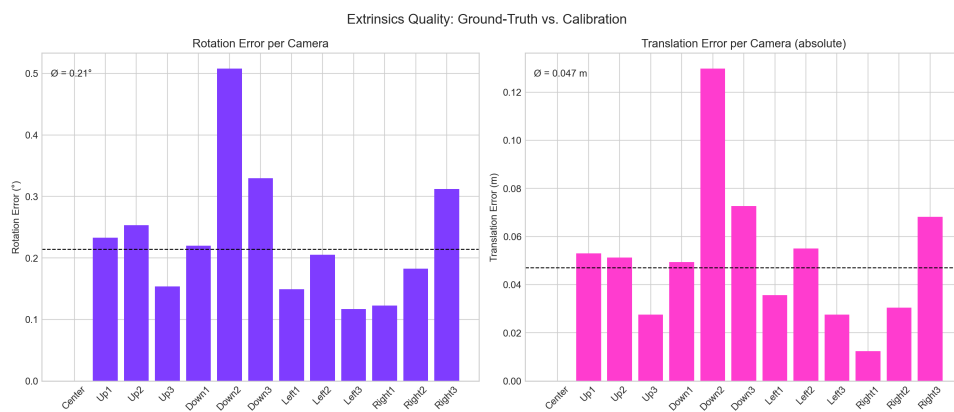


Figure 52: Extrinsic parameter errors of the initial calibration (Imageset7).

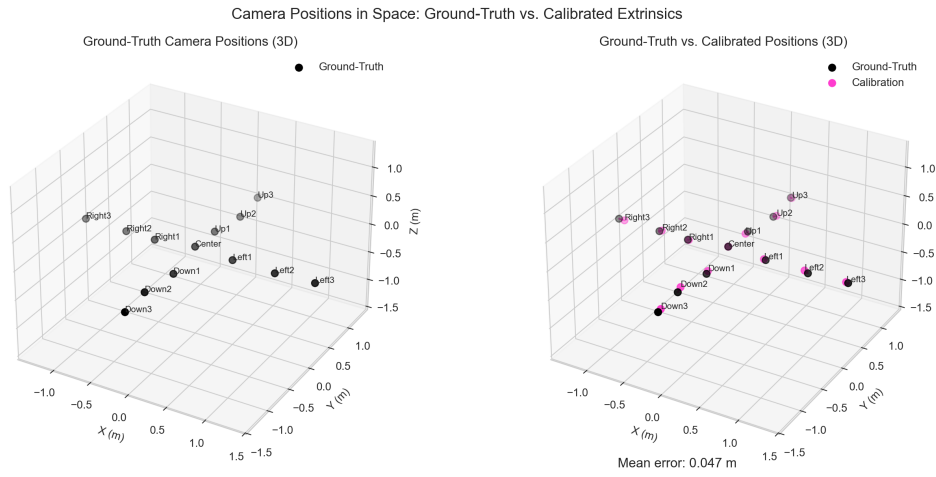


Figure 53: Translation parameter deviations from ground truth and the initial calibration (Imageset7).

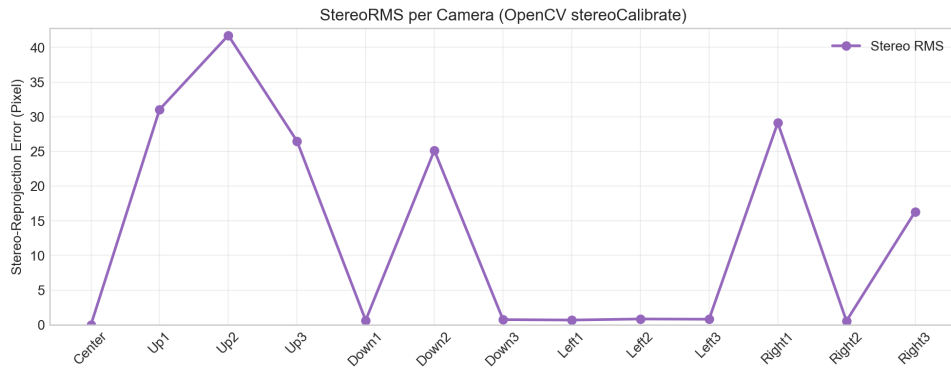


Figure 54: Stereo RMS error of the initial calibration (Imageset7).

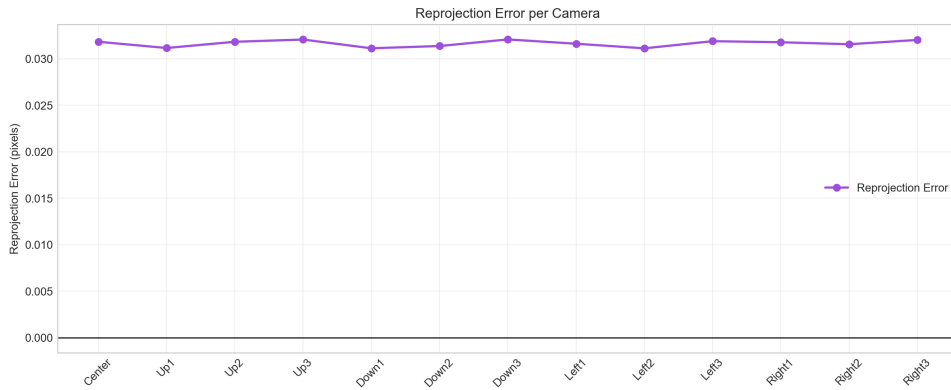


Figure 55: Reprojection error per camera of the initial calibration (Imageset7).

A.2. Blender Scenes

The Blender scenes used to generate the synthetic image data for the evaluations. The first plot shows the scene used for the initial target-based calibration, while the second plot illustrates the scene employed for the LiFCal-based recalibration.

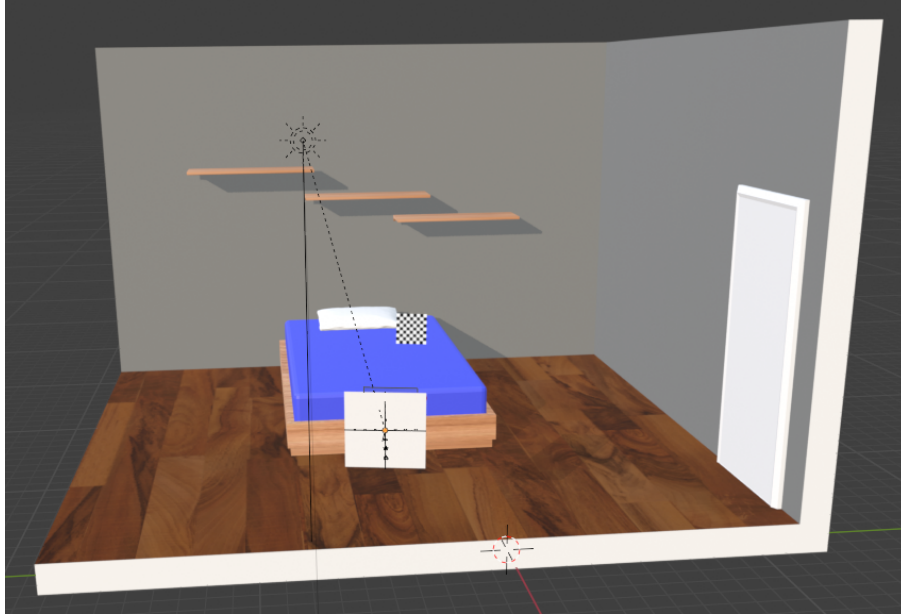


Figure 56: Blender scene for initial target-based calibration with checkerboard.



Figure 57: Blender scene for LiFCal-based recalibration without checkerboard.

Glossary

Light field Information about light rays in a scene, not only their brightness but also their direction. This enables refocusing and better depth cues.

Light field camera system A system built to capture a light field, either with a camera array or with a micro-lens array (MLA).

Multi-camera array Several cameras arranged in a fixed geometry to capture the same scene from different viewpoints.

Multi-camera light field system A light field setup based on a camera array (in this thesis, a cross-shaped layout).

Micro-lens array (MLA) A grid of tiny lenses in front of one sensor that captures angular information in a compact light field camera.

Pinhole camera model A simple camera model that describes how 3D points are projected onto the 2D image plane.

Intrinsic parameters Parameters inside the camera model, mainly focal lengths (f_x, f_y) and principal point (c_x, c_y) (in pixels).

Extrinsic parameters The camera pose in space, usually rotation R and translation t .

Camera matrix (K) Matrix that bundles the intrinsic parameters.

Principal point (c_x, c_y) Where the optical axis hits the image sensor (in pixel coordinates).

Skew coefficient (s) An intrinsic camera parameter that represents the degree of non-orthogonality between the camera's x and y pixel axes; in ideal cameras the axes are perpendicular and the skew is zero.

Radial distortion Lens distortion that bends straight lines, stronger toward the image borders (barrel/pincushion effect).

Tangential distortion Distortion caused by lens-sensor misalignment, modeled with p_1 and p_2 .

Calibration target A physical pattern with known geometry (e.g., a checkerboard) used to calibrate cameras.

Checkerboard calibration A common target-based calibration where checkerboard corners are detected in many images.

Target-free calibration Calibration without a physical pattern, based on scene structure, features, and usually camera motion.

Data-driven calibration It refers to an approach in which decisions and evaluations are based on measured data rather than predefined assumptions. In the context of this thesis, data-driven means that calibration quality is assessed using observed image data and error metrics, which then guide automatic decisions within the system.

Baseline The initial calibration result (from checkerboard) used as a reference for later recalibration attempts.

LiFCal (Online Light Field Camera Calibration via Bundle Adjustment) A target-free calibration method (originally for MLA light field cameras) used here as a method for an automatic recalibration.

Feature detection Finding distinctive points in an image that can be matched across views (needed for many target-free methods).

Depth map An image where each pixel stores depth. In this thesis, depth maps come from Blender and support LiFCal-style processing.

Reprojection error Average pixel distance between detected points and the points projected by the estimated camera model. Lower is better.

Root mean square (RMS) error A single aggregated error value over many points. In OpenCV stereo calibration it reflects joint consistency between two views.

Stereo calibration (stereoCalibrate) OpenCV routine that estimates the relative pose between two cameras and returns an RMS error for that pair.

Bundle adjustment (BA) A global refinement step that improves parameters by minimizing reprojection errors across many observations.

Baseline calibration The stored reference calibration (from the initial checkerboard run) used to judge later recalibration attempts.

Recalibration Repeating calibration after the initial setup to update parameters when the system changes.

Calibration drift Calibration slowly becomes worse over time (e.g., small mechanical shifts), which motivates periodic recalibration.

Health score A single score in $[0, 100]$ used here to summarize calibration quality and decide whether a recalibration is accepted.

Ground truth Reference parameters taken directly from Blender, used to evaluate calibration accuracy in a controlled setting.

Synchronization How well cameras capture images at the same time. Timing offsets can harm multi-camera consistency.

Texture / scene structure Visible detail in a scene (edges, patterns, depth variation) that makes target-free calibration more reliable.

References

- [1] Yuriy Anisimov, Gerd Reis, and Didier Stricker. Calibration and auto-refinement for light field cameras. In *Proceedings of the 29th International Conference in Central Europe on Computer Graphics, Visualization and Computer Vision (WSCG)*, pages 253–261, Pilsen, Czech Republic, 2021. Vaclav Skala - UNION Agency. ISBN 978-80-86943-34-3. URL https://www.dfki.de/fileadmin/user_upload/import/11659_wscg-j53.pdf.
- [2] CamCalib. The reprojection error. <https://www.camcalib.io/post/what-is-the-reprojection-error>, 2022. CamCalib blog, accessed 2026-01-21.
- [3] Blender Online Community. *Blender - a 3D modelling and rendering package*. Blender Foundation, Stichting Blender Foundation, Amsterdam, 2018. URL <http://www.blender.org>.
- [4] Daniel Cremers. Bundle adjustment & nonlinear optimization. https://cvg.cit.tum.de/_media/teaching/ss2019/mvg2019/material/multiviewgeometry7.pdf, 2019. Vorlesungsunterlage, Technische Universität München.
- [5] Elijs Dima, Mårten Sjöström, and Roger Olsson. Assessment of multi-camera calibration algorithms for two-dimensional camera arrays relative to ground truth position and direction. In *2016 3DTV-Conference: The True Vision - Capture, Transmission and Display of 3D Video (3DTV-CON)*, pages 1–4, 2016. doi: 10.1109/3DTV.2016.7548887. URL <https://ieeexplore.ieee.org/document/7548887>.
- [6] Elsevier. Radial distortion. <https://www.sciencedirect.com/topics/computer-science/radial-distortion>, 2022. Definition from ScienceDirect Topics: Computer Science.
- [7] Elsevier. Lens distortion. <https://www.sciencedirect.com/topics/engineering/lens-distortion>, 2024.
- [8] Aymeric Fleith, Doaa Ahmed, Daniel Cremers, and Niclas Zeller. Lifcal: Online light field camera calibration via bundle adjustment. <https://arxiv.org/abs/2408.11682>, 2024.
- [9] Pengxiang Ge, Huanqing Wang, Yonghong Wang, and Biao Wang. Calibration of ring multicamera system with transparent target for panoramic measurement. *IEEE Sensors Journal*, 22(23):23154–23164, 2022. doi: 10.1109/JSEN.2022.3214080. URL <https://ieeexplore.ieee.org/document/9923591>.
- [10] German Aerospace Center (DLR) – Institute of Robotics and Mechatronics. Light field cameras. DLR Research Expertise: 3D Perception, 2026. URL <https://www.dlr.de/en/rm/research/expertise/3d-perception/light-field-cameras>. Accessed January 24, 2026.
- [11] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, Cambridge, UK, 2nd edition, 2003. ISBN 0521540518. URL http://www.r-5.org/files/books/computers/algo-list/image-processing/vision/Richard_Hartley_Andrew_Zisserman-Multiple_View_Geometry_in_Computer_Vision-EN.pdf.

- [12] Jiachun Huang, Shaoli Liu, Jianhua Liu, and Zehua Jian. Camera calibration results: Reprojection error and relative positions. *Scientific Reports*, 2016. URL https://www.researchgate.net/figure/Camera-calibration-results-a-Reprojection-error-and-b-relative-positions-of-the_fig9_301715405. In Fig. 5.
- [13] IEEE. Ieee recommended practice for software requirements specifications. *IEEE Std 830-1998*, pages 1–40, 1998. doi: 10.1109/IEEESTD.1998.88286.
- [14] Mark Hedley Jones. Calibration checkerboard collection. Online project page, 2025. URL <https://markhedleyjones.com/projects/calibration-checkerboard-collection>.
- [15] Dhiraj Kumar. What is bundle adjustment? Baeldung on Computer Science, 2024. URL <https://www.baeldung.com/cs/computer-vision-bundle-adjustment>. Last updated March 18, 2024.
- [16] Philip K. Lee and Anqi R. Ji. Comparison of lightfield reconstruction methods using undersampled data. Technical Report EE367 Winter 2017 Project Report, Stanford University, Department of Electrical Engineering, 2017. URL https://stanford.edu/class/ee367/Winter2017/lee_ji_ee367_win17_report.pdf.
- [17] Fei-Fei Li et al. Camera models. https://web.stanford.edu/class/cs231a/course_notes/01-camera-models.pdf, 2020.
- [18] Shih-Hao Lin, En-Tze Chen, Jia-Jun Xiao, and Ching-Yuan Chang. Stereo digital image correlation (3d-dic) for non-contact measurement and refinement of delta robot arm displacement and jerk. *The International Journal of Advanced Manufacturing Technology*, 133(5-6):2809–2822, July 2024. doi: 10.1007/s00170-024-13957-2. URL <https://doi.org/10.1007/s00170-024-13957-2>.
- [19] MathWorks. Camera calibration. <https://de.mathworks.com/help/vision/ug/camera-calibration.html>, 2024.
- [20] OpenCV Contributors. Camera calibration with opencv. https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html, 2024.
- [21] OpenCV Developers. Camera calibration and 3d reconstruction — opencv documentation. OpenCV 4.x Documentation, calib3d module, 2025. URL https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html.
- [22] rahadianrayhan. Living room 3d model. Free3D online 3D model, 2021. URL <https://free3d.com/3d-model/living-room-997877.html>. Blender, FBX, and OBJ formats, 20 035 vertices, 18 735 faces, 38 514 triangles.
- [23] Carlos Ricolfe-Viala and Antonio-José Sánchez-Salmerón. Lens distortion models evaluation. *Applied Optics*, 49(30):5914–5928, 2010. doi: 10.1364/AO.49.005914. URL https://www.researchgate.net/publication/47510646_Lens_distortion_models_evaluation.
- [24] satriacaster. Isometric room 3d model. Free3D online 3D model, 2021. URL <https://free3d.com/3d-model/isometric-room-525234.html>. Blender (.blend), textures and materials included.

- [25] Chaehyeon Song, Dongjae Lee, Jongwoo Lim, and Ayoung Kim. Camera calibration via circular patterns: A comprehensive framework with measurement uncertainty and unbiased projection model. *arXiv preprint arXiv:2506.16842*, 2025. URL <https://arxiv.org/abs/2506.16842>.
- [26] Stanford Computer Graphics Laboratory. The stanford multi-camera array. Project page, Stanford University, 2011. URL <https://graphics.stanford.edu/projects/array/>.
- [27] Bill Triggs, Philip F. McLauchlan, Richard I. Hartley, and Andrew W. Fitzgibbon. Bundle adjustment – a modern synthesis. In Bill Triggs, Andrew Zisserman, and Richard Szeliski, editors, *Vision Algorithms: Theory and Practice*, volume 1883 of *Lecture Notes in Computer Science*, pages 298–372, London, UK, 2000. Springer-Verlag. doi: 10.1007/3-540-44480-7_21. URL <https://www.cs.jhu.edu/~misha/ReadingSeminar/Papers/Triggs00.pdf>.
- [28] Zhaomin Xie, Mengxiao Zhao, and Yujia He. Research on depth information acquisition of array camera based on depth map fusion. In *2023 5th International Conference on Frontiers Technology of Information and Computer (ICFTIC)*, pages 759–762, 2023. doi: 10.1109/ICFTIC59930.2023.10456113. URL <https://ieeexplore.ieee.org/document/10456113>.
- [29] Yichao Xu, Kazuki Maeno, Hajime Nagahara, and Rin ichiro Taniguchi. Camera array calibration for light field acquisition. *Frontiers of Computer Science*, 9(5): 691–702, 2015. doi: 10.1007/s11704-015-4237-4. URL <https://link.springer.com/article/10.1007/s11704-015-4237-4>.
- [30] Jungang Yang, Chao Xiao, Pu Wang, Yougan Luo, and Chengjin An. Camera array calibration using a simple checkerboard pattern. In *2019 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, pages 1–5, 2019. doi: 10.1109/ICSIDP47821.2019.9172948. URL <https://ieeexplore.ieee.org/document/9172948>.
- [31] Yaning Zhang, Yingqian Wang, Tianhao Wu, Jungang Yang, and Wei An. Fixed relative pose prior for camera array self-calibration. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(1):981–985, 2025. doi: 10.1109/TCSVT.2024.3450706. URL <https://ieeexplore.ieee.org/document/10649654>.
- [32] Z. Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000. doi: 10.1109/34.888718. URL <https://ieeexplore.ieee.org/document/888718>.
- [33] Feng Zhou, Jing Li, Yanran Dai, Lichen Liu, Haidong Qin, Yuqi Jiang, Shikuan Hong, Bo Zhao, and Tao Yang. Time-series fusion-based multicamera self-calibration for free-view video generation in low-texture sports scene. *IEEE Sensors Journal*, 23(3):2956–2969, 2023. doi: 10.1109/JSEN.2022.3230792. URL <https://ieeexplore.ieee.org/document/9998530>.

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C. Source Code

Data-driven camera calibration for light field camera systems

Generated by Doxygen 1.12.0

Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

CalibrationState	??
Controller	??
depthmap	??
evaluate_calibration	??
HealthMonitor	??
ImageSet	??
InitialCalibration	??
lifcal_to_json	??
ResultLogger	??
run_imageset_creation	??
run_initial_calibration	??
run_LiFCal_calibration	??
run_periodic_lifcal	??

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalibrationState.CalibrationState	??
Controller.Controller	??
ImageSet.ImageSet	??
InitialCalibration.InitialCalibration	??
ResultLogger.ResultLogger	??

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

calibrationLFC/ CalibrationState.py	??
calibrationLFC/ Controller.py	??
calibrationLFC/ HealthMonitor.py	??
calibrationLFC/ ImageSet.py	??
calibrationLFC/ InitialCalibration.py	??
calibrationLFC/ ResultLogger.py	??
calibrationLFC/ run_initial_calibration.py	??
calibrationLFC/ run_periodic_lifcal.py	??
calibrationLFC/results/ evaluate_calibration.py	??
external/LiFCal/LiFCal_Data/ depthmap.py	??
external/LiFCal/LiFCal_Data/ lifcal_to_json.py	??
external/LiFCal/LiFCal_Data/ run_imageset_creation.py	??
external/LiFCal/LiFCal_Data/ run_LiFCal_calibration.py	??

Chapter 4

Namespace Documentation

4.1 CalibrationState Namespace Reference

Classes

- class [CalibrationState](#)

4.2 Controller Namespace Reference

Classes

- class [Controller](#)

4.3 depthmap Namespace Reference

Functions

- [build_matchers](#) ()
- bool [imwrite_u16](#) (str path, np.ndarray img_u16)
- np.ndarray [disparity](#) (np.ndarray left_gray, np.ndarray right_gray)
- np.ndarray [to_u16_from_disp](#) (np.ndarray disp)
- np.ndarray [fuse_disparities_median](#) (list disp_list)
- Dict[str, np.ndarray] [generate_depthmaps_for_pose](#) (Dict[str, str] cam_to_focus_path)

Variables

- [_LEFT_MATCHER](#)
- [_RIGHT_MATCHER](#)
- [_WLS](#)

4.3.1 Function Documentation

4.3.1.1 build_matchers()

depthmap.build_matchers ()

@brief Build stereo matchers with SGBM and optional Weighted Least Squares (WLS) filtering.

@return Tuple (left_matcher, right_matcher, wls_filter)

Definition at line 11 of file [depthmap.py](#).

```
00011 def build_matchers():
00012     """
00013     @brief Build stereo matchers with SGBM and optional Weighted Least Squares (WLS) filtering.
00014
00015     @return Tuple (left_matcher, right_matcher, wls_filter)
00016     """
00017     left_matcher = cv2.StereoSGBM_create(
00018         minDisparity=0,
00019         numDisparities=128, # must be divisible by 16
00020         blockSize=7,
00021         P1=8 * 7 * 7,
00022         P2=32 * 7 * 7,
00023         uniquenessRatio=5,
00024         speckleWindowSize=100,
00025         speckleRange=2,
00026         disp12MaxDiff=1,
00027         preFilterCap=31,
00028         mode=cv2.STEREO_SGBM_MODE_SGBM_3WAY
00029     )
00030
00031     right_matcher = None
00032     wls = None
00033
00034     try:
00035         if hasattr(cv2, "ximgproc"):
00036             right_matcher = cv2.ximgproc.createRightMatcher(left_matcher)
00037             wls = cv2.ximgproc.createDisparityWLSFilter(left_matcher)
00038             wls.setLambda(6000)
00039             wls.setSigmaColor(1.0)
00040         except Exception:
00041             right_matcher = None
00042             wls = None
00043
00044     return left_matcher, right_matcher, wls
00045
00046
```

4.3.1.2 disparity()

np.ndarray depthmap.disparity (
 np.ndarray left_gray,
 np.ndarray right_gray)

@brief Compute stereo disparity map with optional WLS filtering.

@param left_gray Left grayscale image

@param right_gray Right grayscale image

@return Float32 disparity map with NaNs for invalid pixels (<=0)

Definition at line 64 of file [depthmap.py](#).

```
00064 def disparity(left_gray: np.ndarray, right_gray: np.ndarray) -> np.ndarray:
00065     """
00066     @brief Compute stereo disparity map with optional WLS filtering.
00067
00068     @param left_gray Left grayscale image
00069     @param right_gray Right grayscale image
00070     @return Float32 disparity map with NaNs for invalid pixels (<=0)
00071     """
```

```

00072     dl = _LEFT_MATCHER.compute(left_gray, right_gray) # int16 scaled by 16
00073
00074     if _RIGHT_MATCHER is not None and _WLS is not None:
00075         dr = _RIGHT_MATCHER.compute(right_gray, left_gray)
00076         d = _WLS.filter(dl, left_gray, None, dr)
00077     else:
00078         d = dl
00079
00080     d = d.astype(np.float32) / 16.0
00081     d[d <= 0] = np.nan
00082     return d
00083
00084

```

4.3.1.3 fuse_disparities_median()

```

np.ndarray depthmap.fuse_disparities_median (
    list disp_list)

```

@brief Fuse multiple disparity maps robustly using median.

@param disp_list List of float32 disparity maps to fuse

@return Float32 fused disparity map with 0 where no valid data exists

Definition at line 103 of file [depthmap.py](#).

```

00103 def fuse_disparities_median(disp_list: list) -> np.ndarray:
00104     """
00105     @brief Fuse multiple disparity maps robustly using median.
00106
00107     @param disp_list List of float32 disparity maps to fuse
00108     @return Float32 fused disparity map with 0 where no valid data exists
00109     """
00110     stack = np.stack(disp_list, axis=0) # (K,H,W)
00111
00112     # Locations where at least one value is valid
00113     valid_any = np.any(~np.isnan(stack), axis=0)
00114
00115     # Initialize result with zeros
00116     disp_final = np.zeros(stack.shape[1:], dtype=np.float32)
00117
00118     # Compute median only where we have valid data
00119     if np.any(valid_any):
00120         disp_final[valid_any] = np.nanmedian(stack[:, valid_any], axis=0)
00121
00122     # Remove non-positive disparities
00123     disp_final[disp_final <= 0] = 0.0
00124     return disp_final
00125
00126

```

4.3.1.4 generate_depthmaps_for_pose()

```

Dict[str, np.ndarray] depthmap.generate_depthmaps_for_pose (
    Dict[str, str] cam_to_focus_path)

```

@brief Generate depth maps for all cameras in a multi-camera pose.

@param cam_to_focus_path Dictionary mapping camera IDs to image file paths

@return Dictionary mapping camera IDs to uint16 depth maps

Definition at line 127 of file [depthmap.py](#).

```

00127 def generate_depthmaps_for_pose(cam_to_focus_path: Dict[str, str]) -> Dict[str, np.ndarray]:
00128     """
00129     @brief Generate depth maps for all cameras in a multi-camera pose.
00130
00131     @param cam_to_focus_path Dictionary mapping camera IDs to image file paths
00132     @return Dictionary mapping camera IDs to uint16 depth maps
00133     """

```

```

00134
00135     # 1) Load images
00136     imgs = {}
00137     for cam, path in cam_to_focus_path.items():
00138         im = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
00139         if im is None:
00140             continue
00141         imgs[cam] = im
00142
00143     if len(imgs) < 2:
00144         return {}
00145
00146     # helper
00147     def opposite_name(name: str) -> Optional[str]:
00148         if name.startswith("Left"):
00149             return name.replace("Left", "Right", 1)
00150         if name.startswith("Right"):
00151             return name.replace("Right", "Left", 1)
00152         if name.startswith("Up"):
00153             return name.replace("Up", "Down", 1)
00154         if name.startswith("Down"):
00155             return name.replace("Down", "Up", 1)
00156         return None
00157
00158     depthmaps = {}
00159
00160     for cam, img in imgs.items():
00161         disp_list = []
00162
00163         if cam == "Center":
00164             # Center uses all other views
00165             for other_cam, other_img in imgs.items():
00166                 if other_cam == "Center":
00167                     continue
00168                 d = disparity(img, other_img)
00169                 # Keep only if there is any valid disparity
00170                 if np.any(~np.isnan(d)):
00171                     disp_list.append(d)
00172
00173         else:
00174             # Opposite direction if available
00175             opp = opposite_name(cam)
00176             if opp is not None and opp in imgs:
00177                 if cam.startswith("Right") or cam.startswith("Down"):
00178                     d = disparity(imgs[opp], img)
00179                 else:
00180                     d = disparity(img, imgs[opp])
00181
00182                 if np.any(~np.isnan(d)):
00183                     disp_list.append(d)
00184
00185             # Add Center view
00186             if "Center" in imgs:
00187                 d = disparity(img, imgs["Center"])
00188                 if np.any(~np.isnan(d)):
00189                     disp_list.append(d)
00190
00191         if not disp_list:
00192             # No valid disparity found; skip
00193             continue
00194
00195         # Robust fusion
00196         disp_final = fuse_disparities_median(disp_list)
00197
00198         # Convert to uint16 depth map
00199         depth16 = to_u16_from_disp(disp_final)
00200         depthmaps[cam] = depth16
00201
00202     return depthmaps

```

4.3.1.5 imwrite_u16()

```

bool depthmap.imwrite_u16 (
    str path,
    np.ndarray img_u16)

```

@brief Write uint16 png, create dirs automatically.

@param path Output file path

@param img_u16 Image array in uint16 format

@return True if write succeeded, False otherwise

Definition at line 50 of file [depthmap.py](#).

```
00050 def imwrite_u16(path: str, img_u16: np.ndarray) -> bool:
00051     """
00052     @brief Write uint16 png, create dirs automatically.
00053
00054     @param path Output file path
00055     @param img_u16 Image array in uint16 format
00056     @return True if write succeeded, False otherwise
00057     """
00058     os.makedirs(os.path.dirname(path), exist_ok=True)
00059     if img_u16.dtype != np.uint16:
00060         img_u16 = img_u16.astype(np.uint16)
00061     return bool(cv2.imwrite(path, img_u16))
00062
00063
```

4.3.1.6 to_u16_from_disp()

```
np.ndarray depthmap.to_u16_from_disp (
    np.ndarray disp)
```

@brief Convert disparity to uint16 depth map to be compatible with LiFCal.

@param disp Float32 disparity map
@return UInt16 depth map

Definition at line 85 of file [depthmap.py](#).

```
00085 def to_u16_from_disp(disp: np.ndarray) -> np.ndarray:
00086     """
00087     @brief Convert disparity to uint16 depth map to be compatible with LiFCal.
00088
00089     @param disp Float32 disparity map
00090     @return UInt16 depth map
00091     """
00092     disp = np.nan_to_num(disp, nan=0.0, posinf=0.0, neginf=0.0)
00093
00094     mx = float(np.percentile(disp, 99))
00095     if not np.isfinite(mx) or mx <= 1e-6:
00096         return np.zeros(disp.shape, dtype=np.uint16)
00097
00098     disp = np.clip(disp, 0.0, mx)
00099     depth16 = (disp / mx * 65535.0).astype(np.uint16)
00100     return depth16
00101
00102
```

4.3.2 Variable Documentation

4.3.2.1 _LEFT_MATCHER

```
depthmap._LEFT_MATCHER [protected]
```

Definition at line 47 of file [depthmap.py](#).

4.3.2.2 _RIGHT_MATCHER

```
depthmap._RIGHT_MATCHER [protected]
```

Definition at line 47 of file [depthmap.py](#).

4.3.2.3 `_WLS`

`depthmap._WLS` [protected]

Definition at line 47 of file [depthmap.py](#).

4.4 `evaluate_calibration` Namespace Reference

Functions

- [rot_angle_deg](#) (R)
- [as_vec3](#) (v)
- [compute_metrics_for_file](#) (Path calib_path, Path gt_path)
- [print_per_file_table](#) (results_by_name)
- [print_scenario_table](#) (results_by_name)
- [main](#) ()

Variables

- [THIS_DIR](#) = Path(__file__).resolve().parent
- [CALIB_LFC_DIR](#) = THIS_DIR.parent
- [PROJECT_ROOT](#) = CALIB_LFC_DIR.parent
- str [RESULTS_DIR](#) = [CALIB_LFC_DIR](#) / "results"
- str [GT_DIR](#) = [PROJECT_ROOT](#) / "TestEnvironment" / "params"
- list [CAM_IDS](#)
- dict [CALIB_FILES](#)
- dict [GT_FILES](#)
- dict [SCENARIOS](#)

4.4.1 Function Documentation

4.4.1.1 `as_vec3()`

```
evaluate_calibration.as_vec3 (
    v)
```

Convert any translation vector to robust 3-element vector.

Definition at line 62 of file [evaluate_calibration.py](#).

```
00062 def as_vec3(v):
00063     """Convert any translation vector to robust 3-element vector."""
00064     arr = np.array(v, dtype=float).reshape(-1)
00065     if arr.size != 3:
00066         raise ValueError(f"Expected 3 entries for translation, got {arr.size}")
00067     return arr
00068
00069
```

4.4.1.2 compute_metrics_for_file()

```
evaluate_calibration.compute_metrics_for_file (
    Path calib_path,
    Path gt_path)
```

Loads a calibration JSON and groundtruth JSON and computes:

- numPoses
- mean T (m)
- mean R (°)
- mean reprojection error (px, from intrinsics)

Definitions as in your extrinsic plot:

```
R_cam = angle(R_est * R_gt^T)
T_cam = || T_est - T_gt ||_2
```

Definition at line 70 of file [evaluate_calibration.py](#).

```
00070 def compute_metrics_for_file(calib_path: Path, gt_path: Path):
00071     """
00072     Loads a calibration JSON and groundtruth JSON and computes:
00073     - numPoses
00074     - mean T (m)
00075     - mean R (°)
00076     - mean reprojection error (px, from intrinsics)
00077
00078     Definitions as in your extrinsic plot:
00079     R_cam = angle(R_est * R_gt^T)
00080     T_cam = || T_est - T_gt ||_2
00081     """
00082     with open(gt_path, "r", encoding="utf-8") as f:
00083         gt = json.load(f)
00084
00085     with open(calib_path, "r", encoding="utf-8") as f:
00086         calib = json.load(f)
00087
00088     meta = calib.get("meta", {})
00089     extr_est = calib["state"]["extrinsics"]
00090     intr_est = calib["state"]["intrinsics"]
00091
00092     bundle_adjust = bool(meta.get("bundleAdjust", False))
00093     num_poses = int(meta.get("numPoses", -1))
00094
00095     rot_err = []
00096     trans_err = []
00097     reproj_err = []
00098
00099     for cam in CAM_IDS:
00100         if cam not in extr_est or cam not in gt:
00101             print(f"[WARN] Camera '{cam}' not in both files for {calib_path.name}, skipping this
cam.")
00102             continue
00103
00104         # Extrinsics
00105         R_est = np.array(extr_est[cam]["rotationMatrix"], dtype=float)
00106         R_gt = np.array(gt[cam]["rotationMatrix"], dtype=float)
00107
00108         T_est = as_vec3(extr_est[cam]["translationVector"])
00109         T_gt = as_vec3(gt[cam]["translationVector"])
00110
00111         # Rotation error as in plot
00112         R_err = R_est @ R_gt.T
00113         ang = rot_angle_deg(R_err)
00114
00115         # Translation error (absolute, m)
00116         d_abs = float(np.linalg.norm(T_est - T_gt))
00117
00118         rot_err.append(ang)
00119         trans_err.append(d_abs)
00120
00121         # Intrinsic reprojection error
00122         if cam in intr_est:
00123             reproj_err.append(float(intr_est[cam]["reprojectionError"]))
00124
00125         # Mean values
00126         mean_dT = float(np.mean(trans_err)) if trans_err else float("nan")
00127         mean_dR = float(np.mean(rot_err)) if rot_err else float("nan")
00128         mean_reproj = float(np.mean(reproj_err)) if reproj_err else float("nan")
00129
00130     return {
```

```

00131         "bundleAdjust": bundle_adjust,
00132         "numPoses": num_poses,
00133         "mean_dT": mean_dT,
00134         "mean_dR": mean_dR,
00135         "mean_reproj": mean_reproj,
00136     }
00137
00138 # Table output (console)

```

4.4.1.3 main()

evaluate_calibration.main ()

Definition at line 217 of file [evaluate_calibration.py](#).

```

00217 def main():
00218     results_by_name = {}
00219
00220     for name, calib_path in CALIB_FILES.items():
00221         if not calib_path.exists():
00222             print(f"[WARN] File not found: {calib_path}")
00223             continue
00224
00225         gt_path = GT_FILES.get(name)
00226         if gt_path is None:
00227             print(f"[WARN] No GT file mapped for {name}, skipping.")
00228             continue
00229         if not gt_path.exists():
00230             print(f"[WARN] GT file not found for {name}: {gt_path}")
00231             continue
00232
00233         metrics = compute_metrics_for_file(calib_path, gt_path)
00234         results_by_name[name] = metrics
00235
00236     if not results_by_name:
00237         print("[ERROR] No valid results computed. Check file paths.")
00238         return
00239
00240     print("Detailed results per calibration:\n")
00241     print_per_file_table(results_by_name)
00242
00243     print_scenario_table(results_by_name)
00244
00245

```

4.4.1.4 print_per_file_table()

evaluate_calibration.print_per_file_table (
 results_by_name)

Prints one line per calibration JSON.

Definition at line 139 of file [evaluate_calibration.py](#).

```

00139 def print_per_file_table(results_by_name):
00140     """
00141     Prints one line per calibration JSON.
00142     """
00143     header = (
00144         f"{'Name':<12} "
00145         f"{'#Poses':>6} "
00146         f"{'BA':<4} "
00147         f"{'mean dT [m]':>12} "
00148         f"{'mean dR [deg]':>13} "
00149         f"{'mean reproj [px]':>16}"
00150     )
00151     print(header)
00152     print("-" * len(header))
00153
00154     for name, res in sorted(results_by_name.items()):
00155         ba_flag = "yes" if res["bundleAdjust"] else "no"
00156         print(
00157             f"{name:<12} "
00158             f"{res['numPoses']:6d} "
00159             f"{ba_flag:<4} "
00160             f"{res['mean_dT']:12.5f} "
00161             f"{res['mean_dR']:12.5f} "
00162             f"{res['mean_reproj']:16.5f}"
00163         )
00164

```

4.4.1.5 print_scenario_table()

```
evaluate_calibration.print_scenario_table (
    results_by_name)
```

Aggregates over scenarios (Robustness / Generalization) and prints a compact table.

Definition at line 165 of file [evaluate_calibration.py](#).

```
00165 def print_scenario_table(results_by_name):
00166     """
00167     Aggregates over scenarios (Robustness / Generalization)
00168     and prints a compact table.
00169     """
00170     print("\n\nScenario Summary:\n")
00171
00172     header = (
00173         f"{'Scenario':<22} "
00174         f"{'#Runs':>6} "
00175         f"{'Avg#Poses':>9} "
00176         f"{'mean dT [m]':>12} "
00177         f"{'mean dR [deg]':>13} "
00178         f"{'mean reproj [px]':>16}"
00179     )
00180     print(header)
00181     print("-" * len(header))
00182
00183     for scen_name, calib_names in SCENARIOS.items():
00184         dT_vals = []
00185         dR_vals = []
00186         reproj_vals = []
00187         pose_counts = []
00188
00189         for cname in calib_names:
00190             res = results_by_name.get(cname)
00191             if res is None:
00192                 continue
00193
00194             dT_vals.append(res["mean_dT"])
00195             dR_vals.append(res["mean_dR"])
00196             reproj_vals.append(res["mean_reproj"])
00197             pose_counts.append(res["numPoses"])
00198
00199         if not dT_vals:
00200             continue
00201
00202         mean_dT = float(np.mean(dT_vals))
00203         mean_dR = float(np.mean(dR_vals))
00204         mean_reproj = float(np.mean(reproj_vals))
00205         mean_poses = float(np.mean(pose_counts))
00206         num_runs = len(dT_vals)
00207
00208         print(
00209             f"{scen_name:<22} "
00210             f"{num_runs:6d} "
00211             f"{mean_poses:9.1f} "
00212             f"{mean_dT:12.5f} "
00213             f"{mean_dR:13.5f} "
00214             f"{mean_reproj:16.5f}"
00215         )
00216
```

4.4.1.6 rot_angle_deg()

```
evaluate_calibration.rot_angle_deg (
    R)
```

Extract rotation angle (degrees) from 3x3 rotation matrix.

Definition at line 55 of file [evaluate_calibration.py](#).

```
00055 def rot_angle_deg(R):
00056     """Extract rotation angle (degrees) from 3x3 rotation matrix."""
00057     tr = np.trace(R)
00058     c = (tr - 1.0) / 2.0
00059     c = np.clip(c, -1.0, 1.0)
00060     return float(np.degrees(np.arccos(c)))
00061
```


4.4.2 Variable Documentation

4.4.2.1 CALIB_FILES

`dict evaluate_calibration.CALIB_FILES`

Initial value:

```
00001 = {
00002     "imageset1": RESULTS_DIR / "calibration_initial_imageset1_20251215_173701.json",
00003     "imageset2": RESULTS_DIR / "calibration_initial_imageset2_20251215_181903.json",
00004     "imageset3": RESULTS_DIR / "calibration_initial_imageset3_20251215_185041.json",
00005     "imageset4": RESULTS_DIR / "calibration_initial_imageset4_20251215_165644.json",
00006     "imageset5": RESULTS_DIR / "calibration_initial_imageset5_20251217_011656.json",
00007     "imageset6": RESULTS_DIR / "calibration_initial_imageset6_20251215_201527.json",
00008     "imageset7": RESULTS_DIR / "calibration_initial_imageset7_20251215_203849.json",
00009 }
```

Definition at line 24 of file [evaluate_calibration.py](#).

4.4.2.2 CALIB_LFC_DIR

`evaluate_calibration.CALIB_LFC_DIR = THIS_DIR.parent`

Definition at line 7 of file [evaluate_calibration.py](#).

4.4.2.3 CAM_IDS

`list evaluate_calibration.CAM_IDS`

Initial value:

```
00001 = [
00002     "Center",
00003     "Up1", "Up2", "Up3",
00004     "Down1", "Down2", "Down3",
00005     "Left1", "Left2", "Left3",
00006     "Right1", "Right2", "Right3",
00007 ]
```

Definition at line 15 of file [evaluate_calibration.py](#).

4.4.2.4 GT_DIR

`str evaluate_calibration.GT_DIR = PROJECT_ROOT / "TestEnvironment" / "params"`

Definition at line 12 of file [evaluate_calibration.py](#).

4.4.2.5 GT_FILES

`dict evaluate_calibration.GT_FILES`

Initial value:

```
00001 = {
00002     # Rig0 for imageset1-4
00003     "imageset1": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00004     "imageset2": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00005     "imageset3": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00006     "imageset4": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00007
00008     # Different rigs/sets:
00009     "imageset5": GT_DIR / "groundtruth_extrinsics_Rig1_set5.json",
00010     "imageset6": GT_DIR / "groundtruth_extrinsics_Rig2_set6.json",
00011     "imageset7": GT_DIR / "groundtruth_extrinsics_Rig3_set7.json",
00012 }
```

Definition at line 35 of file [evaluate_calibration.py](#).

4.4.2.6 PROJECT_ROOT

```
evaluate_calibration.PROJECT_ROOT = CALIB_LFC_DIR.parent
```

Definition at line 8 of file [evaluate_calibration.py](#).

4.4.2.7 RESULTS_DIR

```
str evaluate_calibration.RESULTS_DIR = CALIB_LFC_DIR / "results"
```

Definition at line 9 of file [evaluate_calibration.py](#).

4.4.2.8 SCENARIOS

```
dict evaluate_calibration.SCENARIOS
```

Initial value:

```
00001 = {
00002     "A_Robustheit": ["imageset1", "imageset2", "imageset3", "imageset4"],
00003     "B_Generalisation": ["imageset5", "imageset6", "imageset7"],
00004 }
```

Definition at line 49 of file [evaluate_calibration.py](#).

4.4.2.9 THIS_DIR

```
evaluate_calibration.THIS_DIR = Path(__file__).resolve().parent
```

Definition at line 6 of file [evaluate_calibration.py](#).

4.5 HealthMonitor Namespace Reference

Functions

- float [clamp](#) (float x, float lo=0.0, float hi=1.0)
- [safe_float](#) (v, default=None)
- float [score_from_errors](#) (float E_a, float E_b, float w_a=0.6, float w_b=0.4)
- Dict[str, Any] [compute_initial_health_from_state](#) (Dict[str, Any] state_dict, float r_best=[R_BEST](#), float r_worst=[R_WORST](#), float s_best=[S_BEST](#), float s_worst=[S_WORST](#))
- Dict[str, Optional[float]] [parse_lifcal_protocol](#) (str protocol_path)
- Dict[str, Any] [compute_lifcal_health_from_xy](#) (float x_std, float y_std, float x_mae, float y_mae, float r_best=[R_BEST](#), float r_worst=[R_WORST](#), float s_best=[S_BEST](#), float s_worst=[S_WORST](#))
- Dict[str, Any] [compute_lifcal_health_from_combined_dict](#) (Dict[str, Any] combined)
- Dict[str, Any] [compute_lifcal_health_from_combined_path](#) (str combined_json_path)

Variables

- float [R_BEST](#) = 0.03
- float [R_WORST](#) = 1.0
- float [S_BEST](#) = 0.0
- float [S_WORST](#) = 50.0

4.5.1 Detailed Description

HealthScore Formulas (0..100)

(A) INITIAL Calibration (OpenCV/Controller JSON)
 $\text{HealthScore_initial} = 100 * [1 - (0.6 * E_{\text{reproj}}^2 + 0.4 * E_{\text{rms}}^2)]$

$E_{\text{reproj}} = \text{clamp}((\text{reproj_mean} - R_{\text{BEST}}) / (R_{\text{WORST}} - R_{\text{BEST}}), 0, 1)$
 $E_{\text{rms}} = \text{clamp}((\text{rms_mean} - S_{\text{BEST}}) / (S_{\text{WORST}} - S_{\text{BEST}}), 0, 1)$

Uses:
 - `intrinsics[*].reprojectionError` (per camera)
 - `extrinsics[*].stereoRms` (per camera)

(B) LiFCal Calibration (protocol-based)
 $\text{HealthScore_lifcal} = 100 * [1 - (0.6 * E_{\text{std}}^2 + 0.4 * E_{\text{mae}}^2)]$

$E_{\text{std}} = \text{clamp}((\sqrt{x_{\text{std}}^2 + y_{\text{std}}^2} - S_{\text{BEST}}) / (S_{\text{WORST}} - S_{\text{BEST}}), 0, 1)$
 $E_{\text{mae}} = \text{clamp}((\sqrt{x_{\text{mae}}^2 + y_{\text{mae}}^2} - R_{\text{BEST}}) / (R_{\text{WORST}} - R_{\text{BEST}}), 0, 1)$

Uses:
 - `std.Dev. x/y` and `mae x/y` from LiFCal `calibrationProtocol.txt`

Shared thresholds (for BOTH methods):

`R_BEST = 0.03, R_WORST = 1.0, S_BEST = 0.0, S_WORST = 50.0`

4.5.2 Function Documentation

4.5.2.1 clamp()

```
float HealthMonitor.clamp (
    float x,
    float lo = 0.0,
    float hi = 1.0)
```

@brief Clamp value between lower and upper bounds.

@param x Value to clamp
 @param lo Lower bound (default: 0.0)
 @param hi Upper bound (default: 1.0)
 @return Clamped value

Definition at line 44 of file [HealthMonitor.py](#).

```
00044 def clamp(x: float, lo: float = 0.0, hi: float = 1.0) -> float:
00045     """
00046     @brief Clamp value between lower and upper bounds.
00047
00048     @param x Value to clamp
00049     @param lo Lower bound (default: 0.0)
00050     @param hi Upper bound (default: 1.0)
00051     @return Clamped value
00052     """
00053     if x < lo:
00054         return lo
00055     if x > hi:
00056         return hi
00057     return x
00058
00059
```

4.5.2.2 compute_initial_health_from_state()

```
Dict[str, Any] HealthMonitor.compute_initial_health_from_state (
    Dict[str, Any] state_dict,
    float r_best = R_BEST,
    float r_worst = R_WORST,
    float s_best = S_BEST,
    float s_worst = S_WORST)

@brief Compute health score from initial calibration state.

@param state_dict Dictionary containing calibration state with intrinsics and extrinsics
@param r_best Best reprojection error threshold (default: R_BEST)
@param r_worst Worst reprojection error threshold (default: R_WORST)
@param s_best Best stereo RMS threshold (default: S_BEST)
@param s_worst Worst stereo RMS threshold (default: S_WORST)
@return Dictionary with health score, method name, and detailed metrics
```

Definition at line 90 of file [HealthMonitor.py](#).

```
00096 ) -> Dict[str, Any]:
00097     """
00098     @brief Compute health score from initial calibration state.
00099
00100     @param state_dict Dictionary containing calibration state with intrinsics and extrinsics
00101     @param r_best Best reprojection error threshold (default: R_BEST)
00102     @param r_worst Worst reprojection error threshold (default: R_WORST)
00103     @param s_best Best stereo RMS threshold (default: S_BEST)
00104     @param s_worst Worst stereo RMS threshold (default: S_WORST)
00105     @return Dictionary with health score, method name, and detailed metrics
00106     """
00107     st = state_dict.get("state", state_dict)
00108     intr = st.get("intrinsics", {}) or {}
00109     extr = st.get("extrinsics", {}) or {}
00110
00111     reproj_vals = []
00112     rms_vals = []
00113
00114     for cam, intr_data in intr.items():
00115         reproj = safe_float(intr_data.get("reprojectionError"), None)
00116         if reproj is not None:
00117             reproj_vals.append(reproj)
00118
00119         ex = extr.get(cam, {})
00120         rms = safe_float(ex.get("stereoRms"), None)
00121         if rms is not None:
00122             rms_vals.append(rms)
00123
00124     if not reproj_vals and not rms_vals:
00125         return {
00126             "health": 0.0,
00127             "method": "initial",
00128             "details": {"reason": "no reprojectionError and no stereoRms found"},
00129         }
00130
00131     reproj_mean = (sum(reproj_vals) / len(reproj_vals)) if reproj_vals else r_worst
00132     rms_mean = (sum(rms_vals) / len(rms_vals)) if rms_vals else s_worst
00133
00134     E_reproj = clamp((reproj_mean - r_best) / (r_worst - r_best), 0.0, 1.0)
00135     E_rms = clamp((rms_mean - s_best) / (s_worst - s_best), 0.0, 1.0)
00136
00137     score = score_from_errors(E_reproj, E_rms)
00138
00139     return {
00140         "health": float(score),
00141         "method": "initial",
00142         "details": {
00143             "reproj_mean": float(reproj_mean),
00144             "rms_mean": float(rms_mean),
00145             "E_reproj": float(E_reproj),
00146             "E_rms": float(E_rms),
00147             "thresholds": {
00148                 "R_BEST": float(r_best),
00149                 "R_WORST": float(r_worst),
00150                 "S_BEST": float(s_best),
00151                 "S_WORST": float(s_worst),
00152             },
00153         },
00154     }
00155
00156
```

4.5.2.3 compute_lifcal_health_from_combined_dict()

```
Dict[str, Any] HealthMonitor.compute_lifcal_health_from_combined_dict (
    Dict[str, Any] combined)
```

@brief Compute LiFCal health from combined.json structure.

@param combined Dictionary from combined.json with parameters for all cameras
 @return Dictionary with global health score, method name, and per-camera metrics

Definition at line 228 of file [HealthMonitor.py](#).

```
00228 def compute_lifcal_health_from_combined_dict(combined: Dict[str, Any]) -> Dict[str, Any]:
00229     """
00230     @brief Compute LiFCal health from combined.json structure.
00231
00232     @param combined Dictionary from combined.json with parameters for all cameras
00233     @return Dictionary with global health score, method name, and per-camera metrics
00234     """
00235     params_all = combined.get("state", {}).get("parameters", {})
00236     if not isinstance(params_all, dict) or len(params_all) == 0:
00237         return {"health": 0.0, "method": "lifcal", "perCam": {}}
00238
00239     per_cam = {}
00240     scores = []
00241
00242     for cam, p in params_all.items():
00243         rep = (p or {}).get("reprojection", {}) or {}
00244
00245         x_std = rep.get("x_std", None)
00246         y_std = rep.get("y_std", None)
00247         x_mae = rep.get("x_mae", None)
00248         y_mae = rep.get("y_mae", None)
00249
00250         # Skip camera if any field is missing
00251         if any(v is None for v in [x_std, y_std, x_mae, y_mae]):
00252             per_cam[cam] = {"health": 0.0, "reason": "missing reprojection fields"}
00253             continue
00254
00255         std_mag = math.sqrt(float(x_std) ** 2 + float(y_std) ** 2)
00256         mae_mag = math.sqrt(float(x_mae) ** 2 + float(y_mae) ** 2)
00257
00258         E_std = clamp((std_mag - S_BEST) / (S_WORST - S_BEST), 0.0, 1.0)
00259         E_mae = clamp((mae_mag - R_BEST) / (R_WORST - R_BEST), 0.0, 1.0)
00260
00261         health = 100.0 * (1.0 - (0.6 * (E_std ** 2) + 0.4 * (E_mae ** 2)))
00262         health = clamp(health, 0.0, 100.0)
00263
00264         per_cam[cam] = {
00265             "health": float(health),
00266             "std_mag": float(std_mag),
00267             "mae_mag": float(mae_mag),
00268             "E_std": float(E_std),
00269             "E_mae": float(E_mae),
00270         }
00271         scores.append(float(health))
00272
00273     if len(scores) == 0:
00274         return {"health": 0.0, "method": "lifcal", "perCam": per_cam}
00275
00276     # Global health: average across all cameras
00277     global_health = sum(scores) / len(scores)
00278
00279     return {"health": float(global_health), "method": "lifcal", "perCam": per_cam}
00280
00281
```

4.5.2.4 compute_lifcal_health_from_combined_path()

```
Dict[str, Any] HealthMonitor.compute_lifcal_health_from_combined_path (
    str combined_json_path)
```

@brief Load combined.json from file and compute LiFCal health.

@param combined_json_path Path to the combined.json file
 @return Dictionary with global health score, method name, and per-camera metrics

Definition at line 282 of file [HealthMonitor.py](#).

```
00282 def compute_lifcal_health_from_combined_path(combined_json_path: str) -> Dict[str, Any]:
00283     """
00284     @brief Load combined.json from file and compute LiFCal health.
00285
00286     @param combined_json_path Path to the combined.json file
00287     @return Dictionary with global health score, method name, and per-camera metrics
00288     """
00289     with open(combined_json_path, "r", encoding="utf-8") as f:
00290         combined = json.load(f)
00291     return compute_lifcal_health_from_combined_dict(combined)
```

4.5.2.5 compute_lifcal_health_from_xy()

```
Dict[str, Any] HealthMonitor.compute_lifcal_health_from_xy (
    float x_std,
    float y_std,
    float x_mae,
    float y_mae,
    float r_best = R_BEST,
    float r_worst = R_WORST,
    float s_best = S_BEST,
    float s_worst = S_WORST)

@brief Compute LiFCal health score from x/y standard deviation and MAE values.

@param x_std Standard deviation in x direction
@param y_std Standard deviation in y direction
@param x_mae Mean absolute error in x direction
@param y_mae Mean absolute error in y direction
@param r_best Best reprojection error threshold (default: R_BEST)
@param r_worst Worst reprojection error threshold (default: R_WORST)
@param s_best Best stereo RMS threshold (default: S_BEST)
@param s_worst Worst stereo RMS threshold (default: S_WORST)
@return Dictionary with health score, method name, and detailed metrics
```

Definition at line 180 of file [HealthMonitor.py](#).

```
00189 ) -> Dict[str, Any]:
00190     """
00191     @brief Compute LiFCal health score from x/y standard deviation and MAE values.
00192
00193     @param x_std Standard deviation in x direction
00194     @param y_std Standard deviation in y direction
00195     @param x_mae Mean absolute error in x direction
00196     @param y_mae Mean absolute error in y direction
00197     @param r_best Best reprojection error threshold (default: R_BEST)
00198     @param r_worst Worst reprojection error threshold (default: R_WORST)
00199     @param s_best Best stereo RMS threshold (default: S_BEST)
00200     @param s_worst Worst stereo RMS threshold (default: S_WORST)
00201     @return Dictionary with health score, method name, and detailed metrics
00202     """
00203     std_norm = (x_std * x_std + y_std * y_std) ** 0.5
00204     mae_norm = (x_mae * x_mae + y_mae * y_mae) ** 0.5
00205
00206     E_std = clamp((std_norm - s_best) / (s_worst - s_best), 0.0, 1.0)
00207     E_mae = clamp((mae_norm - r_best) / (r_worst - r_best), 0.0, 1.0)
00208
00209     score = score_from_errors(E_std, E_mae)
00210
00211     return {
00212         "health": float(score),
00213         "method": "lifcal",
00214         "details": {
00215             "std_norm": float(std_norm),
00216             "mae_norm": float(mae_norm),
00217             "E_std": float(E_std),
00218             "E_mae": float(E_mae),
00219             "thresholds": {
00220                 "R_BEST": float(r_best),
00221                 "R_WORST": float(r_worst),
00222                 "S_BEST": float(s_best),
00223                 "S_WORST": float(s_worst),
00224             },
00225         },
00226     }
00227
```

4.5.2.6 parse_lifcal_protocol()

```
Dict[str, Optional[float]] HealthMonitor.parse_lifcal_protocol (
    str protocol_path)
```

@brief Parse LiFCal calibration protocol text file for error metrics.

@param protocol_path Path to the calibration protocol text file

@return Dictionary with x_std, y_std, x_mae, y_mae values or None if not found

Definition at line 157 of file [HealthMonitor.py](#).

```
00157 def parse_lifcal_protocol(protocol_path: str) -> Dict[str, Optional[float]]:
00158     """
00159     @brief Parse LiFCal calibration protocol text file for error metrics.
00160
00161     @param protocol_path Path to the calibration protocol text file
00162     @return Dictionary with x_std, y_std, x_mae, y_mae values or None if not found
00163     """
00164     txt = open(protocol_path, "r", encoding="utf-8", errors="ignore").read()
00165
00166     def find_float(pattern: str) -> Optional[float]:
00167         m = re.search(pattern, txt, re.MULTILINE)
00168         if not m:
00169             return None
00170         return safe_float(m.group(1), None)
00171
00172     x_std = find_float(r"std\\.\\s*Dev\\.\\s*x:\\s*([0-9.+-eE]+)")
00173     y_std = find_float(r"std\\.\\s*Dev\\.\\s*y:\\s*([0-9.+-eE]+)")
00174     x_mae = find_float(r"mae\\s*x:\\s*([0-9.+-eE]+)")
00175     y_mae = find_float(r"mae\\s*y:\\s*([0-9.+-eE]+)")
00176
00177     return {"x_std": x_std, "y_std": y_std, "x_mae": x_mae, "y_mae": y_mae}
00178
00179
```

4.5.2.7 safe_float()

```
HealthMonitor.safe_float (
    v,
    default = None)
```

@brief Convert value to float safely, return default on error.

@param v Value to convert

@param default Default value to return on conversion error

@return Converted float value or default

Definition at line 60 of file [HealthMonitor.py](#).

```
00060 def safe_float(v, default=None):
00061     """
00062     @brief Convert value to float safely, return default on error.
00063
00064     @param v Value to convert
00065     @param default Default value to return on conversion error
00066     @return Converted float value or default
00067     """
00068     try:
00069         if v is None:
00070             return default
00071         return float(v)
00072     except Exception:
00073         return default
00074
00075
```

4.5.2.8 score_from_errors()

```
float HealthMonitor.score_from_errors (
    float E_a,
    float E_b,
    float w_a = 0.6,
    float w_b = 0.4)

@brief Compute health score from two normalized error metrics.

@param E_a First normalized error metric (0..1)
@param E_b Second normalized error metric (0..1)
@param w_a Weight for first error metric (default: 0.6)
@param w_b Weight for second error metric (default: 0.4)
@return Health score between 0 and 100
```

Definition at line 76 of file [HealthMonitor.py](#).

```
00076 def score_from_errors(E_a: float, E_b: float, w_a: float = 0.6, w_b: float = 0.4) -> float:
00077     """
00078     @brief Compute health score from two normalized error metrics.
00079
00080     @param E_a First normalized error metric (0..1)
00081     @param E_b Second normalized error metric (0..1)
00082     @param w_a Weight for first error metric (default: 0.6)
00083     @param w_b Weight for second error metric (default: 0.4)
00084     @return Health score between 0 and 100
00085     """
00086     val = 100.0 * (1.0 - (w_a * (E_a ** 2) + w_b * (E_b ** 2)))
00087     return clamp(val / 100.0, 0.0, 1.0) * 100.0
00088
00089
```

4.5.3 Variable Documentation

4.5.3.1 R_BEST

```
float HealthMonitor.R_BEST = 0.03
```

Definition at line 38 of file [HealthMonitor.py](#).

4.5.3.2 R_WORST

```
float HealthMonitor.R_WORST = 1.0
```

Definition at line 39 of file [HealthMonitor.py](#).

4.5.3.3 S_BEST

```
float HealthMonitor.S_BEST = 0.0
```

Definition at line 40 of file [HealthMonitor.py](#).

4.5.3.4 S_WORST

```
float HealthMonitor.S_WORST = 50.0
```

Definition at line 41 of file [HealthMonitor.py](#).

4.6 ImageSet Namespace Reference

Classes

- class [ImageSet](#)

4.7 InitialCalibration Namespace Reference

Classes

- class [InitialCalibration](#)

4.8 lifcal_to_json Namespace Reference

Functions

- dict [parse_protocol](#) (str protocol_path)
- Dict[str, Any] [parse_extrinsics_xml](#) (str xml_path)
- str [export_lifcal_parameters_json_in_place](#) (str result_dir, str cam_id, str image_dir, str run_type="recalib", float pixel_size_mm_fallback=0.0055, str protocol_name="calibrationProtocol.txt", str extr_name="extrinsicOrientations.xml", str out_name="parameters.json", Optional[float] pixel_size_mm=None, Optional[float] fallback_cx=None, Optional[float] fallback_cy=None)
- str [build_combined_parameters_json](#) (str run_root, list camera_ids, str out_name="combined.json")

4.8.1 Function Documentation

4.8.1.1 build_combined_parameters_json()

```
str lifcal_to_json.build_combined_parameters_json (
    str run_root,
    list camera_ids,
    str out_name = "combined.json")
```

@brief Combine per-camera parameters.json files into a single combined.json.

@param run_root Root directory containing per-camera subdirectories
 @param camera_ids List of camera identifiers
 @param out_name Output filename (default: "combined.json")
 @return Path to written combined.json file

Definition at line 206 of file [lifcal_to_json.py](#).

```
00206 def build_combined_parameters_json(run_root: str, camera_ids: list, out_name: str = "combined.json")
-> str:
00207     """
00208     @brief Combine per-camera parameters.json files into a single combined.json.
00209
00210     @param run_root Root directory containing per-camera subdirectories
00211     @param camera_ids List of camera identifiers
00212     @param out_name Output filename (default: "combined.json")
00213     @return Path to written combined.json file
00214     """
00215     combined = {
00216         "meta": {
00217             "runType": "combined",
```

```

00218         "cameraIds": camera_ids,
00219     },
00220     "state": {
00221         "parameters": {},
00222         "extrinsicsByPose": {},
00223         "timeStamp": None
00224     }
00225 }
00226
00227 found_any = False
00228
00229 for cam in camera_ids:
00230     cam_dir = os.path.join(run_root, cam)
00231     jpath = os.path.join(cam_dir, "parameters.json")
00232
00233     if not os.path.isfile(jpath):
00234         continue
00235
00236     with open(jpath, "r", encoding="utf-8") as f:
00237         data = json.load(f)
00238
00239     st = data.get("state", {})
00240     params = st.get("parameters", {})
00241     extr_pose = st.get("extrinsicsByPose", {})
00242
00243     if cam in params:
00244         combined["state"]["parameters"][cam] = params[cam]
00245         found_any = True
00246
00247     if cam in extr_pose:
00248         combined["state"]["extrinsicsByPose"][cam] = extr_pose[cam]
00249         found_any = True
00250
00251     if not found_any:
00252         raise RuntimeError("No parameters.json found to combine.")
00253
00254     out_path = os.path.join(run_root, out_name)
00255     with open(out_path, "w", encoding="utf-8") as f:
00256         json.dump(combined, f, indent=2)
00257
00258     return out_path

```

4.8.1.2 export_lifcal_parameters_json_in_place()

```

str lifcal_to_json.export_lifcal_parameters_json_in_place (
    str result_dir,
    str cam_id,
    str image_dir,
    str run_type = "recalib",
    float pixel_size_mm_fallback = 0.0055,
    str protocol_name = "calibrationProtocol.txt",
    str extr_name = "extrinsicOrientations.xml",
    str out_name = "parameters.json",
    Optional[float] pixel_size_mm = None,
    Optional[float] fallback_cx = None,
    Optional[float] fallback_cy = None)

@brief Write LiFCal parameters.json for a single camera.

@param result_dir Directory containing LiFCal output files
@param cam_id Camera identifier
@param image_dir Directory containing input images
@param run_type Type of calibration run (default: "recalib")
@param pixel_size_mm_fallback Fallback pixel size in mm (default: 0.0055)
@param protocol_name Protocol filename (default: "calibrationProtocol.txt")
@param extr_name Extrinsics filename (default: "extrinsicOrientations.xml")
@param out_name Output JSON filename (default: "parameters.json")
@param pixel_size_mm Optional pixel size override
@param fallback_cx Optional fallback cx (compatibility only, not used)
@param fallback_cy Optional fallback cy (compatibility only, not used)
@return Path to written parameters.json file

```

Definition at line 112 of file `lifcal_to_json.py`.

```

00125 ) -> str:
00126     """
00127     @brief Write LiFCal parameters.json for a single camera.
00128
00129     @param result_dir Directory containing LiFCal output files
00130     @param cam_id Camera identifier
00131     @param image_dir Directory containing input images
00132     @param run_type Type of calibration run (default: "recalib")
00133     @param pixel_size_mm_fallback Fallback pixel size in mm (default: 0.0055)
00134     @param protocol_name Protocol filename (default: "calibrationProtocol.txt")
00135     @param extr_name Extrinsics filename (default: "extrinsicOrientations.xml")
00136     @param out_name Output JSON filename (default: "parameters.json")
00137     @param pixel_size_mm Optional pixel size override
00138     @param fallback_cx Optional fallback cx (compatibility only, not used)
00139     @param fallback_cy Optional fallback cy (compatibility only, not used)
00140     @return Path to written parameters.json file
00141     """
00142     if pixel_size_mm is not None:
00143         pixel_size_mm_fallback = float(pixel_size_mm)
00144
00145     protocol_path = os.path.join(result_dir, protocol_name)
00146     extr_path = os.path.join(result_dir, extr_name)
00147
00148     if not os.path.isfile(protocol_path):
00149         raise FileNotFoundError(protocol_path)
00150     if not os.path.isfile(extr_path):
00151         raise FileNotFoundError(extr_path)
00152
00153     protocol = parse_protocol(protocol_path)
00154     extr_by_pose = parse_extrinsics_xml(extr_path)
00155
00156     # Ensure pixel size is available
00157     px = protocol.get("pixelSizeMm", None)
00158     if px is None or (not np.isfinite(px)) or px <= 0:
00159         protocol["pixelSizeMm"] = float(pixel_size_mm_fallback)
00160
00161     # Compact reprojection error (average MAE)
00162     rep = protocol.get("reprojection", {})
00163     x_mae = rep.get("x_mae", None)
00164     y_mae = rep.get("y_mae", None)
00165     avg_mae = None
00166     if x_mae is not None and y_mae is not None and np.isfinite(x_mae) and np.isfinite(y_mae):
00167         avg_mae = float((x_mae + y_mae) / 2.0)
00168
00169     data = {
00170         "meta": {
00171             "runType": run_type,
00172             "numPoses": int(len(extr_by_pose)),
00173             "imageDir": image_dir,
00174             "cameraIds": [cam_id],
00175             "bundleAdjust": True,
00176         },
00177         "state": {
00178             "parameters": {
00179                 cam_id: {
00180                     "pixelSizeMm": protocol.get("pixelSizeMm"),
00181                     "fL_mm": protocol.get("fL_mm"),
00182                     "bL0_mm": protocol.get("bL0_mm"),
00183                     "B_mm": protocol.get("B_mm"),
00184                     "cx_px": protocol.get("cx_px"),
00185                     "cy_px": protocol.get("cy_px"),
00186                     "a0": protocol.get("a0"),
00187                     "a1": protocol.get("a1"),
00188                     "b0": protocol.get("b0"),
00189                     "b1": protocol.get("b1"),
00190                     "reprojection": rep,
00191                     "reprojectionAvgMae": avg_mae,
00192                 }
00193             },
00194             "extrinsicsByPose": {cam_id: extr_by_pose},
00195             "timeStamp": None
00196         }
00197     }
00198
00199     out_path = os.path.join(result_dir, out_name)
00200     with open(out_path, "w", encoding="utf-8") as f:
00201         json.dump(data, f, indent=2)
00202
00203     return out_path
00204
00205

```

4.8.1.3 parse_extrinsics_xml()

```
Dict[str, Any] lifcal_to_json.parse_extrinsics_xml (
    str xml_path)

@brief Parse LiFCal extrinsic orientations XML file.

@param xml_path Path to extrinsicOrientations.xml file
@return Dictionary mapping pose keys to rotation and translation data
```

Definition at line 70 of file [lifcal_to_json.py](#).

```
00070 def parse_extrinsics_xml(xml_path: str) -> Dict[str, Any]:
00071     """
00072     @brief Parse LiFCal extrinsic orientations XML file.
00073
00074     @param xml_path Path to extrinsicOrientations.xml file
00075     @return Dictionary mapping pose keys to rotation and translation data
00076     """
00077     root = ET.parse(xml_path).getroot()
00078     out: Dict[str, Any] = {}
00079
00080     for frame in root.findall("Frame"):
00081         fid = int(frame.attrib.get("id"))
00082         pose_key = "pose%03d" % fid
00083
00084         rot = frame.find("Rotation")
00085         tra = frame.find("Translation")
00086
00087         rvec = [0.0, 0.0, 0.0]
00088         tvec = [0.0, 0.0, 0.0]
00089
00090         if rot is not None:
00091             for c in rot.findall("Coeff"):
00092                 i = int(c.attrib["i"])
00093                 rvec[i] = float(c.text.strip())
00094
00095         if tra is not None:
00096             for c in tra.findall("Coeff"):
00097                 i = int(c.attrib["i"])
00098                 tvec[i] = float(c.text.strip())
00099
00100         rvec_np = np.array(rvec, dtype=np.float64).reshape(3, 1)
00101         R, _ = cv2.Rodrigues(rvec_np)
00102
00103         out[pose_key] = {
00104             "rotationMatrix": R.astype(float).tolist(),
00105             "rotationVector": [[float(rvec[0])], [float(rvec[1])], [float(rvec[2])]],
00106             "translationVector": [[float(tvec[0])], [float(tvec[1])], [float(tvec[2])]],
00107         }
00108
00109     return out
00110
00111
```

4.8.1.4 parse_protocol()

```
dict lifcal_to_json.parse_protocol (
    str protocol_path)

@brief Parse LiFCal calibration protocol text file.

@return Dictionary containing parsed calibration parameters and reprojection errors
```

Definition at line 15 of file [lifcal_to_json.py](#).

```
00015 def parse_protocol(protocol_path: str) -> dict:
00016     """
00017     @brief Parse LiFCal calibration protocol text file.
00018
00019     @return Dictionary containing parsed calibration parameters and reprojection errors
00020     """
00021     txt = open(protocol_path, "r", encoding="utf-8", errors="ignore").read()
```

```

00022
00023     def find_float(pattern: str, default=None) -> Optional[float]:
00024         m = re.search(pattern, txt, re.MULTILINE)
00025         if not m:
00026             return default
00027         try:
00028             return float(m.group(1))
00029         except Exception:
00030             return default
00031
00032     pixel_size = find_float(r"Pixel Size:\s*([0-9.+-eE]+)\s*mm", None)
00033
00034     fL = find_float(r"^\s*fL\s*:\s*([0-9.+-eE]+)\s*$", None)
00035     bL0 = find_float(r"^\s*bL0\s*:\s*([0-9.+-eE]+)\s*$", None)
00036     B = find_float(r"^\s*B\s*:\s*([0-9.+-eE]+)\s*$", None)
00037     cx = find_float(r"^\s*cx\s*:\s*([0-9.+-eE]+)\s*$", None)
00038     cy = find_float(r"^\s*cy\s*:\s*([0-9.+-eE]+)\s*$", None)
00039
00040     a0 = find_float(r"^\s*a0\s*:\s*([0-9.+-eE]+)\s*$", 0.0)
00041     a1 = find_float(r"^\s*a1\s*:\s*([0-9.+-eE]+)\s*$", 0.0)
00042     b0 = find_float(r"^\s*b0\s*:\s*([0-9.+-eE]+)\s*$", 0.0)
00043     b1 = find_float(r"^\s*b1\s*:\s*([0-9.+-eE]+)\s*$", 0.0)
00044
00045     x_std = find_float(r"std\. \s*Dev\. \s*x:\s*([0-9.+-eE]+)", None)
00046     y_std = find_float(r"std\. \s*Dev\. \s*y:\s*([0-9.+-eE]+)", None)
00047     x_mae = find_float(r"mae\s*x:\s*([0-9.+-eE]+)", None)
00048     y_mae = find_float(r"mae\s*y:\s*([0-9.+-eE]+)", None)
00049
00050     return {
00051         "pixelSizeMm": pixel_size,
00052         "fL_mm": fL,
00053         "bL0_mm": bL0,
00054         "B_mm": B,
00055         "cx_px": cx,
00056         "cy_px": cy,
00057         "a0": a0,
00058         "a1": a1,
00059         "b0": b0,
00060         "b1": b1,
00061         "reprojection": {
00062             "x_std": x_std,
00063             "y_std": y_std,
00064             "x_mae": x_mae,
00065             "y_mae": y_mae
00066         }
00067     }
00068
00069

```

4.9 ResultLogger Namespace Reference

Classes

- class [ResultLogger](#)

4.10 run_imageset_creation Namespace Reference

Functions

- [ensure_camera_subfolders](#) (str base_dir)
- [group_focus_images_by_pose](#) ()
- [copy_focus_into_camera_folders](#) (str pose, dict cam_to_path)
- [cleanup_focus_root](#) ()
- [main](#) ()

Variables

- str `ROOT` = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
- `FOCUS_DIR` = `os.path.join(ROOT, "focus")`
- `DEPTH_DIR` = `os.path.join(ROOT, "depth")`
- bool `CLEAN_FOCUS_ROOT_AFTER` = True
- list `CAMERA_IDS`
- `PATTERN`

4.10.1 Function Documentation

4.10.1.1 `cleanup_focus_root()`

```
run_imageset_creation.cleanup_focus_root ()
```

@brief Delete only loose files directly in focus/ (not in subdirectories).

Definition at line 70 of file [run_imageset_creation.py](#).

```
00070 def cleanup_focus_root():
00071     """
00072     @brief Delete only loose files directly in focus/ (not in subdirectories).
00073     """
00074     for fn in os.listdir(FOCUS_DIR):
00075         p = os.path.join(FOCUS_DIR, fn)
00076         if os.path.isfile(p) and PATTERN.match(fn):
00077             os.remove(p)
00078
```

4.10.1.2 `copy_focus_into_camera_folders()`

```
run_imageset_creation.copy_focus_into_camera_folders (
    str pose,
    dict cam_to_path)
```

@brief Copy focus images into camera-specific subdirectories.

@param pose Pose identifier string (e.g., "pose000")

@param cam_to_path Dictionary mapping camera IDs to source image paths

Definition at line 57 of file [run_imageset_creation.py](#).

```
00057 def copy_focus_into_camera_folders(pose: str, cam_to_path: dict):
00058     """
00059     @brief Copy focus images into camera-specific subdirectories.
00060
00061     @param pose Pose identifier string (e.g., "pose000")
00062     @param cam_to_path Dictionary mapping camera IDs to source image paths
00063     """
00064     for cam, src in cam_to_path.items():
00065         ext = os.path.splitext(src)[1].lower()
00066         dst = os.path.join(FOCUS_DIR, cam, f"{pose}_{cam}{ext}")
00067         if os.path.abspath(src) != os.path.abspath(dst):
00068             shutil.copy2(src, dst)
00069
```

4.10.1.3 ensure_camera_subfolders()

```
run_imageset_creation.ensure_camera_subfolders (
    str base_dir)
```

@brief Create subdirectory for each camera ID under base_dir.

@param base_dir Base directory path

Definition at line 32 of file [run_imageset_creation.py](#).

```
00032 def ensure_camera_subfolders(base_dir: str):
00033     """
00034     @brief Create subdirectory for each camera ID under base_dir.
00035
00036     @param base_dir Base directory path
00037     """
00038     os.makedirs(base_dir, exist_ok=True)
00039     for cid in CAMERA_IDS:
00040         os.makedirs(os.path.join(base_dir, cid), exist_ok=True)
00041
```

4.10.1.4 group_focus_images_by_pose()

```
run_imageset_creation.group_focus_images_by_pose ()
```

@brief Group focus images by pose ID from flat directory structure.

@return Dictionary mapping pose IDs to camera-path dictionaries

Definition at line 42 of file [run_imageset_creation.py](#).

```
00042 def group_focus_images_by_pose():
00043     """
00044     @brief Group focus images by pose ID from flat directory structure.
00045
00046     @return Dictionary mapping pose IDs to camera-path dictionaries
00047     """
00048     pose_to_cam = defaultdict(dict)
00049     for fn in sorted(os.listdir(FOCUS_DIR)):
00050         m = PATTERN.match(fn)
00051         if not m:
00052             continue
00053         pose, cam, ext = m.group(1), m.group(2), m.group(3)
00054         pose_to_cam[pose][cam] = os.path.join(FOCUS_DIR, fn)
00055     return pose_to_cam
00056
```

4.10.1.5 main()

```
run_imageset_creation.main ()
```

@brief Main pipeline: organize focus images and generate depth maps.

Definition at line 79 of file [run_imageset_creation.py](#).

```
00079 def main():
00080     """
00081     @brief Main pipeline: organize focus images and generate depth maps.
00082     """
00083     if not os.path.isdir(FOCUS_DIR):
00084         raise SystemExit(f"focus folder not found: {FOCUS_DIR}")
00085
00086     ensure_camera_subfolders(FOCUS_DIR)
00087     ensure_camera_subfolders(DEPTH_DIR)
00088
```

```

00089     pose_to_cam = group_focus_images_by_pose()
00090     if not pose_to_cam:
00091         raise SystemExit("No focus images found. Expected: pose000_Center.png etc.")
00092
00093     poses = sorted(pose_to_cam.keys())
00094     print("Found poses:", len(poses))
00095
00096     for i, pose in enumerate(poses, start=1):
00097         cam_to_focus_path = pose_to_cam[pose]
00098
00099         # Sort focus images into camera folders
00100         copy_focus_into_camera_folders(pose, cam_to_focus_path)
00101
00102         # Generate depth maps for this pose
00103         depthmaps = depthmap.generate_depthmaps_for_pose(cam_to_focus_path)
00104
00105         # Save depth maps: depth/<CamId>/<pose>_<CamId>_depth.png
00106         for cam, depth_ul6 in depthmaps.items():
00107             out_path = os.path.join(DEPTH_DIR, cam, f"{pose}_{cam}_depth.png")
00108             ok = depthmap.imwrite_ul6(out_path, depth_ul6)
00109             if not ok:
00110                 print("FAILED write:", out_path)
00111
00112         print(f"[{i}/{len(poses)}] {pose}: views={len(cam_to_focus_path)} depthmaps={len(depthmaps)}")
00113
00114     if CLEAN_FOCUS_ROOT_AFTER:
00115         cleanup_focus_root()
00116
00117     print("DONE.")
00118     print("Focus by camera:", FOCUS_DIR)
00119     print("Depth by camera:", DEPTH_DIR)
00120

```

4.10.2 Variable Documentation

4.10.2.1 CAMERA_IDS

list run_imageset_creation.CAMERA_IDS

Initial value:

```

00001 = [
00002     "Center", "Up1", "Up2", "Up3",
00003     "Down1", "Down2", "Down3",
00004     "Left1", "Left2", "Left3",
00005     "Right1", "Right2", "Right3"
00006 ]

```

Definition at line 19 of file [run_imageset_creation.py](#).

4.10.2.2 CLEAN_FOCUS_ROOT_AFTER

bool run_imageset_creation.CLEAN_FOCUS_ROOT_AFTER = True

Definition at line 17 of file [run_imageset_creation.py](#).

4.10.2.3 DEPTH_DIR

run_imageset_creation.DEPTH_DIR = os.path.join(ROOT, "depth")

Definition at line 14 of file [run_imageset_creation.py](#).

4.10.2.4 FOCUS_DIR

run_imageset_creation.FOCUS_DIR = os.path.join(ROOT, "focus")

Definition at line 13 of file [run_imageset_creation.py](#).

4.10.2.5 PATTERN

```
run_imageset_creation.PATTERN
```

Initial value:

```
00001 = re.compile(
00002     r"^(pose\d+)_ (Center|Up[123]|Down[123]|Left[123]|Right[123]) \. (png|jpg|jpeg) $",
00003     re.IGNORECASE
00004 )
```

Definition at line 27 of file [run_imageset_creation.py](#).

4.10.2.6 ROOT

```
str run_imageset_creation.ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_↵
Imageset"
```

Definition at line 12 of file [run_imageset_creation.py](#).

4.11 run_initial_calibration Namespace Reference

Functions

- [parse_args](#) ()
- [main](#) (args)

Variables

- str [DEFAULT_BASE_DIR](#) = "/data/calibrationLFC/sets/imageset5"
- int [DEFAULT_NUM_POSES](#) = 28
- list [CAM_IDS](#)
- [repoRoot](#) = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))

4.11.1 Detailed Description

Test runner for initial calibration.

Usage:

```
cd calibrationLFC
python3 run_initial_calibration.py --base_dir /path/to/imageset --num_poses 28 --bundle-adjust
python3 run_initial_calibration.py --base_dir /path/to/imageset --num_poses 28 --no-bundle-adjust
```

Performs validation checks (file existence, corner detection on pose 0)
then runs the complete initial calibration pipeline. Outputs summary statistics
for intrinsics, extrinsics, and health score.

4.11.2 Function Documentation

4.11.2.1 main()

```
run_initial_calibration.main (
    args)
```

@brief Run complete initial calibration pipeline with validation and summary output.

@param args

Definition at line 59 of file [run_initial_calibration.py](#).

```
00059 def main(args):
00060     """
00061     @brief Run complete initial calibration pipeline with validation and summary output.
00062
00063     @param args
00064     """
00065     print("Creating ImageSet...")
00066     imageSet = ImageSet(args.base_dir, args.num_poses, CAM_IDS)
00067
00068     print("Instantiating Controller and InitialCalibration...")
00069     controller = Controller(bundleAdjust=args.bundle_adjust)
00070     logger = controller.resultLogger
00071
00072     print("\nQuick checks (pose 0): file exists / corners detected")
00073     for cam in CAM_IDS:
00074         path = imageSet.getImagePath(0, cam)
00075         exists = os.path.exists(path)
00076         cornersFound = None
00077
00078         try:
00079             cornersFound, _, _ = controller.initialCalibration.detectCorners(path)
00080         except Exception as e:
00081             cornersFound = f"error: {e.__class__.__name__}"
00082         print(f" {cam}: {path} - exists={exists} - corners={cornersFound}")
00083
00084     print("\nRunning initial calibration (this may take time depending on data)...")
00085     try:
00086         calibrationState = controller.runInitialCalibration(imageSet)
00087     except Exception as e:
00088         logger.logger.error(f"Calibration failed with exception: {e}", exc_info=True)
00089         print("Calibration failed, see calibration.log for details.")
00090         raise
00091
00092     print("\nCalibration finished. Attempting to print summary...")
00093     try:
00094         stateDict = calibrationState.__getState__()
00095     except Exception:
00096         stateDict = None
00097
00098     if stateDict is None:
00099         print("Could not extract a state dictionary.")
00100         return
00101
00102     intr = stateDict.get("intrinsics", {})
00103     extr = stateDict.get("extrinsics", {})
00104
00105     # Print calibration results per camera
00106     print("\nCalibration Results by Camera")
00107     print("=" * 50)
00108
00109     for cam in CAM_IDS:
00110         print(f"\n--- {cam} ---")
00111
00112         # Intrinsic parameters
00113         if cam in intr:
00114             data = intr[cam]
00115             K = data["cameraMatrix"]
00116             dist = data["distortionCoeffs"].ravel()
00117             err = data["reprojectionError"]
00118
00119             print("Intrinsics:")
00120             print(" K =")
00121             print(f"   {K[0][0]:.3f} {K[0][1]:.3f} {K[0][2]:.3f}")
00122             print(f"   {K[1][0]:.3f} {K[1][1]:.3f} {K[1][2]:.3f}")
00123             print(f"   {K[2][0]:.3f} {K[2][1]:.3f} {K[2][2]:.3f}")
00124             print(" distCoeffs =", dist)
00125             print(f" reprojectionError = {err:.6f}")
```

```

00126         else:
00127             print("Intrinsics: (no data)")
00128
00129         # Extrinsic parameters
00130         if cam in extr:
00131             data = extr[cam]
00132             R = data["rotationMatrix"]
00133             T = data["translationVector"].ravel()
00134             rms = data["stereoRms"]
00135
00136             print("Extrinsics:")
00137             print(" R =")
00138             print(f" {R[0][0]:.5f} {R[0][1]:.5f} {R[0][2]:.5f}")
00139             print(f" {R[1][0]:.5f} {R[1][1]:.5f} {R[1][2]:.5f}")
00140             print(f" {R[2][0]:.5f} {R[2][1]:.5f} {R[2][2]:.5f}")
00141             print(" T =", T)
00142             print(f" stereoRMS = {rms:.6f}")
00143         else:
00144             print(" Extrinsics: (no data)")
00145
00146         # Print health score
00147         print("\nOverall Health Score")
00148         print("=" * 50)
00149         score = controller.selfHealthCheck(calibrationState)
00150         print(f"Global HealthScore = {score:.2f} / 100")
00151

```

4.11.2.2 parse_args()

run_initial_calibration.parse_args ()

@brief Parse command line arguments for calibration configuration.

@return Parsed arguments with base_dir, num_poses, and bundle_adjust settings

Definition at line 34 of file [run_initial_calibration.py](#).

```

00034 def parse_args():
00035     """
00036     @brief Parse command line arguments for calibration configuration.
00037
00038     @return Parsed arguments with base_dir, num_poses, and bundle_adjust settings
00039     """
00040     parser = argparse.ArgumentParser(description="Run initial camera calibration")
00041     parser.add_argument("--base_dir", default=DEFAULT_BASE_DIR, help="Path to imageset directory")
00042     parser.add_argument("--num_poses", type=int, default=DEFAULT_NUM_POSES, help="Number of poses in
the imageset")
00043     parser.add_argument(
00044         "--bundle-adjust",
00045         dest="bundle_adjust",
00046         action="store_true",
00047         help="Enable bundle adjustment (default)",
00048     )
00049     parser.add_argument(
00050         "--no-bundle-adjust",
00051         dest="bundle_adjust",
00052         action="store_false",
00053         help="Disable bundle adjustment",
00054     )
00055     parser.set_defaults(bundle_adjust=True)
00056     return parser.parse_args()
00057
00058

```

4.11.3 Variable Documentation

4.11.3.1 CAM_IDS

list run_initial_calibration.CAM_IDS

Initial value:

```

00001 = [
00002     "Center",
00003     "Up1", "Up2", "Up3",
00004     "Down1", "Down2", "Down3",
00005     "Left1", "Left2", "Left3",
00006     "Right1", "Right2", "Right3",
00007 ]

```

Definition at line 25 of file [run_initial_calibration.py](#).

4.11.3.2 DEFAULT_BASE_DIR

```
str run_initial_calibration.DEFAULT_BASE_DIR = "/data/calibrationLFC/sets/imageset5"
```

Definition at line 23 of file [run_initial_calibration.py](#).

4.11.3.3 DEFAULT_NUM_POSES

```
int run_initial_calibration.DEFAULT_NUM_POSES = 28
```

Definition at line 24 of file [run_initial_calibration.py](#).

4.11.3.4 repoRoot

```
run_initial_calibration.repoRoot = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
```

Definition at line 154 of file [run_initial_calibration.py](#).

4.12 run_LiFCal_calibration Namespace Reference

Functions

- str [read_text](#) (str path)
- None [write_text](#) (str path, str s)
- str [replace_yaml_key_line](#) (str text, str key, str new_val)
- str [patch_settings_yaml](#) (str template_text, str focus_dir, str depth_dir, str mla_xml)
- int [run_lifcal_recalib](#) (str settings_path, str fixed_params_path, str store_dir)
- str [find_results_folder](#) (str store_dir)
- None [copy_essentials](#) (str result_dir, str out_cam_dir)
- [main](#) ()

Variables

- str [LIFCAL_BIN](#) = "/data/external/LiFCal/build/bin/LiFCal"
- str [SETTINGS](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/Settings.yaml"
- str [FIXED_PARAMS](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/Fixed_Parameters.txt"
- str [MLA_CALIB_XML](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/MLA_Calibration.xml"
- str [ROOT_IMAGESET](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
- [FOCUS_ROOT](#) = os.path.join([ROOT_IMAGESET](#), "focus")
- [DEPTH_ROOT](#) = os.path.join([ROOT_IMAGESET](#), "depth")
- str [OUT_ROOT](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/CalibrationByCamera"
- list [CAMERA_IDS](#)

4.12.1 Function Documentation

4.12.1.1 `copy_essentials()`

```
None run_LiFCal_calibration.copy_essentials (
    str result_dir,
    str out_cam_dir)
```

@brief Copy calibrationProtocol.txt and extrinsicOrientations.xml to output directory.

@param result_dir Source directory containing LiFCal results

@param out_cam_dir Destination directory for essential files

Definition at line 137 of file [run_LiFCal_calibration.py](#).

```
00137 def copy_essentials(result_dir: str, out_cam_dir: str) -> None:
00138     """
00139     @brief Copy calibrationProtocol.txt and extrinsicOrientations.xml to output directory.
00140
00141     @param result_dir Source directory containing LiFCal results
00142     @param out_cam_dir Destination directory for essential files
00143     """
00144     proto = os.path.join(result_dir, "calibrationProtocol.txt")
00145     extr = os.path.join(result_dir, "extrinsicOrientations.xml")
00146
00147     missing = []
00148     if not os.path.isfile(proto):
00149         missing.append("calibrationProtocol.txt")
00150     if not os.path.isfile(extr):
00151         missing.append("extrinsicOrientations.xml")
00152     if missing:
00153         raise RuntimeError(f"Missing files in {result_dir}: {missing}")
00154
00155     shutil.copy2(proto, os.path.join(out_cam_dir, "calibrationProtocol.txt"))
00156     shutil.copy2(extr, os.path.join(out_cam_dir, "extrinsicOrientations.xml"))
00157
00158
```

4.12.1.2 `find_results_folder()`

```
str run_LiFCal_calibration.find_results_folder (
    str store_dir)
```

@brief Find LiFCal results folder.

@param store_dir LiFCal output directory

@return Path to results folder (typically Calibration_Results_... subdirectory)

Definition at line 114 of file [run_LiFCal_calibration.py](#).

```
00114 def find_results_folder(store_dir: str) -> str:
00115     """
00116     @brief Find LiFCal results folder.
00117
00118     @param store_dir LiFCal output directory
00119     @return Path to results folder (typically Calibration_Results_... subdirectory)
00120     """
00121     if not os.path.isdir(store_dir):
00122         raise RuntimeError(f"store_dir does not exist: {store_dir}")
00123
00124     subdirs = []
00125     for name in os.listdir(store_dir):
00126         p = os.path.join(store_dir, name)
00127         if os.path.isdir(p):
00128             subdirs.append(p)
00129
00130     if not subdirs:
00131         return store_dir
00132
00133     subdirs.sort(key=lambda p: os.path.getmtime(p))
00134     return subdirs[-1]
00135
00136
```

4.12.1.3 main()

```
run_LiFCal_calibration.main ()
```

@brief Main pipeline: run LiFCal for each camera, export JSON, and log health.

Definition at line 159 of file run_LiFCal_calibration.py.

```
00159 def main():
00160     """
00161     @brief Main pipeline: run LiFCal for each camera, export JSON, and log health.
00162     """
00163     if not os.path.isfile(LIFCAL_BIN):
00164         raise SystemExit(f"LiFCal binary not found: {LIFCAL_BIN}")
00165     if not os.path.isfile(SETTINGS):
00166         raise SystemExit(f"Settings template not found: {SETTINGS}")
00167     if not os.path.isfile(FIXED_PARAMS):
00168         raise SystemExit(f"Fixed parameters file not found: {FIXED_PARAMS}")
00169     if not os.path.isfile(MLA_CALIB_XML):
00170         raise SystemExit(f"MLA_Calibration.xml not found: {MLA_CALIB_XML}")
00171
00172     templ = read_text(SETTINGS)
00173
00174     timestamp = datetime.now().strftime("%Y_%m_%d_%H%M%S")
00175     session_out = os.path.join(OUT_ROOT, f"run_{timestamp}")
00176     os.makedirs(session_out, exist_ok=True)
00177
00178     print("Output:", session_out)
00179
00180     for cam in CAMERA_IDS:
00181         focus_dir = os.path.join(FOCUS_ROOT, cam)
00182         depth_dir = os.path.join(DEPTH_ROOT, cam)
00183
00184         if not os.path.isdir(focus_dir):
00185             print(f"[SKIP] {cam}: focus dir missing: {focus_dir}")
00186             continue
00187         if not os.path.isdir(depth_dir):
00188             print(f"[SKIP] {cam}: depth dir missing: {depth_dir}")
00189             continue
00190
00191         cam_out = os.path.join(session_out, cam)
00192         os.makedirs(cam_out, exist_ok=True)
00193
00194         settings_cam = os.path.join(cam_out, "Settings.yaml")
00195         patched = patch_settings_yaml(templ, focus_dir, depth_dir, MLA_CALIB_XML)
00196         write_text(settings_cam, patched)
00197
00198         store_dir = os.path.join(cam_out, "lifcal_store")
00199         rc = run_lifcal_recalib(settings_cam, FIXED_PARAMS, store_dir)
00200
00201         if rc != 0:
00202             print(f"[FAIL] {cam}: LiFCal returncode={rc} (siehe {store_dir}/lifcal_stdout.log)")
00203             continue
00204
00205         result_dir = find_results_folder(store_dir)
00206
00207         try:
00208             copy_essentials(result_dir, cam_out) # ???iert protocol + xml nach cam_out
00209         except Exception as e:
00210             print(f"[FAIL] {cam}: {e}")
00211             print(f"check log: {store_dir}/lifcal_stdout.log")
00212             continue
00213
00214         # store dir wegräumen
00215         shutil.rmtree(store_dir, ignore_errors=True)
00216
00217         # ? JSON EXPORT: result_dir muss cam_out sein, weil DA liegen jetzt die 2 Dateien
00218         try:
00219             json_path = lifcal_to_json.export_lifcal_parameters_json_in_place(
00220                 result_dir=cam_out,
00221                 cam_id=cam,
00222                 image_dir="LiFCal_Imageset",
00223                 run_type="recalib",
00224                 pixel_size_mm=0.0055,
00225                 fallback_cx=640.0, # compatibility only
00226                 fallback_cy=360.0, # compatibility only
00227                 out_name="parameters.json"
00228             )
00229         except Exception as e:
00230             print(f"[WARN] {cam}: JSON export failed: {e}")
00231
00232     # Build combined JSON from all cameras
```

```

00233     try:
00234         combined_path = lifcal_to_json.build_combined_parameters_json(
00235             run_root=session_out,
00236             camera_ids=CAMERA_IDS,
00237             out_name="combined.json"
00238         )
00239         print("[OK] COMBINED:", combined_path)
00240     except Exception as e:
00241         print("[WARN] combined.json not created:", e)
00242
00243     # Log health score and update baseline
00244     try:
00245         import sys
00246         CALIB_ROOT = "/data/calibrationLFC"
00247         if CALIB_ROOT not in sys.path:
00248             sys.path.insert(0, CALIB_ROOT)
00249
00250         from ResultLogger import ResultLogger
00251         logger = ResultLogger(baseDir="/data/baseline")
00252         score = logger.logRecalibration(
00253             combined_json_path=combined_path,
00254             meta={
00255                 "runType": "recalib",
00256                 "imageDir": "LiFCal_Imageset",
00257                 "bundleAdjust": True
00258             }
00259         )
00260         print(f"[OK] HealthScore (lifcal) = {score:.2f}")
00261     except Exception as e:
00262         print("[WARN] LiFCal logging/health failed:", e)
00263
00264

```

4.12.1.4 patch_settings_yaml()

```

str run_LiFCal_calibration.patch_settings_yaml (
    str template_text,
    str focus_dir,
    str depth_dir,
    str mla_xml)

```

@brief Update YAML settings with camera-specific paths.

@param template_text Original YAML template content
 @param focus_dir Path to focus images directory
 @param depth_dir Path to depth maps directory
 @param mla_xml Path to MLA calibration XML file
 @return Modified YAML text with updated paths

Definition at line 69 of file [run_LiFCal_calibration.py](#).

```

00069 def patch_settings_yaml(template_text: str, focus_dir: str, depth_dir: str, mla_xml: str) -> str:
00070     """
00071     @brief Update YAML settings with camera-specific paths.
00072
00073     @param template_text Original YAML template content
00074     @param focus_dir Path to focus images directory
00075     @param depth_dir Path to depth maps directory
00076     @param mla_xml Path to MLA calibration XML file
00077     @return Modified YAML text with updated paths
00078     """
00079     out = template_text
00080     out = replace_yaml_key_line(out, "Path.totalFocusImages", focus_dir)
00081     out = replace_yaml_key_line(out, "Path.virtualDepthData", depth_dir)
00082     out = replace_yaml_key_line(out, "Path.microLensCalibration", mla_xml)
00083     return out
00084
00085

```

4.12.1.5 read_text()

```

str run_LiFCal_calibration.read_text (
    str path)

```

```
@brief Read text file with UTF-8 encoding.

@param path Path to the text file
@return File content as string
```

Definition at line 33 of file [run_LiFCal_calibration.py](#).

```
00033 def read_text(path: str) -> str:
00034     """
00035     @brief Read text file with UTF-8 encoding.
00036
00037     @param path Path to the text file
00038     @return File content as string
00039     """
00040     with open(path, "r", encoding="utf-8", errors="replace") as f:
00041         return f.read()
00042
00043
```

4.12.1.6 replace_yaml_key_line()

```
str run_LiFCal_calibration.replace_yaml_key_line (
    str text,
    str key,
    str new_val)
```

```
@brief Replace YAML key-value lines.
```

```
@param text YAML text content
@param key YAML key to find and replace
@param new_val New value to set (will be quoted)
@return Modified YAML text
```

Definition at line 56 of file [run_LiFCal_calibration.py](#).

```
00056 def replace_yaml_key_line(text: str, key: str, new_val: str) -> str:
00057     """
00058     @brief Replace YAML key-value lines.
00059
00060     @param text YAML text content
00061     @param key YAML key to find and replace
00062     @param new_val New value to set (will be quoted)
00063     @return Modified YAML text
00064     """
00065     pattern = re.compile(rf"^{(re.escape(key))}\s*:\s*(.*)$", re.MULTILINE)
00066     return pattern.sub(rf'\1"{new_val}"', text)
00067
00068
```

4.12.1.7 run_lifcal_recalib()

```
int run_LiFCal_calibration.run_lifcal_recalib (
    str settings_path,
    str fixed_params_path,
    str store_dir)
```

```
@brief Run LiFCal recalibration and return exit code.
```

```
@param settings_path Path to Settings.yaml file
@param fixed_params_path Path to fixed parameters file
@param store_dir Directory where LiFCal will store results
@return Exit code from LiFCal process
```


Definition at line 86 of file [run_LiFCal_calibration.py](#).

```
00086 def run_lifcal_recalib(settings_path: str, fixed_params_path: str, store_dir: str) -> int:
00087     """
00088     @brief Run LiFCal recalibration and return exit code.
00089
00090     @param settings_path Path to Settings.yaml file
00091     @param fixed_params_path Path to fixed parameters file
00092     @param store_dir Directory where LiFCal will store results
00093     @return Exit code from LiFCal process
00094     """
00095     os.makedirs(store_dir, exist_ok=True)
00096
00097     cmd = [LIFCAL_BIN, "recalib", settings_path, fixed_params_path]
00098
00099     proc = subprocess.Popen(
00100         cmd,
00101         stdin=subprocess.PIPE,
00102         stdout=subprocess.PIPE,
00103         stderr=subprocess.STDOUT,
00104         text=True,
00105         cwd=os.path.dirname(LIFCAL_BIN),
00106     )
00107
00108     out, _ = proc.communicate(input=f"y\n{store_dir}\n")
00109     write_text(os.path.join(store_dir, "lifcal_stdout.log"), out)
00110
00111     return proc.returncode
00112
00113
```

4.12.1.8 write_text()

```
None run_LiFCal_calibration.write_text (
    str path,
    str s)
```

@brief Write text to file, creating directories if needed.

@param path Output file path
 @param s Text content to write

Definition at line 44 of file [run_LiFCal_calibration.py](#).

```
00044 def write_text(path: str, s: str) -> None:
00045     """
00046     @brief Write text to file, creating directories if needed.
00047
00048     @param path Output file path
00049     @param s Text content to write
00050     """
00051     os.makedirs(os.path.dirname(path), exist_ok=True)
00052     with open(path, "w", encoding="utf-8") as f:
00053         f.write(s)
00054
00055
```

4.12.2 Variable Documentation

4.12.2.1 CAMERA_IDS

```
list run_LiFCal_calibration.CAMERA_IDS
```

Initial value:

```
00001 = [
00002     "Center", "Up1", "Up2", "Up3",
00003     "Down1", "Down2", "Down3",
00004     "Left1", "Left2", "Left3",
00005     "Right1", "Right2", "Right3"
00006 ]
```

Definition at line 25 of file [run_LiFCal_calibration.py](#).

4.12.2.2 DEPTH_ROOT

```
run_LiFCal_calibration.DEPTH_ROOT = os.path.join(ROOT_IMAGESET, "depth")
```

Definition at line 21 of file [run_LiFCal_calibration.py](#).

4.12.2.3 FIXED_PARAMS

```
str run_LiFCal_calibration.FIXED_PARAMS = "/data/external/LiFCal/LiFCal_Data/Recalibration/Fixed↵↵_Parameters.txt"
```

Definition at line 16 of file [run_LiFCal_calibration.py](#).

4.12.2.4 FOCUS_ROOT

```
run_LiFCal_calibration.FOCUS_ROOT = os.path.join(ROOT_IMAGESET, "focus")
```

Definition at line 20 of file [run_LiFCal_calibration.py](#).

4.12.2.5 LIFCAL_BIN

```
str run_LiFCal_calibration.LIFCAL_BIN = "/data/external/LiFCal/build/bin/LiFCal"
```

Definition at line 13 of file [run_LiFCal_calibration.py](#).

4.12.2.6 MLA_CALIB_XML

```
str run_LiFCal_calibration.MLA_CALIB_XML = "/data/external/LiFCal/LiFCal_Data/Recalibration/MLA↵↵_Calibration.xml"
```

Definition at line 17 of file [run_LiFCal_calibration.py](#).

4.12.2.7 OUT_ROOT

```
str run_LiFCal_calibration.OUT_ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/Calibration↵↵ByCamera"
```

Definition at line 23 of file [run_LiFCal_calibration.py](#).

4.12.2.8 ROOT_IMAGESET

```
str run_LiFCal_calibration.ROOT_IMAGESET = "/data/external/LiFCal/LiFCal_Data/Recalibration/Li↵↵FCal_ImageSet"
```

Definition at line 19 of file [run_LiFCal_calibration.py](#).

4.12.2.9 SETTINGS

```
str run_LiFCal_calibration.SETTINGS = "/data/external/LiFCal/LiFCal_Data/Recalibration/Settings.↵
yaml"
```

Definition at line 15 of file [run_LiFCal_calibration.py](#).

4.13 run_periodic_lifcal Namespace Reference

Functions

- [setup_logger](#) ()
- bool [acquire_lock](#) (logger)
- [release_lock](#) (logger)
- int [run_cmd](#) (logger, cmd, cwd=None)
- int [count_images_in_dir](#) (str d, exts=(".png", ".jpg", ".jpeg"))
- bool [needs_imageset_creation](#) (logger)
- str [find_latest_run_dir](#) (str out_root)
- str [combined_json_path](#) (str run_dir)
- [try_log_lifcal_with_resultlogger](#) (logger, str combined_path)
- [run_cycle](#) (logger)
- [main](#) ()

Variables

- int [INTERVAL_SECONDS](#) = 5 * 60
- str [RUN_IMAGESET_CREATION](#) = "/data/external/LiFCal/LiFCal_Data/run_imageset_creation.py"
- str [RUN_LIFCAL_CALIB](#) = "/data/external/LiFCal/LiFCal_Data/run_LiFCal_calibration.py"
- str [LIFCAL_OUT_ROOT](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/CalibrationByCamera"
- str [BASELINE_DIR](#) = "/data/baseline"
- [LOG_PATH](#) = os.path.join([BASELINE_DIR](#), "calibration.log")
- str [LOCK_PATH](#) = "/tmp/lifcal_periodic.lock"
- str [ROOT_IMAGESET](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_ImageSet"
- [FOCUS_ROOT](#) = os.path.join([ROOT_IMAGESET](#), "focus")
- [DEPTH_ROOT](#) = os.path.join([ROOT_IMAGESET](#), "depth")
- list [CAMERA_IDS](#)
- int [MIN_POSES_PER_CAM](#) = 1

4.13.1 Function Documentation

4.13.1.1 [acquire_lock\(\)](#)

```
bool run_periodic_lifcal.acquire_lock (
    logger)
```

@brief Create lock file to prevent overlapping scheduler cycles.

@param logger Logger instance for status messages
 @return True if lock was acquired, False otherwise

Definition at line 63 of file [run_periodic_lifcal.py](#).

```
00063 def acquire_lock(logger) -> bool:
00064     """
00065     @brief Create lock file to prevent overlapping scheduler cycles.
00066
00067     @param logger Logger instance for status messages
00068     @return True if lock was acquired, False otherwise
00069     """
00070     if os.path.exists(LOCK_PATH):
00071         logger.warning(f"Lock exists, skipping this cycle: {LOCK_PATH}")
00072         return False
00073     try:
00074         with open(LOCK_PATH, "w") as f:
00075             f.write(str(os.getpid()))
00076             return True
00077     except Exception as e:
00078         logger.error(f"Could not create lock: {e}")
00079         return False
00080
00081
```

4.13.1.2 combined_json_path()

```
str run_periodic_lifcal.combined_json_path (
    str run_dir)

@brief Return path to combined.json inside run directory if present.

@param run_dir Path to run directory
@return Path to combined.json file, or empty string if not found
```

Definition at line 192 of file [run_periodic_lifcal.py](#).

```
00192 def combined_json_path(run_dir: str) -> str:
00193     """
00194     @brief Return path to combined.json inside run directory if present.
00195
00196     @param run_dir Path to run directory
00197     @return Path to combined.json file, or empty string if not found
00198     """
00199     if not run_dir:
00200         return ""
00201     p = os.path.join(run_dir, "combined.json")
00202     return p if os.path.isfile(p) else ""
00203
00204
```

4.13.1.3 count_images_in_dir()

```
int run_periodic_lifcal.count_images_in_dir (
    str d,
    exts = (".png", ".jpg", ".jpeg"))

@brief Count images in directory matching expected extensions.

@param d Directory path to scan
@param exts Tuple of file extensions to match (default: (".png", ".jpg", ".jpeg"))
@return Number of matching image files
```

Definition at line 124 of file [run_periodic_lifcal.py](#).

```
00124 def count_images_in_dir(d: str, exts=(".png", ".jpg", ".jpeg")) -> int:
00125     """
00126     @brief Count images in directory matching expected extensions.
00127
00128     @param d Directory path to scan
00129     @param exts Tuple of file extensions to match (default: (".png", ".jpg", ".jpeg"))
00130     @return Number of matching image files
00131     """
00132     if not os.path.isdir(d):
00133         return 0
00134     cnt = 0
00135     for fn in os.listdir(d):
00136         if fn.lower().endswith(exts):
00137             cnt += 1
00138     return cnt
00139
00140
```

4.13.1.4 find_latest_run_dir()

```
str run_periodic_lifcal.find_latest_run_dir (
    str out_root)
```

@brief Return latest run_* directory inside output root, or empty string if none.

@param out_root Root directory containing run_* folders
 @return Path to the latest run directory, or empty string

Definition at line 172 of file [run_periodic_lifcal.py](#).

```
00172 def find_latest_run_dir(out_root: str) -> str:
00173     """
00174     @brief Return latest run_* directory inside output root, or empty string if none.
00175
00176     @param out_root Root directory containing run_* folders
00177     @return Path to the latest run directory, or empty string
00178     """
00179     if not os.path.isdir(out_root):
00180         return ""
00181     runs = []
00182     for name in os.listdir(out_root):
00183         p = os.path.join(out_root, name)
00184         if os.path.isdir(p) and name.startswith("run_"):
00185             runs.append(p)
00186     if not runs:
00187         return ""
00188     runs.sort(key=lambda p: os.path.getmtime(p))
00189     return runs[-1]
00190
00191
```

4.13.1.5 main()

```
run_periodic_lifcal.main ()
```

@brief Periodic loop that runs LiFCal recalibration on a schedule.

Definition at line 277 of file [run_periodic_lifcal.py](#).

```
00277 def main():
00278     """
00279     @brief Periodic loop that runs LiFCal recalibration on a schedule.
00280     """
00281     logger = setup_logger()
00282     logger.info("Starting periodic LiFCal scheduler.")
00283     logger.info(f"Interval: {INTERVAL_SECONDS}s")
00284     logger.info(f"Lock: {LOCK_PATH}")
00285
00286     while True:
00287         start = time.time()
00288
00289         if acquire_lock(logger):
00290             try:
00291                 logger.info("=== CYCLE START ===")
00292                 run_cycle(logger)
00293                 logger.info("=== CYCLE END ===")
00294             finally:
00295                 release_lock(logger)
00296
00297         elapsed = time.time() - start
00298         sleep_s = max(1.0, INTERVAL_SECONDS - elapsed)
00299         logger.info(f"Sleeping {sleep_s:.1f}s\n")
00300         time.sleep(sleep_s)
00301
00302
```

4.13.1.6 needs_imageset_creation()

```
bool run_periodic_lifcal.needs_imageset_creation (
    logger)

@brief Check if imageset exists and has enough images per camera; trigger creation if not.

@param logger Logger instance for status messages
@return True if imageset needs to be created, False if it already exists
```

Definition at line 141 of file [run_periodic_lifcal.py](#).

```
00141 def needs_imageset_creation(logger) -> bool:
00142     """
00143     @brief Check if imageset exists and has enough images per camera; trigger creation if not.
00144
00145     @param logger Logger instance for status messages
00146     @return True if imageset needs to be created, False if it already exists
00147     """
00148     if not os.path.isdir(ROOT_IMAGESET):
00149         logger.info("Imageset missing: ROOT_IMAGESET not found.")
00150         return True
00151     if not os.path.isdir(FOCUS_ROOT) or not os.path.isdir(DEPTH_ROOT):
00152         logger.info("Imageset missing: focus/depth folders not found.")
00153         return True
00154
00155     for cam in CAMERA_IDS:
00156         fdir = os.path.join(FOCUS_ROOT, cam)
00157         ddir = os.path.join(DEPTH_ROOT, cam)
00158
00159         fcnt = count_images_in_dir(fdir, exts=(".png", ".jpg", ".jpeg"))
00160         dcnt = count_images_in_dir(ddir, exts=(".png", ".jpg", ".jpeg"))
00161
00162         if fcnt < MIN_POSES_PER_CAM:
00163             logger.info(f"Imageset not ready: focus/{cam} has {fcnt} images (<{MIN_POSES_PER_CAM}).")
00164             return True
00165         if dcnt < MIN_POSES_PER_CAM:
00166             logger.info(f"Imageset not ready: depth/{cam} has {dcnt} images (<{MIN_POSES_PER_CAM}).")
00167             return True
00168     logger.info("Imageset looks ready. Skipping run_imageset_creation.py.")
00169     return False
00170
00171
```

4.13.1.7 release_lock()

```
run_periodic_lifcal.release_lock (
    logger)

@brief Remove lock file after a cycle completes.

@param logger Logger instance for error messages
```

Definition at line 82 of file [run_periodic_lifcal.py](#).

```
00082 def release_lock(logger):
00083     """
00084     @brief Remove lock file after a cycle completes.
00085
00086     @param logger Logger instance for error messages
00087     """
00088     try:
00089         if os.path.exists(LOCK_PATH):
00090             os.remove(LOCK_PATH)
00091     except Exception as e:
00092         logger.error(f"Could not remove lock: {e}")
00093
00094
```

4.13.1.8 run_cmd()

```
int run_periodic_lifcal.run_cmd (
    logger,
    cmd,
    cwd = None)
```

@brief Run a subprocess, log combined stdout/stderr, and return exit code.

@param logger Logger instance for command output

@param cmd Command as list of strings

@param cwd Working directory for the command (default: None)

@return Exit code of the subprocess

Definition at line 95 of file [run_periodic_lifcal.py](#).

```
00095 def run_cmd(logger, cmd, cwd=None) -> int:
00096     """
00097     @brief Run a subprocess, log combined stdout/stderr, and return exit code.
00098
00099     @param logger Logger instance for command output
00100     @param cmd Command as list of strings
00101     @param cwd Working directory for the command (default: None)
00102     @return Exit code of the subprocess
00103     """
00104     logger.info(f"RUN: {' '.join(cmd)}")
00105     try:
00106         p = subprocess.run(
00107             cmd,
00108             cwd=cwd,
00109             stdout=subprocess.PIPE,
00110             stderr=subprocess.STDOUT,
00111             text=True
00112         )
00113         out = p.stdout or ""
00114         if len(out) > 5000:
00115             logger.info(out[:5000] + "\n... (truncated) ...")
00116         else:
00117             logger.info(out)
00118         return int(p.returncode)
00119     except Exception as e:
00120         logger.error(f"Command failed: {e}", exc_info=True)
00121         return 1
00122
00123
```

4.13.1.9 run_cycle()

```
run_periodic_lifcal.run_cycle (
    logger)
```

@brief Execute one scheduler cycle: prepare imageset, run LiFCal, log health.

@param logger Logger instance for cycle logging

Definition at line 234 of file [run_periodic_lifcal.py](#).

```
00234 def run_cycle(logger):
00235     """
00236     @brief Execute one scheduler cycle: prepare imageset, run LiFCal, log health.
00237
00238     @param logger Logger instance for cycle logging
00239     """
00240     # 1) Only prepare imageset if needed
00241     if needs_imageset_creation(logger):
00242         if not os.path.isfile(RUN_IMAGESET_CREATION):
00243             logger.error(f"Missing: {RUN_IMAGESET_CREATION}")
00244             return
00245
00246     rc = run_cmd(logger, [sys.executable, RUN_IMAGESET_CREATION])
00247     if rc != 0:
00248         logger.error(f"run_imageset_creation failed (rc={rc}). Skipping LiFCal.")
```

```

00249         return
00250
00251     # 2) LiFCal calibration
00252     if not os.path.isfile(RUN_LIFCAL_CALIB):
00253         logger.error(f"Missing: {RUN_LIFCAL_CALIB}")
00254         return
00255
00256     rc = run_cmd(logger, [sys.executable, RUN_LIFCAL_CALIB])
00257     if rc != 0:
00258         logger.error(f"run_LiFCal_calibration failed (rc={rc}).")
00259         return
00260
00261     # 3) locate combined.json (latest run)
00262     latest = find_latest_run_dir(LIFCAL_OUT_ROOT)
00263     cpath = combined_json_path(latest)
00264
00265     if not cpath:
00266         logger.warning("No combined.json found after LiFCal run.")
00267         return
00268
00269     logger.info(f"Latest combined.json: {cpath}")
00270
00271     # 4) log health + baseline update using your ResultLogger
00272     ok = try_log_lifcal_with_resultlogger(logger, cpath)
00273     if not ok:
00274         logger.info("LiFCal finished, but no baseline/health logging was executed.")
00275
00276

```

4.13.1.10 setup_logger()

```
run_periodic_lifcal.setup_logger ()
```

@brief Create logger that writes to file and stdout.

@return Logger instance configured for calibration logging

Definition at line 41 of file [run_periodic_lifcal.py](#).

```

00041 def setup_logger():
00042     """
00043     @brief Create logger that writes to file and stdout.
00044
00045     @return Logger instance configured for calibration logging
00046     """
00047     os.makedirs(BASELINE_DIR, exist_ok=True)
00048     logger = logging.getLogger("lifcal_periodic")
00049     logger.setLevel(logging.INFO)
00050
00051     if not logger.handlers:
00052         fh = logging.FileHandler(LOG_PATH, encoding="utf-8")
00053         fh.setFormatter(logging.Formatter("%(asctime)s [% (levelname)s] %(message)s"))
00054         logger.addHandler(fh)
00055
00056         sh = logging.StreamHandler(sys.stdout)
00057         sh.setFormatter(logging.Formatter("%(asctime)s [% (levelname)s] %(message)s"))
00058         logger.addHandler(sh)
00059
00060     return logger
00061
00062

```

4.13.1.11 try_log_lifcal_with_resultlogger()

```
run_periodic_lifcal.try_log_lifcal_with_resultlogger (
    logger,
    str combined_path)
```

@brief Use ResultLogger to record LiFCal health score and update baseline.

@param logger Logger instance for status messages
 @param combined_path Path to combined.json from LiFCal
 @return True if logging succeeded, False otherwise

Definition at line 205 of file [run_periodic_lifcal.py](#).

```
00205 def try_log_lifcal_with_resultlogger(logger, combined_path: str):
00206     """
00207     @brief Use ResultLogger to record LiFCal health score and update baseline.
00208
00209     @param logger Logger instance for status messages
00210     @param combined_path Path to combined.json from LiFCal
00211     @return True if logging succeeded, False otherwise
00212     """
00213     try:
00214         repo_dir = "/data/calibrationLFC"
00215         if repo_dir not in sys.path:
00216             sys.path.insert(0, repo_dir)
00217
00218         from ResultLogger import ResultLogger
00219         rl = ResultLogger(baseDir=BASELINE_DIR)
00220
00221         meta = {
00222             "runType": "recalib",
00223             "source": "periodic_scheduler",
00224             "timestamp": datetime.now().isoformat(timespec="seconds"),
00225         }
00226         score = rl.logRecalibration(combined_path, meta=meta)
00227         logger.info(f"LiFCal HealthScore logged via ResultLogger: {score:.2f}")
00228         return True
00229     except Exception as e:
00230         logger.warning(f"Could not use ResultLogger for LiFCal logging: {e}")
00231         return False
00232
00233
```

4.13.2 Variable Documentation

4.13.2.1 BASELINE_DIR

str run_periodic_lifcal.BASELINE_DIR = "/data/baseline"

Definition at line 20 of file [run_periodic_lifcal.py](#).

4.13.2.2 CAMERA_IDS

list run_periodic_lifcal.CAMERA_IDS

Initial value:

```
00001 = [
00002     "Center", "Up1", "Up2", "Up3",
00003     "Down1", "Down2", "Down3",
00004     "Left1", "Left2", "Left3",
00005     "Right1", "Right2", "Right3"
00006 ]
```

Definition at line 30 of file [run_periodic_lifcal.py](#).

4.13.2.3 DEPTH_ROOT

run_periodic_lifcal.DEPTH_ROOT = os.path.join(ROOT_IMAGESET, "depth")

Definition at line 28 of file [run_periodic_lifcal.py](#).

4.13.2.4 FOCUS_ROOT

run_periodic_lifcal.FOCUS_ROOT = os.path.join(ROOT_IMAGESET, "focus")

Definition at line 27 of file [run_periodic_lifcal.py](#).

4.13.2.5 INTERVAL_SECONDS

```
int run_periodic_lifcal.INTERVAL_SECONDS = 5 * 60
```

Definition at line 14 of file [run_periodic_lifcal.py](#).

4.13.2.6 LIFCAL_OUT_ROOT

```
str run_periodic_lifcal.LIFCAL_OUT_ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/Calibration↵  
ByCamera"
```

Definition at line 19 of file [run_periodic_lifcal.py](#).

4.13.2.7 LOCK_PATH

```
str run_periodic_lifcal.LOCK_PATH = "/tmp/lifcal_periodic.lock"
```

Definition at line 23 of file [run_periodic_lifcal.py](#).

4.13.2.8 LOG_PATH

```
run_periodic_lifcal.LOG_PATH = os.path.join(BASELINE_DIR, "calibration.log")
```

Definition at line 21 of file [run_periodic_lifcal.py](#).

4.13.2.9 MIN_POSES_PER_CAM

```
int run_periodic_lifcal.MIN_POSES_PER_CAM = 1
```

Definition at line 38 of file [run_periodic_lifcal.py](#).

4.13.2.10 ROOT_IMAGESET

```
str run_periodic_lifcal.ROOT_IMAGESET = "/data/external/LiFCal/LiFCal_Data/Recalibration/Li↵  
FCal_ImageSet"
```

Definition at line 26 of file [run_periodic_lifcal.py](#).

4.13.2.11 RUN_IMAGESET_CREATION

```
str run_periodic_lifcal.RUN_IMAGESET_CREATION = "/data/external/LiFCal/LiFCal_Data/run_↵  
imageset_creation.py"
```

Definition at line 16 of file [run_periodic_lifcal.py](#).

4.13.2.12 RUN_LIFCAL_CALIB

```
str run_periodic_lifcal.RUN_LIFCAL_CALIB = "/data/external/LiFCal/LiFCal_Data/run_LiFCal_↵  
calibration.py"
```

Definition at line 17 of file [run_periodic_lifcal.py](#).

Chapter 5

Class Documentation

5.1 CalibrationState.CalibrationState Class Reference

Public Member Functions

- [__init__](#) (self)
- [setIntrinsics](#) (self, camId, cameraMatrix, distortionCoeffs, repError)
- [setExtrinsics](#) (self, camId, rotationMatrix, translationVector, stereoRMS)
- [setTimeStamp](#) (self, timestamp)
- [__getState__](#) (self)

Public Attributes

- [timeStamp](#) = timestamp
- [intrinsics](#)
- [extrinsics](#)

5.1.1 Detailed Description

@brief Stores the complete calibration state for a multi-camera system.

@details This class manages intrinsic and extrinsic parameters for all cameras in the system. It provides methods to set and retrieve calibration data including camera matrices, distortion coefficients, rotation matrices, and translation vectors.

Definition at line 1 of file [CalibrationState.py](#).

5.1.2 Constructor & Destructor Documentation

5.1.2.1 `__init__()`

```
CalibrationState.CalibrationState.__init__ (
    self)
```

@brief Constructor for CalibrationState.

@details Initializes empty dictionaries for intrinsic and extrinsic parameters and sets the timestamp to None.

Definition at line 10 of file [CalibrationState.py](#).

```
00010     def __init__(self):
00011         """
00012         @brief Constructor for CalibrationState.
00013
00014         @details Initializes empty dictionaries for intrinsic and extrinsic parameters
00015         and sets the timestamp to None.
00016         """
00017         self.intrinsics = {}
00018         self.extrinsics = {}
00019         self.timeStamp = None
00020
```

5.1.3 Member Function Documentation

5.1.3.1 `__getState__()`

```
CalibrationState.CalibrationState.__getState__ (
    self)
```

@brief Return the complete calibration state as dictionary.

@return Dictionary containing intrinsics, extrinsics, and timestamp

Definition at line 59 of file [CalibrationState.py](#).

```
00059     def __getState__(self):
00060         """
00061         @brief Return the complete calibration state as dictionary.
00062
00063         @return Dictionary containing intrinsics, extrinsics, and timestamp
00064         """
00065         return {
00066             "intrinsics": self.intrinsics,
00067             "extrinsics": self.extrinsics,
00068             "timeStamp": self.timeStamp
00069         }
```

5.1.3.2 `setExtrinsics()`

```
CalibrationState.CalibrationState.setExtrinsics (
    self,
    camId,
    rotationMatrix,
    translationVector,
    stereoRMS)
```

@brief Store extrinsic calibration parameters for a camera.

@param camId Camera identifier
 @param rotationMatrix 3x3 rotation matrix representing camera orientation
 @param translationVector 3x1 translation vector representing camera position
 @param stereoRMS Root mean square error of the stereo calibration

Definition at line 36 of file [CalibrationState.py](#).

```
00036     def setExtrinsics(self, camId, rotationMatrix, translationVector, stereoRMS):
00037         """
00038         @brief Store extrinsic calibration parameters for a camera.
00039
00040         @param camId Camera identifier
00041         @param rotationMatrix 3x3 rotation matrix representing camera orientation
00042         @param translationVector 3x1 translation vector representing camera position
00043         @param stereoRMS Root mean square error of the stereo calibration
00044         """
00045         self.extrinsics[camId] = {
00046             "rotationMatrix": rotationMatrix,
00047             "translationVector": translationVector,
00048             "stereoRms": stereoRMS
00049         }
00050
```

5.1.3.3 setIntrinsics()

```
CalibrationState.CalibrationState.setIntrinsics (
    self,
    camId,
    cameraMatrix,
    distortionCoeffs,
    repError)
```

@brief Store intrinsic calibration parameters for a camera.

@param camId Camera identifier
 @param cameraMatrix 3x3 camera matrix containing focal lengths and principal point
 @param distortionCoeffs Distortion coefficients vector
 @param repError Reprojection error of the calibration

Definition at line 21 of file [CalibrationState.py](#).

```
00021     def setIntrinsics(self, camId, cameraMatrix, distortionCoeffs, repError):
00022         """
00023         @brief Store intrinsic calibration parameters for a camera.
00024
00025         @param camId Camera identifier
00026         @param cameraMatrix 3x3 camera matrix containing focal lengths and principal point
00027         @param distortionCoeffs Distortion coefficients vector
00028         @param repError Reprojection error of the calibration
00029         """
00030         self.intrinsics[camId] = {
00031             "cameraMatrix": cameraMatrix,
00032             "distortionCoeffs": distortionCoeffs,
00033             "reprojectionError": repError
00034         }
00035
```

5.1.3.4 setTimeStamp()

```
CalibrationState.CalibrationState.setTimeStamp (
    self,
    timestamp)
```

@brief Set the calibration timestamp.

@param timestamp Timestamp indicating when the calibration was performed

Definition at line 51 of file [CalibrationState.py](#).

```
00051     def setTimeStamp(self, timestamp):
00052         """
00053         @brief Set the calibration timestamp.
00054
00055         @param timestamp Timestamp indicating when the calibration was performed
00056         """
00057         self.timeStamp = timestamp
00058
```

5.1.4 Member Data Documentation

5.1.4.1 extrinsics

`CalibrationState.CalibrationState.extrinsics`

Definition at line 67 of file [CalibrationState.py](#).

5.1.4.2 intrinsics

`CalibrationState.CalibrationState.intrinsics`

Definition at line 66 of file [CalibrationState.py](#).

5.1.4.3 timeStamp

`CalibrationState.CalibrationState.timeStamp = timestamp`

Definition at line 57 of file [CalibrationState.py](#).

The documentation for this class was generated from the following file:

- [calibrationLFC/CalibrationState.py](#)

5.2 Controller.Controller Class Reference

Public Member Functions

- [__init__](#) (self, [bundleAdjust](#))
- [runInitialCalibration](#) (self, [imageSet](#))
- float [selfHealthCheck](#) (self, [calibrationState](#))

Public Attributes

- [initialCalibration](#) = [InitialCalibration](#)()
- list [scoreHistory](#) = []
- [currentState](#) = None
- [resultLogger](#) = [ResultLogger](#)()
- [bundleAdjust](#) = [bundleAdjust](#)

5.2.1 Detailed Description

@brief Main controller for managing calibration workflow.

@details Orchestrates the complete calibration process including initial calibration, health monitoring, and result logging. Manages the calibration state and maintains a history of health scores.

Definition at line 6 of file [Controller.py](#).

5.2.2 Constructor & Destructor Documentation

5.2.2.1 __init__()

```
Controller.Controller.__init__ (
    self,
    bundleAdjust)
```

@brief Constructor for Controller.

@param bundleAdjust Boolean flag indicating whether to apply bundle adjustment

@details Initializes the controller with an InitialCalibration instance, empty score history, result logger, and bundle adjustment setting.

Definition at line 15 of file [Controller.py](#).

```
00015     def __init__(self, bundleAdjust):
00016         """
00017         @brief Constructor for Controller.
00018
00019         @param bundleAdjust Boolean flag indicating whether to apply bundle adjustment
00020
00021         @details Initializes the controller with an InitialCalibration instance,
00022         empty score history, result logger, and bundle adjustment setting.
00023         """
00024         self.initialCalibration = InitialCalibration()
00025         self.scoreHistory = []
00026         self.currentState = None
00027         self.resultLogger = ResultLogger()
00028         self.bundleAdjust = bundleAdjust
00029
```

5.2.3 Member Function Documentation

5.2.3.1 runInitialCalibration()

```
Controller.Controller.runInitialCalibration (
    self,
    imageSet)
```

@brief Execute initial calibration for all cameras in the image set.

@param imageSet ImageSet object containing calibration images and camera information

@return CalibrationState object with computed intrinsic and extrinsic parameters

Definition at line 30 of file [Controller.py](#).

```
00030     def runInitialCalibration(self, imageSet):
00031         """
00032         @brief Execute initial calibration for all cameras in the image set.
00033
00034         @param imageSet ImageSet object containing calibration images and camera information
00035         @return CalibrationState object with computed intrinsic and extrinsic parameters
00036
00037         """
00038         print("Controller: starting initial calibration...")
00039         calibrationState = self.initialCalibration.run(imageSet, self.bundleAdjust)
00040
00041         # Prepare metadata for logging
00042         meta = {
00043             "runType": "initial",
00044             "numPoses": imageSet.numPoses,
00045             "imageDir": imageSet.baseDir,
00046             "cameraIds": imageSet.cameraIds,
00047             "bundleAdjust": self.bundleAdjust,
00048         }
00049         self.resultLogger.logInitialCalibration(calibrationState, meta=meta)
00050
00051         return calibrationState
00052
```

5.2.3.2 selfHealthCheck()

```
float Controller.Controller.selfHealthCheck (
    self,
    calibrationState)
```

@brief Computes a global health score for the calibration quality.

@param calibrationState CalibrationState object to evaluate

@return Health score as float between 0 and 100, or 0.0 if evaluation fails

Definition at line 53 of file [Controller.py](#).

```
00053     def selfHealthCheck(self, calibrationState) -> float:
00054         """
00055         @brief Computes a global health score for the calibration quality.
00056
00057         @param calibrationState CalibrationState object to evaluate
00058         @return Health score as float between 0 and 100, or 0.0 if evaluation fails
00059
00060         """
00061         # Extract state dictionary
00062         try:
00063             state_dict = calibrationState.__getState__()
00064         except Exception as e:
00065             self.resultLogger.logger.error(
00066                 f"HealthCheck: could not extract state dict: {e}", exc_info=True
00067             )
00068             return 0.0
00069
00070         # Compute health score from calibration metrics
00071         try:
00072             report = compute_initial_health_from_state(state_dict)
00073             score = float(report["health"])
00074             self.resultLogger.logger.info(
00075                 f"HealthScore = {score:.2f} (method={report.get('method')})"
00076             )
00077             return score
00078         except Exception as e:
00079             self.resultLogger.logger.error(f"HealthCheck failed: {e}", exc_info=True)
00080             return 0.0
```

5.2.4 Member Data Documentation

5.2.4.1 bundleAdjust

```
Controller.Controller.bundleAdjust = bundleAdjust
```

Definition at line 28 of file [Controller.py](#).

5.2.4.2 currentState

```
Controller.Controller.currentState = None
```

Definition at line 26 of file [Controller.py](#).

5.2.4.3 initialCalibration

```
Controller.Controller.initialCalibration = InitialCalibration()
```

Definition at line 24 of file [Controller.py](#).

5.2.4.4 resultLogger

```
Controller.Controller.resultLogger = ResultLogger()
```

Definition at line 27 of file [Controller.py](#).

5.2.4.5 scoreHistory

```
list Controller.Controller.scoreHistory = []
```

Definition at line 25 of file [Controller.py](#).

The documentation for this class was generated from the following file:

- calibrationLFC/[Controller.py](#)

5.3 ImageSet.ImageSet Class Reference

Public Member Functions

- [__init__](#) (self, str [baseDir](#), int [numPoses](#), List[str] [cameralds](#))
- str [getImagePath](#) (self, int [poseIndex](#), str [camerald](#))

Public Attributes

- [baseDir](#) = baseDir
- [numPoses](#) = numPoses
- [cameralds](#) = cameralds

5.3.1 Detailed Description

@brief Represents a set of calibration images organized by pose and camera.

Definition at line 3 of file [ImageSet.py](#).

5.3.2 Constructor & Destructor Documentation

5.3.2.1 `__init__()`

```
ImageSet.ImageSet.__init__ (
    self,
    str baseDir,
    int numPoses,
    List[str] cameraIds)
```

@brief Constructor for ImageSet.

@param baseDir Base directory containing calibration images
 @param numPoses Total number of poses in the calibration set
 @param cameraIds List of camera identifiers

Definition at line 8 of file [ImageSet.py](#).

```
00008     def __init__(self, baseDir: str, numPoses: int, cameraIds: List[str]):
00009         """
00010         @brief Constructor for ImageSet.
00011
00012         @param baseDir Base directory containing calibration images
00013         @param numPoses Total number of poses in the calibration set
00014         @param cameraIds List of camera identifiers
00015         """
00016         self.baseDir = baseDir
00017         self.numPoses = numPoses
00018         self.cameraIds = cameraIds
00019
```

5.3.3 Member Function Documentation

5.3.3.1 `getImagePath()`

```
str ImageSet.ImageSet.getImagePath (
    self,
    int poseIndex,
    str cameraId)
```

@brief Construct the file path for a specific pose and camera.

@param poseIndex Zero-based pose index
 @param cameraId Camera identifier string
 @return String path in format: baseDir/pose_XXX_cameraId.png

Definition at line 20 of file [ImageSet.py](#).

```
00020     def getImagePath(self, poseIndex: int, cameraId: str) -> str:
00021         """
00022         @brief Construct the file path for a specific pose and camera.
00023
00024         @param poseIndex Zero-based pose index
00025         @param cameraId Camera identifier string
00026         @return String path in format: baseDir/pose_XXX_cameraId.png
00027         """
00028         poseStr = f"pose_{poseIndex:03d}"
00029         return f"{self.baseDir}/{poseStr}_{cameraId}.png"
00030
00031         """
00032         Beispiel:
00033         baseDir = "/data/images"
00034         poseIndex = 5 → poseStr = "pose_005"
00035         cameraId = "center"
00036
00037         Ergebnis: "/data/images/pose_005_center.png"
00038         """
```

5.3.4 Member Data Documentation

5.3.4.1 baseDir

```
ImageSet.ImageSet.baseDir = baseDir
```

Definition at line 16 of file [ImageSet.py](#).

5.3.4.2 cameraIds

```
ImageSet.ImageSet.cameraIds = cameraIds
```

Definition at line 18 of file [ImageSet.py](#).

5.3.4.3 numPoses

```
ImageSet.ImageSet.numPoses = numPoses
```

Definition at line 17 of file [ImageSet.py](#).

The documentation for this class was generated from the following file:

- [calibrationLFC/ImageSet.py](#)

5.4 InitialCalibration.InitialCalibration Class Reference

Public Member Functions

- [__init__](#) (self, [chessboardSize](#)=(8, 8), [squareSize](#)=2/9)
- [createObjectPoints](#) (self)
- [detectCorners](#) (self, imagePath)
- [calibrateIntrinsics](#) (self, imageSet)
- [calibrateExtrinsics](#) (self, imageSet, state, refCamId)
- [bundleAdjustExtrinsics](#) (self, imageSet, state, refCamId)
- [run](#) (self, imageSet, bundleAdjust=True)

Public Attributes

- [chessboardSize](#) = chessboardSize
- [squareSize](#) = squareSize
- tuple [criteria](#) = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001)

5.4.1 Detailed Description

@brief Multi-camera calibration using chessboard patterns and bundle adjustment.

Definition at line 7 of file [InitialCalibration.py](#).

5.4.2 Constructor & Destructor Documentation

5.4.2.1 `__init__()`

```
InitialCalibration.InitialCalibration.__init__ (
    self,
    chessboardSize = (8,8),
    squareSize = 2/9)

@brief Constructor for InitialCalibration.

@param chessboardSize Tuple (cols, rows) defining chessboard dimensions (default: (8,8))
@param squareSize Physical size of one chessboard square in meters (default: 2/9)
```

Definition at line 12 of file [InitialCalibration.py](#).

```
00012     def __init__(self, chessboardSize=(8,8), squareSize=2/9):
00013         """
00014         @brief Constructor for InitialCalibration.
00015
00016         @param chessboardSize Tuple (cols, rows) defining chessboard dimensions (default: (8,8))
00017         @param squareSize Physical size of one chessboard square in meters (default: 2/9)
00018         """
00019         self.chessboardSize = chessboardSize
00020         self.squareSize = squareSize
00021         self.criteria = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001)
00022
```

5.4.3 Member Function Documentation

5.4.3.1 `bundleAdjustExtrinsics()`

```
InitialCalibration.InitialCalibration.bundleAdjustExtrinsics (
    self,
    imageSet,
    state,
    refCamId)

@brief Global bundle adjustment optimizing extrinsics and board poses.

@param imageSet ImageSet object containing calibration images
@param state CalibrationState with initial intrinsics and extrinsics
@param refCamId Camera ID to use as reference coordinate system
@return Updated CalibrationState with refined extrinsics after bundle adjustment
```

Definition at line 155 of file [InitialCalibration.py](#).

```
00155     def bundleAdjustExtrinsics(self, imageSet, state, refCamId):
00156         """
00157         @brief Global bundle adjustment optimizing extrinsics and board poses.
00158
00159         @param imageSet ImageSet object containing calibration images
00160         @param state CalibrationState with initial intrinsics and extrinsics
00161         @param refCamId Camera ID to use as reference coordinate system
00162         @return Updated CalibrationState with refined extrinsics after bundle adjustment
00163         """
00164         objp = self.createObjectPoints()
00165         cameraIds = list(imageSet.cameraIds)
00166         if refCamId not in cameraIds:
00167             raise ValueError("refCamId not in imageSet.cameraIds")
00168
00169         ref_idx = cameraIds.index(refCamId)
00170
00171         # Initialize board poses via PnP-RANSAC in reference camera
00172         K_ref = state.intrinsics[refCamId]["cameraMatrix"]
00173         dist_ref = state.intrinsics[refCamId]["distortionCoeffs"]
```

```

00174
00175     pose_rvecs = {}
00176     pose_tvecs = {}
00177     valid_poses = []
00178
00179     for pose in range(imageSet.numPoses):
00180         img_path = imageSet.getImagePath(pose, refCamId)
00181         ok, corners, size = self.detectCorners(img_path)
00182         if not ok:
00183             continue
00184
00185         # Use PnP-RANSAC for robust board pose estimation
00186         ok_pnp, rvec, tvec, inliers = cv2.solvePnP(
00187             objp,
00188             corners,
00189             K_ref,
00190             dist_ref,
00191             reprojectionError=2.0,
00192             confidence=0.99,
00193             flags=cv2.SOLVEPNP_ITERATIVE
00194         )
00195
00196         if not ok_pnp or inliers is None or len(inliers) < 10:
00197             # Skip pose if too few reliable points
00198             continue
00199
00200         pose_rvecs[pose] = rvec.astype(np.float64).reshape(3)
00201         pose_tvecs[pose] = tvec.astype(np.float64).reshape(3)
00202         valid_poses.append(pose)
00203
00204         if len(valid_poses) < 3:
00205             print("[BA] Not enough valid poses for bundle adjustment.")
00206             return state
00207
00208         # Initialize extrinsics from state
00209         # Only optimize non-reference cameras
00210         optim_cams = []
00211         cam_param_index = {}
00212
00213         for ci, cam in enumerate(cameraIds):
00214             if cam == refCamId:
00215                 continue
00216             if cam not in state.extrinsics:
00217                 continue
00218
00219             optim_cams.append(ci)
00220             cam_param_index[ci] = len(cam_param_index)
00221
00222         num_optim_cams = len(optim_cams)
00223         if num_optim_cams == 0:
00224             print("[BA] No extrinsic data to optimize, skipping BA.")
00225             return state
00226
00227         # Collect measurements (all cameras, all valid poses)
00228         # Each measurement: (cam_idx, pose, 2D points)
00229         measurements = []
00230
00231         for ci, cam in enumerate(cameraIds):
00232             for pose in valid_poses:
00233                 img_path = imageSet.getImagePath(pose, cam)
00234                 ok, corners, size = self.detectCorners(img_path)
00235                 if not ok:
00236                     continue
00237                 pts2d = corners.reshape(-1, 2).astype(np.float64)
00238                 measurements.append((ci, pose, pts2d))
00239
00240         if len(measurements) == 0:
00241             print("[BA] No overlapping detections found, skipping BA.")
00242             return state
00243
00244         # Initialize parameter vector
00245         # Layout: [cams (without ref): (r_cam(3), t_cam(3))... , poses: (r_p(3), t_p(3))...]
00246
00247         param_cam = []
00248         for ci in optim_cams:
00249             cam = cameraIds[ci]
00250             extr = state.extrinsics[cam]
00251             R = extr["rotationMatrix"]
00252             T = extr["translationVector"].reshape(3)
00253             rvec_cam, _ = cv2.Rodrigues(R)
00254             param_cam.append(rvec_cam.reshape(3))
00255             param_cam.append(T)
00256         param_cam = np.concatenate(param_cam).astype(np.float64)
00257
00258         param_pose = []
00259         for pose in valid_poses:
00260             param_pose.append(pose_rvecs[pose])

```

```

00261         param_pose.append(pose_tvecs[pose])
00262     param_pose = np.concatenate(param_pose).astype(np.float64)
00263
00264     x0 = np.concatenate([param_cam, param_pose])
00265
00266     # Helper functions for parameter decoding
00267     def decode_params(params):
00268         # Cameras
00269         cam_rvecs = {}
00270         cam_tvecs = {}
00271         idx = 0
00272         for ci in optim_cams:
00273             r = params[idx:idx+3]; idx += 3
00274             t = params[idx:idx+3]; idx += 3
00275             cam_rvecs[ci] = r
00276             cam_tvecs[ci] = t
00277
00278         # Poses
00279         pose_r = {}
00280         pose_t = {}
00281         for pose in valid_poses:
00282             r = params[idx:idx+3]; idx += 3
00283             t = params[idx:idx+3]; idx += 3
00284             pose_r[pose] = r
00285             pose_t[pose] = t
00286
00287         return cam_rvecs, cam_tvecs, pose_r, pose_t
00288
00289     # Residuals function
00290     def residuals(params):
00291         cam_rvecs, cam_tvecs, pose_r, pose_t = decode_params(params)
00292
00293         # Precompute rotation matrices
00294         R_cam = {}
00295         for ci, r in cam_rvecs.items():
00296             R_cam[ci], _ = cv2.Rodrigues(r.astype(np.float64))
00297
00298         R_pose = {}
00299         for pose, r in pose_r.items():
00300             R_pose[pose], _ = cv2.Rodrigues(r.astype(np.float64))
00301
00302         residual_list = []
00303
00304         for ci, pose, pts2d in measurements:
00305             cam_id = cameraIds[ci]
00306             K = state.intrinsics[cam_id]["cameraMatrix"]
00307             dist = state.intrinsics[cam_id]["distortionCoeffs"]
00308
00309             # Board pose in reference frame
00310             r_p = pose_r[pose]
00311             t_p = pose_t[pose]
00312             R_p = R_pose[pose]
00313
00314             if ci == ref_idx:
00315                 # Reference camera: use board pose directly
00316                 r_total = r_p
00317                 t_total = t_p
00318             else:
00319                 if ci not in cam_rvecs:
00320                     # For non-optimized cameras, use static extrinsics
00321                     extr = state.extrinsics[cam_id]
00322                     R_c = extr["rotationMatrix"]
00323                     t_c = extr["translationVector"].reshape(3)
00324                 else:
00325                     R_c = R_cam[ci]
00326                     t_c = cam_tvecs[ci]
00327
00328                 # Composition:  $X_{cam} = R_c * (R_p * X + t_p) + t_c$ 
00329                 R_total = R_c @ R_p
00330                 t_total = R_c @ t_p + t_c
00331                 r_total, _ = cv2.Rodrigues(R_total.astype(np.float64))
00332                 r_total = r_total.reshape(3)
00333
00334             # Projection
00335             proj, _ = cv2.projectPoints(
00336                 objp,
00337                 r_total.astype(np.float64),
00338                 t_total.astype(np.float64),
00339                 K,
00340                 dist
00341             )
00342             proj = proj.reshape(-1, 2)
00343
00344             res = (proj - pts2d).reshape(-1)
00345             residual_list.append(res)
00346
00347     if not residual_list:

```

```

00348         return np.zeros(0, dtype=np.float64)
00349
00350     return np.concatenate(residual_list)
00351
00352     # Optimization (robust, soft_l1)
00353     result = least_squares(
00354         residuals,
00355         x0,
00356         verbose=2,
00357         method="trf",
00358         loss="soft_l1",
00359         f_scale=1.0
00360     )
00361
00362     print(
00363         f"[BA] Done. Final cost: {result.cost:.4f}, "
00364         f"RMS per coord: {np.sqrt(2*result.cost/len(result.fun)):.4f} px"
00365     )
00366
00367     # Write results back to state.extrinsics
00368     cam_rvecs_opt, cam_tvecs_opt, pose_r_opt, pose_t_opt = decode_params(result.x)
00369
00370     # Reference camera remains identity
00371     state.setExtrinsics(
00372         refCamId,
00373         np.eye(3, dtype=np.float64),
00374         np.zeros((3, 1), dtype=np.float64),
00375         0.0
00376     )
00377
00378     # Store optimized cameras
00379     for ci in optim_cams:
00380         cam_id = cameraIds[ci]
00381         r = cam_rvecs_opt[ci]
00382         t = cam_tvecs_opt[ci]
00383         R, _ = cv2.Rodrigues(r.astype(np.float64))
00384         T = t.reshape(3, 1).astype(np.float64)
00385
00386         # Approximate RMS for this camera from global residuals
00387         # (simple average across all measurements)
00388         cam_residuals = []
00389         for m_ci, pose, pts2d in measurements:
00390             if m_ci != ci:
00391                 continue
00392             # Reproject with final parameters
00393             R_p, _ = cv2.Rodrigues(pose_r_opt[pose].astype(np.float64))
00394             t_p = pose_t_opt[pose].astype(np.float64)
00395
00396             R_total = R @ R_p
00397             t_total = R @ t_p + t.flatten()
00398             r_total, _ = cv2.Rodrigues(R_total)
00399             K = state.intrinsics[cam_id]["cameraMatrix"]
00400             dist = state.intrinsics[cam_id]["distortionCoeffs"]
00401
00402             proj, _ = cv2.projectPoints(
00403                 objp, r_total, t_total, K, dist
00404             )
00405             proj = proj.reshape(-1, 2)
00406             res = (proj - pts2d).reshape(-1)
00407             cam_residuals.append(res)
00408
00409         if cam_residuals:
00410             cam_residuals = np.concatenate(cam_residuals)
00411             stereo_rms = float(np.sqrt(np.mean(cam_residuals**2)))
00412         else:
00413             stereo_rms = state.extrinsics[cam_id].get("stereoRms", 0.0)
00414
00415         state.setExtrinsics(cam_id, R, T, stereo_rms)
00416
00417     return state
00418

```

5.4.3.2 calibrateExtrinsics()

```

InitialCalibration.InitialCalibration.calibrateExtrinsics (
    self,
    imageSet,
    state,
    refCamId)

```



```

@brief Compute pairwise stereo calibration between reference and each other camera.

@param imageSet ImageSet object containing calibration images
@param state CalibrationState with intrinsics already computed
@param refCamId Camera ID to use as reference coordinate system
@return Updated CalibrationState with extrinsics for all cameras

```

Definition at line 91 of file [InitialCalibration.py](#).

```

00091     def calibrateExtrinsics(self, imageSet, state, refCamId):
00092         """
00093         @brief Compute pairwise stereo calibration between reference and each other camera.
00094
00095         @param imageSet ImageSet object containing calibration images
00096         @param state CalibrationState with intrinsics already computed
00097         @param refCamId Camera ID to use as reference coordinate system
00098         @return Updated CalibrationState with extrinsics for all cameras
00099         """
00100         objectPoints = self.createObjectPoints()
00101         K_ref = state.intrinsics[refCamId]["cameraMatrix"]
00102         dist_ref = state.intrinsics[refCamId]["distortionCoeffs"]
00103
00104         for cam in imageSet.cameraIds:
00105             if cam == refCamId:
00106                 state.setExtrinsics(cam, np.eye(3), np.zeros((3,1)), 0.0)
00107                 continue
00108
00109             if cam not in state.intrinsics:
00110                 continue
00111
00112             K_cam = state.intrinsics[cam]["cameraMatrix"]
00113             dist_cam = state.intrinsics[cam]["distortionCoeffs"]
00114
00115             objList = []
00116             imgRef = []
00117             imgCam = []
00118             imageSize = None
00119
00120             for pose in range(imageSet.numPoses):
00121                 pRef = imageSet.getImagePath(pose, refCamId)
00122                 pCam = imageSet.getImagePath(pose, cam)
00123
00124                 okR, cornersR, sizeR = self.detectCorners(pRef)
00125                 okC, cornersC, sizeC = self.detectCorners(pCam)
00126
00127                 if okR and okC:
00128                     objList.append(objectPoints)
00129                     imgRef.append(cornersR)
00130                     imgCam.append(cornersC)
00131                     imageSize = sizeR
00132
00133             if len(objList) < 3:
00134                 continue
00135
00136             flags = cv2.CALIB_FIX_INTRINSIC
00137
00138             stereo_rms, KR0, d0, KC1, d1, R, T, E, F = cv2.stereoCalibrate(
00139                 objList,
00140                 imgRef,
00141                 imgCam,
00142                 K_ref,
00143                 dist_ref,
00144                 K_cam,
00145                 dist_cam,
00146                 imageSize,
00147                 criteria=self.criteria,
00148                 flags=flags
00149             )
00150
00151             state.setExtrinsics(cam, R, T, stereo_rms)
00152
00153         return state
00154

```

5.4.3.3 calibrateIntrinsics()

```

InitialCalibration.InitialCalibration.calibrateIntrinsics (
    self,
    imageSet)

```

```
@brief Calibrate intrinsic parameters independently for each camera.

@param imageSet ImageSet object containing calibration images
@return CalibrationState with populated intrinsics for all cameras
```

Definition at line 54 of file [InitialCalibration.py](#).

```
00054     def calibrateIntrinsics(self, imageSet):
00055         """
00056         @brief Calibrate intrinsic parameters independently for each camera.
00057
00058         @param imageSet ImageSet object containing calibration images
00059         @return CalibrationState with populated intrinsics for all cameras
00060         """
00061         state = CalibrationState()
00062         objp = self.createObjectPoints()
00063
00064         objpoints = {cam: [] for cam in imageSet.cameraIds}
00065         imgpoints = {cam: [] for cam in imageSet.cameraIds}
00066         imgsize = {}
00067
00068         # Collect corner points from all poses for each camera
00069         for pose in range(imageSet.numPoses):
00070             for cam in imageSet.cameraIds:
00071                 path = imageSet.getImagePath(pose, cam)
00072                 ok, corners, size = self.detectCorners(path)
00073                 if ok:
00074                     objpoints[cam].append(objp)
00075                     imgpoints[cam].append(corners)
00076                     imgsize[cam] = size
00077
00078         # Calibrate each camera separately
00079         for cam in imageSet.cameraIds:
00080             if len(objpoints[cam]) < 5:
00081                 print(f"[WARN] Not enough samples for camera {cam}")
00082                 continue
00083
00084             err, K, dist, rvecs, tvecs = cv2.calibrateCamera(
00085                 objpoints[cam], imgpoints[cam], imgsize[cam], None, None)
00086
00087             state.setIntrinsics(cam, K, dist, err)
00088
00089         return state
00090
```

5.4.3.4 createObjectPoints()

```
InitialCalibration.InitialCalibration.createObjectPoints (
    self)
```

```
@brief Generate 3D coordinates of chessboard corners.

@return Numpy array of shape (N, 3) with 3D corner coordinates
```

Definition at line 23 of file [InitialCalibration.py](#).

```
00023     def createObjectPoints(self):
00024         """
00025         @brief Generate 3D coordinates of chessboard corners.
00026
00027         @return Numpy array of shape (N, 3) with 3D corner coordinates
00028         """
00029         cols, rows = self.chessboardSize
00030         objp = np.zeros((cols * rows, 3), np.float32)
00031         objp[:, :2] = np.mgrid[0:cols, 0:rows].T.reshape(-1, 2)
00032         objp *= self.squareSize
00033         return objp
00034
```

5.4.3.5 detectCorners()

```
InitialCalibration.InitialCalibration.detectCorners (
    self,
    imagePath)
```

@brief Detect and refine chessboard corner positions in image.

@param imagePath Path to the image file

@return Tuple (success, corners, imageSize) where success is bool, corners are refined 2D points, and imageSize

Definition at line 35 of file [InitialCalibration.py](#).

```
00035     def detectCorners(self, imagePath):
00036         """
00037         @brief Detect and refine chessboard corner positions in image.
00038
00039         @param imagePath Path to the image file
00040         @return Tuple (success, corners, imageSize) where success is bool, corners are refined 2D
00041         points, and imageSize is (width, height)
00042         """
00042         img = cv2.imread(imagePath)
00043         if img is None:
00044             return False, None, None
00045         gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
00046
00047         found, corners = cv2.findChessboardCorners(gray, self.chessboardSize)
00048         if not found:
00049             return False, None, None
00050
00051         refined = cv2.cornerSubPix(gray, corners, (11, 11), (-1, -1), self.criteria)
00052         return True, refined, gray.shape[::-1]
00053
```

5.4.3.6 run()

```
InitialCalibration.InitialCalibration.run (
    self,
    imageSet,
    bundleAdjust = True)
```

@brief Main entry point for calibration pipeline.

@param imageSet ImageSet object containing calibration images

@param bundleAdjust Whether to apply bundle adjustment (default: True)

@return CalibrationState with complete intrinsic and extrinsic calibration

Definition at line 419 of file [InitialCalibration.py](#).

```
00419     def run(self, imageSet, bundleAdjust=True):
00420         """
00421         @brief Main entry point for calibration pipeline.
00422
00423         @param imageSet ImageSet object containing calibration images
00424         @param bundleAdjust Whether to apply bundle adjustment (default: True)
00425         @return CalibrationState with complete intrinsic and extrinsic calibration
00426         """
00427         print("Running intrinsic calibration...")
00428         state = self.calibrateIntrinsics(imageSet)
00429
00430         print("\nRunning pairwise extrinsic calibration...")
00431         state = self.calibrateExtrinsics(imageSet, state, refCamId=imageSet.cameraIds[0])
00432
00433         if bundleAdjust:
00434             print("\nRunning global bundle adjustment on extrinsics...")
00435             state = self.bundleAdjustExtrinsics(imageSet, state, refCamId=imageSet.cameraIds[0])
00436
00437         return state
```

5.4.4 Member Data Documentation

5.4.4.1 chessboardSize

```
InitialCalibration.InitialCalibration.chessboardSize = chessboardSize
```

Definition at line 19 of file [InitialCalibration.py](#).

5.4.4.2 criteria

```
tuple InitialCalibration.InitialCalibration.criteria = (cv2.TERM_CRITERIA_EPS + cv2.TERM_↵
CRITERIA_MAX_ITER, 30, 0.001)
```

Definition at line 21 of file [InitialCalibration.py](#).

5.4.4.3 squareSize

```
InitialCalibration.InitialCalibration.squareSize = squareSize
```

Definition at line 20 of file [InitialCalibration.py](#).

The documentation for this class was generated from the following file:

- calibrationLFC/[InitialCalibration.py](#)

5.5 ResultLogger.ResultLogger Class Reference

Public Member Functions

- [__init__](#) (self, [baseDir](#)="/data/baseline")
- [to_serializable](#) (self, obj)
- float [read_baseline_health](#) (self, str baseline_path)
- None [update_baseline](#) (self, dict candidate_obj, float candidate_score, str method)
- [logInitialCalibration](#) (self, calibrationState, meta=None)
- [logRecalibration](#) (self, str combined_json_path, meta=None)

Public Attributes

- [baseDir](#) = baseDir
- [logger](#) = logging.getLogger("calibration")

5.5.1 Detailed Description

@brief Manages calibration result logging and baseline versioning.

Definition at line 8 of file [ResultLogger.py](#).

5.5.2 Constructor & Destructor Documentation

5.5.2.1 `__init__()`

```
ResultLogger.ResultLogger.__init__ (
    self,
    baseDir = "/data/baseline")
```

@brief Constructor for ResultLogger.

@param baseDir Directory for storing calibration results and baseline (default: "/data/baseline")

Definition at line 13 of file [ResultLogger.py](#).

```
00013     def __init__(self, baseDir="/data/baseline"):
00014         """
00015         @brief Constructor for ResultLogger.
00016
00017         @param baseDir Directory for storing calibration results and baseline (default:
00018         "/data/baseline")
00019         """
00019         self.baseDir = baseDir
00020         os.makedirs(baseDir, exist_ok=True)
00021
00022         self.logger = logging.getLogger("calibration")
00023         self.logger.setLevel(logging.INFO)
00024
00025         if not self.logger.handlers:
00026             log_path = os.path.join(baseDir, "calibration.log")
00027             file_handler = logging.FileHandler(log_path, encoding="utf-8")
00028             file_handler.setFormatter(logging.Formatter("%(asctime)s [%(levelname)s] %(message)s"))
00029             self.logger.addHandler(file_handler)
00030
```

5.5.3 Member Function Documentation

5.5.3.1 `logInitialCalibration()`

```
ResultLogger.ResultLogger.logInitialCalibration (
    self,
    calibrationState,
    meta = None)
```

@brief Store initial calibration result with health score.

@param calibrationState CalibrationState object with intrinsic and extrinsic parameters

@param meta Optional dictionary with metadata about the calibration run

@return Health score as float

Definition at line 125 of file [ResultLogger.py](#).

```
00125     def logInitialCalibration(self, calibrationState, meta=None):
00126         """
00127         @brief Store initial calibration result with health score.
00128
00129         @param calibrationState CalibrationState object with intrinsic and extrinsic parameters
00130         @param meta Optional dictionary with metadata about the calibration run
00131         @return Health score as float
00132         """
00133         if meta is None:
00134             meta = {}
00135
00136         # Extract calibration state
00137         try:
00138             state_dict = calibrationState.__getState__()
00139         except Exception as e:
00140             self.logger.error(f"Could not extract state dict from CalibrationState: {e}")
```

```

00141         return 0.0
00142
00143     safe_state = self.to_serializable(state_dict)
00144     safe_meta = self.to_serializable(meta)
00145
00146     # Compute health score from metrics
00147     try:
00148         from HealthMonitor import compute_initial_health_from_state
00149         report = compute_initial_health_from_state(state_dict)
00150         score = float(report.get("health", 0.0))
00151     except Exception as e:
00152         self.logger.error(f"Initial HealthScore computation failed: {e}", exc_info=True)
00153         score = 0.0
00154
00155     # Create timestamped filename
00156     timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
00157     imagesetnumber = meta.get("imageDir", "unknown")
00158     if isinstance(imagesetnumber, str) and "imageset" in imagesetnumber:
00159         imagesetnumber = imagesetnumber.split("imageset")[-1]
00160
00161     json_name = f"calibration_initial_imageset{imagesetnumber}_{timestamp}.json"
00162     json_path = os.path.join(self.baseDir, json_name)
00163
00164     # Prepare output structure
00165     out = {
00166         "meta": safe_meta,
00167         "state": safe_state
00168     }
00169
00170     # Attach health score to metadata
00171     out.setdefault("meta", {})
00172     out["meta"]["health"] = score
00173     out["meta"]["method"] = "initial"
00174     out["meta"]["runType"] = out["meta"].get("runType", "initial")
00175
00176     # Save to file
00177     try:
00178         with open(json_path, "w", encoding="utf-8") as f:
00179             json.dump(out, f, indent=2)
00180     except Exception as e:
00181         self.logger.error(f"Failed to write JSON result file '{json_path}': {e}")
00182         return 0.0
00183
00184     intr = state_dict.get("intrinsic", {})
00185     extr = state_dict.get("extrinsic", {})
00186     self.logger.info(
00187         f"Stored initial calibration to '{json_name}' "
00188         f"(intrinsic for {len(intr)} cams, extrinsic for {len(extr)} cams, bundleAdjustf: {meta.get('bundleAdjust', True)})."
00189     )
00190     self.logger.info(f"HealthScore = {score:.2f} (method=initial)")
00191
00192     # Update baseline (creates baseline on first run)
00193     self.update_baseline(out, score, method="initial")
00194
00195     return score
00196

```

5.5.3.2 logRecalibration()

```

ResultLogger.ResultLogger.logRecalibration (
    self,
    str combined_json_path,
    meta = None)

```

@brief Store LiFCal recalibration result with health score.

@param combined_json_path Path to the combined.json file from LiFCal
 @param meta Optional dictionary with metadata about the recalibration run
 @return Health score as float

Definition at line 197 of file [ResultLogger.py](#).

```

00197     def logRecalibration(self, combined_json_path: str, meta=None):
00198         """
00199         @brief Store LiFCal recalibration result with health score.
00200

```

```

00201         @param combined_json_path Path to the combined.json file from LiFCal
00202         @param meta Optional dictionary with metadata about the recalibration run
00203         @return Health score as float
00204         """
00205         if meta is None:
00206             meta = {}
00207
00208         # Import locally to avoid errors when not needed
00209         from HealthMonitor import compute_lifcal_health_from_combined_path
00210
00211         # Read combined.json
00212         try:
00213             with open(combined_json_path, "r", encoding="utf-8") as f:
00214                 combined = json.load(f)
00215         except Exception as e:
00216             self.logger.error(f"Could not read combined.json '{combined_json_path}': {e}")
00217             return 0.0
00218
00219         # Compute health score
00220         try:
00221             report = compute_lifcal_health_from_combined_path(combined_json_path)
00222             score = float(report.get("health", 0.0))
00223         except Exception as e:
00224             self.logger.error(f"LiFCal HealthScore computation failed: {e}", exc_info=True)
00225             score = 0.0
00226
00227         # Create timestamped copy
00228         timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
00229         json_name = f"calibration_lifcal_{timestamp}.json"
00230         out_path = os.path.join(self.baseDir, json_name)
00231
00232         combined.setdefault("meta", {})
00233         combined["meta"].update(self.to_serializable(meta))
00234         combined["meta"]["health"] = score
00235         combined["meta"]["method"] = "lifcal"
00236         combined["meta"]["runType"] = combined["meta"].get("runType", "recalib")
00237
00238         # Save to file
00239         try:
00240             with open(out_path, "w", encoding="utf-8") as f:
00241                 json.dump(combined, f, indent=2)
00242         except Exception as e:
00243             self.logger.error(f"Failed to write LiFCal result '{out_path}': {e}")
00244             return 0.0
00245
00246         num_cams = len(combined.get("state", {}).get("parameters", {}))
00247         self.logger.info(f"Stored lifcal recalibration to '{json_name}' (cams={num_cams}).")
00248         self.logger.info(f"HealthScore = {score:.2f} (method=lifcal)")
00249
00250         # Update baseline only if better
00251         self.update_baseline(combined, score, method="lifcal")
00252
00253         return score

```

5.5.3.3 read_baseline_health()

```

float ResultLogger.ResultLogger.read_baseline_health (
    self,
    str baseline_path)

```

@brief Extract health score from baseline.json.

@param baseline_path Path to the baseline.json file
 @return Health score as float, or -1.0 if not found or cannot be determined

Definition at line 50 of file [ResultLogger.py](#).

```

00050     def read_baseline_health(self, baseline_path: str) -> float:
00051         """
00052         @brief Extract health score from baseline.json.
00053
00054         @param baseline_path Path to the baseline.json file
00055         @return Health score as float, or -1.0 if not found or cannot be determined
00056         """
00057         if not os.path.isfile(baseline_path):
00058             return -1.0
00059
00060         try:

```

```

00061         with open(baseline_path, "r", encoding="utf-8") as f:
00062             base = json.load(f)
00063         except Exception:
00064             return -1.0
00065
00066         # Try direct health value from metadata
00067         try:
00068             h = float(base.get("meta", {}).get("health", -1.0))
00069             if np.isfinite(h) and h >= 0:
00070                 return h
00071         except Exception:
00072             pass
00073
00074         # Fallback: compute based on calibration type
00075         try:
00076             from HealthMonitor import (
00077                 compute_initial_health_from_state,
00078                 compute_lifcal_health_from_combined_dict,
00079             )
00080         except Exception:
00081             return -1.0
00082
00083         run_type = (base.get("meta", {}) or {}).get("runType", "")
00084         state = base.get("state", {})
00085
00086         try:
00087             if run_type == "initial":
00088                 rep = compute_initial_health_from_state(state)
00089                 return float(rep.get("health", -1.0))
00090             else:
00091                 # Treat as lifcal/combined
00092                 rep = compute_lifcal_health_from_combined_dict(base)
00093                 return float(rep.get("health", -1.0))
00094         except Exception:
00095             return -1.0
00096

```

5.5.3.4 to_serializable()

```

ResultLogger.ResultLogger.to_serializable (
    self,
    obj)

```

@brief Convert numpy arrays and types to JSON-serializable format.

@param obj Object to convert (numpy array, dict, list, or primitive type)
 @return JSON-serializable version of the object

Definition at line 31 of file [ResultLogger.py](#).

```

00031     def to_serializable(self, obj):
00032         """
00033         @brief Convert numpy arrays and types to JSON-serializable format.
00034
00035         @param obj Object to convert (numpy array, dict, list, or primitive type)
00036         @return JSON-serializable version of the object
00037         """
00038         if isinstance(obj, np.ndarray):
00039             return obj.tolist()
00040         if isinstance(obj, (np.floating, np.float32, np.float64)):
00041             return float(obj)
00042         if isinstance(obj, (np.integer, np.int32, np.int64)):
00043             return int(obj)
00044         if isinstance(obj, dict):
00045             return {k: self.to_serializable(v) for k, v in obj.items()}
00046         if isinstance(obj, (list, tuple)):
00047             return [self.to_serializable(v) for v in obj]
00048         return obj
00049

```

5.5.3.5 update_baseline()

```

None ResultLogger.ResultLogger.update_baseline (
    self,

```



```
dict candidate_obj,
float candidate_score,
str method)
```

@brief Update baseline.json with new calibration if health improved.

@param candidate_obj Dictionary containing calibration data
 @param candidate_score Health score of the candidate calibration
 @param method Calibration method identifier ("initial" or "lifcal")

Definition at line 97 of file [ResultLogger.py](#).

```
00097     def update_baseline(self, candidate_obj: dict, candidate_score: float, method: str) -> None:
00098         """
00099         @brief Update baseline.json with new calibration if health improved.
00100
00101         @param candidate_obj Dictionary containing calibration data
00102         @param candidate_score Health score of the candidate calibration
00103         @param method Calibration method identifier ("initial" or "lifcal")
00104         """
00105         baseline_path = os.path.join(self.baseDir, "baseline.json")
00106         old_score = self.read_baseline_health(baseline_path)
00107
00108         if (not os.path.isfile(baseline_path)) or (candidate_score > old_score):
00109             try:
00110                 with open(baseline_path, "w", encoding="utf-8") as f:
00111                     json.dump(candidate_obj, f, indent=2)
00112                     if old_score < 0:
00113                         self.logger.info(f"Baseline created: baseline.json (new={candidate_score:.2f},
method={method})")
00114             except:
00115                 self.logger.info(
00116                     f"Baseline updated: baseline.json (old={old_score:.2f},
new={candidate_score:.2f}, method={method})"
00117                 )
00118             except Exception as e:
00119                 self.logger.error(f"Failed to update baseline.json: {e}")
00120         else:
00121             self.logger.info(
00122                 f"Baseline kept: baseline.json (old={old_score:.2f} >= new={candidate_score:.2f},
method={method})"
00123             )
00124
```

5.5.4 Member Data Documentation

5.5.4.1 baseDir

ResultLogger.ResultLogger.baseDir = baseDir

Definition at line 19 of file [ResultLogger.py](#).

5.5.4.2 logger

ResultLogger.ResultLogger.logger = logging.getLogger("calibration")

Definition at line 22 of file [ResultLogger.py](#).

The documentation for this class was generated from the following file:

- calibrationLFC/[ResultLogger.py](#)

Chapter 6

File Documentation

6.1 calibrationLFC/CalibrationState.py File Reference

Classes

- class [CalibrationState.CalibrationState](#)

Namespaces

- namespace [CalibrationState](#)

6.2 CalibrationState.py

[Go to the documentation of this file.](#)

```
00001 class CalibrationState:
00002     """
00003     @brief Stores the complete calibration state for a multi-camera system.
00004
00005     @details This class manages intrinsic and extrinsic parameters for all cameras
00006     in the system. It provides methods to set and retrieve calibration data including
00007     camera matrices, distortion coefficients, rotation matrices, and translation vectors.
00008     """
00009
00010     def __init__(self):
00011         """
00012         @brief Constructor for CalibrationState.
00013
00014         @details Initializes empty dictionaries for intrinsic and extrinsic parameters
00015         and sets the timestamp to None.
00016         """
00017         self.intrinsics = {}
00018         self.extrinsics = {}
00019         self.timeStamp = None
00020
00021     def setIntrinsics(self, camId, cameraMatrix, distortionCoeffs, repError):
00022         """
00023         @brief Store intrinsic calibration parameters for a camera.
00024
00025         @param camId Camera identifier
00026         @param cameraMatrix 3x3 camera matrix containing focal lengths and principal point
00027         @param distortionCoeffs Distortion coefficients vector
00028         @param repError Reprojection error of the calibration
00029         """
00030         self.intrinsics[camId] = {
00031             "cameraMatrix": cameraMatrix,
00032             "distortionCoeffs": distortionCoeffs,
00033             "reprojectionError": repError
00034         }
```

```

00035
00036     def setExtrinsics(self, camId, rotationMatrix, translationVector, stereoRMS):
00037         """
00038         @brief Store extrinsic calibration parameters for a camera.
00039
00040         @param camId Camera identifier
00041         @param rotationMatrix 3x3 rotation matrix representing camera orientation
00042         @param translationVector 3x1 translation vector representing camera position
00043         @param stereoRMS Root mean square error of the stereo calibration
00044         """
00045         self.extrinsics[camId] = {
00046             "rotationMatrix": rotationMatrix,
00047             "translationVector": translationVector,
00048             "stereoRms": stereoRMS
00049         }
00050
00051     def setTimeStamp(self, timestamp):
00052         """
00053         @brief Set the calibration timestamp.
00054
00055         @param timestamp Timestamp indicating when the calibration was performed
00056         """
00057         self.timeStamp = timestamp
00058
00059     def __getState__(self):
00060         """
00061         @brief Return the complete calibration state as dictionary.
00062
00063         @return Dictionary containing intrinsics, extrinsics, and timestamp
00064         """
00065         return {
00066             "intrinsics": self.intrinsics,
00067             "extrinsics": self.extrinsics,
00068             "timeStamp": self.timeStamp
00069         }

```

6.3 calibrationLFC/Controller.py File Reference

Classes

- class [Controller.Controller](#)

Namespaces

- namespace [Controller](#)

6.4 Controller.py

[Go to the documentation of this file.](#)

```

00001 from InitialCalibration import InitialCalibration
00002 from ResultLogger import ResultLogger
00003 from HealthMonitor import compute_initial_health_from_state
00004
00005
00006 class Controller:
00007     """
00008     @brief Main controller for managing calibration workflow.
00009
00010     @details Orchestrates the complete calibration process including initial calibration,
00011     health monitoring, and result logging. Manages the calibration state and maintains
00012     a history of health scores.
00013     """
00014
00015     def __init__(self, bundleAdjust):
00016         """
00017         @brief Constructor for Controller.
00018
00019         @param bundleAdjust Boolean flag indicating whether to apply bundle adjustment
00020
00021         @details Initializes the controller with an InitialCalibration instance,

```

```

00022         empty score history, result logger, and bundle adjustment setting.
00023         """
00024         self.initialCalibration = InitialCalibration()
00025         self.scoreHistory = []
00026         self.currentState = None
00027         self.resultLogger = ResultLogger()
00028         self.bundleAdjust = bundleAdjust
00029
00030     def runInitialCalibration(self, imageSet):
00031         """
00032         @brief Execute initial calibration for all cameras in the image set.
00033
00034         @param imageSet ImageSet object containing calibration images and camera information
00035         @return CalibrationState object with computed intrinsic and extrinsic parameters
00036
00037         """
00038         print("Controller: starting initial calibration...")
00039         calibrationState = self.initialCalibration.run(imageSet, self.bundleAdjust)
00040
00041         # Prepare metadata for logging
00042         meta = {
00043             "runType": "initial",
00044             "numPoses": imageSet.numPoses,
00045             "imageDir": imageSet.baseDir,
00046             "cameraIds": imageSet.cameraIds,
00047             "bundleAdjust": self.bundleAdjust,
00048         }
00049         self.resultLogger.logInitialCalibration(calibrationState, meta=meta)
00050
00051         return calibrationState
00052
00053     def selfHealthCheck(self, calibrationState) -> float:
00054         """
00055         @brief Computes a global health score for the calibration quality.
00056
00057         @param calibrationState CalibrationState object to evaluate
00058         @return Health score as float between 0 and 100, or 0.0 if evaluation fails
00059
00060         """
00061         # Extract state dictionary
00062         try:
00063             state_dict = calibrationState.__getState__()
00064         except Exception as e:
00065             self.resultLogger.logger.error(
00066                 f"HealthCheck: could not extract state dict: {e}", exc_info=True
00067             )
00068             return 0.0
00069
00070         # Compute health score from calibration metrics
00071         try:
00072             report = compute_initial_health_from_state(state_dict)
00073             score = float(report["health"])
00074             self.resultLogger.logger.info(
00075                 f"HealthScore = {score:.2f} (method={report.get('method')})"
00076             )
00077             return score
00078         except Exception as e:
00079             self.resultLogger.logger.error(f"HealthCheck failed: {e}", exc_info=True)
00080             return 0.0

```

6.5 calibrationLFC/HealthMonitor.py File Reference

Namespaces

- namespace [HealthMonitor](#)

Functions

- float [HealthMonitor.clamp](#) (float x, float lo=0.0, float hi=1.0)
- [HealthMonitor.safe_float](#) (v, default=None)
- float [HealthMonitor.score_from_errors](#) (float E_a, float E_b, float w_a=0.6, float w_b=0.4)
- Dict[str, Any] [HealthMonitor.compute_initial_health_from_state](#) (Dict[str, Any] state_dict, float r_↵ best=R_BEST, float r_worst=R_WORST, float s_best=S_BEST, float s_worst=S_WORST)
- Dict[str, Optional[float]] [HealthMonitor.parse_lifcal_protocol](#) (str protocol_path)

- Dict[str, Any] [HealthMonitor.compute_lifcal_health_from_xy](#) (float x_std, float y_std, float x_mae, float y_mae, float r_best=[R_BEST](#), float r_worst=[R_WORST](#), float s_best=[S_BEST](#), float s_worst=[S_WORST](#))
- Dict[str, Any] [HealthMonitor.compute_lifcal_health_from_combined_dict](#) (Dict[str, Any] combined)
- Dict[str, Any] [HealthMonitor.compute_lifcal_health_from_combined_path](#) (str combined_json_path)

Variables

- float [HealthMonitor.R_BEST](#) = 0.03
- float [HealthMonitor.R_WORST](#) = 1.0
- float [HealthMonitor.S_BEST](#) = 0.0
- float [HealthMonitor.S_WORST](#) = 50.0

6.6 HealthMonitor.py

[Go to the documentation of this file.](#)

```
00001 """
00002 HealthScore Formulas (0..100)
00003
00004 (A) INITIAL Calibration (OpenCV/Controller JSON)
00005     HealthScore_initial = 100 * [ 1 - (0.6 * E_reproj^2 + 0.4 * E_rms^2) ]
00006
00007     E_reproj = clamp( (reproj_mean - R_BEST) / (R_WORST - R_BEST), 0, 1 )
00008     E_rms     = clamp( (rms_mean     - S_BEST) / (S_WORST - S_BEST), 0, 1 )
00009
00010     Uses:
00011         - intrinsics[*].reprojectionError (per camera)
00012         - extrinsics[*].stereoRms        (per camera)
00013
00014 (B) LiFCal Calibration (protocol-based)
00015     HealthScore_lifcal = 100 * [ 1 - (0.6 * E_std^2 + 0.4 * E_mae^2) ]
00016
00017     E_std = clamp( ( sqrt(x_std^2 + y_std^2) - S_BEST ) / (S_WORST - S_BEST), 0, 1 )
00018     E_mae = clamp( ( sqrt(x_mae^2 + y_mae^2) - R_BEST ) / (R_WORST - R_BEST), 0, 1 )
00019
00020     Uses:
00021         - std.Dev. x/y and mae x/y from LiFCal calibrationProtocol.txt
00022
00023 Shared thresholds (for BOTH methods):
00024     R_BEST = 0.03, R_WORST = 1.0, S_BEST = 0.0, S_WORST = 50.0
00025 """
00026
00027 import re
00028 import json
00029 import math
00030 from typing import Dict, Any, Optional
00031
00032 """
00033 @brief HealthMonitor module for computing calibration health scores.
00034
00035 """
00036
00037 # Shared thresholds
00038 R_BEST = 0.03
00039 R_WORST = 1.0
00040 S_BEST = 0.0
00041 S_WORST = 50.0
00042
00043
00044 def clamp(x: float, lo: float = 0.0, hi: float = 1.0) -> float:
00045     """
00046     @brief Clamp value between lower and upper bounds.
00047
00048     @param x Value to clamp
00049     @param lo Lower bound (default: 0.0)
00050     @param hi Upper bound (default: 1.0)
00051     @return Clamped value
00052     """
00053     if x < lo:
00054         return lo
00055     if x > hi:
00056         return hi
00057     return x
00058
00059
```

```

00060 def safe_float(v, default=None):
00061     """
00062     @brief Convert value to float safely, return default on error.
00063
00064     @param v Value to convert
00065     @param default Default value to return on conversion error
00066     @return Converted float value or default
00067     """
00068     try:
00069         if v is None:
00070             return default
00071         return float(v)
00072     except Exception:
00073         return default
00074
00075
00076 def score_from_errors(E_a: float, E_b: float, w_a: float = 0.6, w_b: float = 0.4) -> float:
00077     """
00078     @brief Compute health score from two normalized error metrics.
00079
00080     @param E_a First normalized error metric (0..1)
00081     @param E_b Second normalized error metric (0..1)
00082     @param w_a Weight for first error metric (default: 0.6)
00083     @param w_b Weight for second error metric (default: 0.4)
00084     @return Health score between 0 and 100
00085     """
00086     val = 100.0 * (1.0 - (w_a * (E_a ** 2) + w_b * (E_b ** 2)))
00087     return clamp(val / 100.0, 0.0, 1.0) * 100.0
00088
00089
00090 def compute_initial_health_from_state(
00091     state_dict: Dict[str, Any],
00092     r_best: float = R_BEST,
00093     r_worst: float = R_WORST,
00094     s_best: float = S_BEST,
00095     s_worst: float = S_WORST,
00096 ) -> Dict[str, Any]:
00097     """
00098     @brief Compute health score from initial calibration state.
00099
00100     @param state_dict Dictionary containing calibration state with intrinsics and extrinsics
00101     @param r_best Best reprojection error threshold (default: R_BEST)
00102     @param r_worst Worst reprojection error threshold (default: R_WORST)
00103     @param s_best Best stereo RMS threshold (default: S_BEST)
00104     @param s_worst Worst stereo RMS threshold (default: S_WORST)
00105     @return Dictionary with health score, method name, and detailed metrics
00106     """
00107     st = state_dict.get("state", state_dict)
00108     intr = st.get("intrinsics", {}) or {}
00109     extr = st.get("extrinsics", {}) or {}
00110
00111     reproj_vals = []
00112     rms_vals = []
00113
00114     for cam, intr_data in intr.items():
00115         reproj = safe_float(intr_data.get("reprojectionError"), None)
00116         if reproj is not None:
00117             reproj_vals.append(reproj)
00118
00119         ex = extr.get(cam, {})
00120         rms = safe_float(ex.get("stereoRms"), None)
00121         if rms is not None:
00122             rms_vals.append(rms)
00123
00124     if not reproj_vals and not rms_vals:
00125         return {
00126             "health": 0.0,
00127             "method": "initial",
00128             "details": {"reason": "no reprojectionError and no stereoRms found"},
00129         }
00130
00131     reproj_mean = (sum(reproj_vals) / len(reproj_vals)) if reproj_vals else r_worst
00132     rms_mean = (sum(rms_vals) / len(rms_vals)) if rms_vals else s_worst
00133
00134     E_reproj = clamp((reproj_mean - r_best) / (r_worst - r_best), 0.0, 1.0)
00135     E_rms = clamp((rms_mean - s_best) / (s_worst - s_best), 0.0, 1.0)
00136
00137     score = score_from_errors(E_reproj, E_rms)
00138
00139     return {
00140         "health": float(score),
00141         "method": "initial",
00142         "details": {
00143             "reproj_mean": float(reproj_mean),
00144             "rms_mean": float(rms_mean),
00145             "E_reproj": float(E_reproj),
00146             "E_rms": float(E_rms),

```

```

00147         "thresholds": {
00148             "R_BEST": float(r_best),
00149             "R_WORST": float(r_worst),
00150             "S_BEST": float(s_best),
00151             "S_WORST": float(s_worst),
00152         },
00153     },
00154 }
00155
00156
00157 def parse_lifcal_protocol(protocol_path: str) -> Dict[str, Optional[float]]:
00158     """
00159     @brief Parse LiFCal calibration protocol text file for error metrics.
00160
00161     @param protocol_path Path to the calibration protocol text file
00162     @return Dictionary with x_std, y_std, x_mae, y_mae values or None if not found
00163     """
00164     txt = open(protocol_path, "r", encoding="utf-8", errors="ignore").read()
00165
00166     def find_float(pattern: str) -> Optional[float]:
00167         m = re.search(pattern, txt, re.MULTILINE)
00168         if not m:
00169             return None
00170         return safe_float(m.group(1), None)
00171
00172     x_std = find_float(r"std\\.\\s*Dev\\.\\s*x:\\s*([0-9.+-eE]+)")
00173     y_std = find_float(r"std\\.\\s*Dev\\.\\s*y:\\s*([0-9.+-eE]+)")
00174     x_mae = find_float(r"mae\\s*x:\\s*([0-9.+-eE]+)")
00175     y_mae = find_float(r"mae\\s*y:\\s*([0-9.+-eE]+)")
00176
00177     return {"x_std": x_std, "y_std": y_std, "x_mae": x_mae, "y_mae": y_mae}
00178
00179
00180 def compute_lifcal_health_from_xy(
00181     x_std: float,
00182     y_std: float,
00183     x_mae: float,
00184     y_mae: float,
00185     r_best: float = R_BEST,
00186     r_worst: float = R_WORST,
00187     s_best: float = S_BEST,
00188     s_worst: float = S_WORST,
00189 ) -> Dict[str, Any]:
00190     """
00191     @brief Compute LiFCal health score from x/y standard deviation and MAE values.
00192
00193     @param x_std Standard deviation in x direction
00194     @param y_std Standard deviation in y direction
00195     @param x_mae Mean absolute error in x direction
00196     @param y_mae Mean absolute error in y direction
00197     @param r_best Best reprojection error threshold (default: R_BEST)
00198     @param r_worst Worst reprojection error threshold (default: R_WORST)
00199     @param s_best Best stereo RMS threshold (default: S_BEST)
00200     @param s_worst Worst stereo RMS threshold (default: S_WORST)
00201     @return Dictionary with health score, method name, and detailed metrics
00202     """
00203     std_norm = (x_std * x_std + y_std * y_std) ** 0.5
00204     mae_norm = (x_mae * x_mae + y_mae * y_mae) ** 0.5
00205
00206     E_std = clamp((std_norm - s_best) / (s_worst - s_best), 0.0, 1.0)
00207     E_mae = clamp((mae_norm - r_best) / (r_worst - r_best), 0.0, 1.0)
00208
00209     score = score_from_errors(E_std, E_mae)
00210
00211     return {
00212         "health": float(score),
00213         "method": "lifcal",
00214         "details": {
00215             "std_norm": float(std_norm),
00216             "mae_norm": float(mae_norm),
00217             "E_std": float(E_std),
00218             "E_mae": float(E_mae),
00219             "thresholds": {
00220                 "R_BEST": float(r_best),
00221                 "R_WORST": float(r_worst),
00222                 "S_BEST": float(s_best),
00223                 "S_WORST": float(s_worst),
00224             },
00225         },
00226     }
00227
00228 def compute_lifcal_health_from_combined_dict(combined: Dict[str, Any]) -> Dict[str, Any]:
00229     """
00230     @brief Compute LiFCal health from combined.json structure.
00231
00232     @param combined Dictionary from combined.json with parameters for all cameras
00233     @return Dictionary with global health score, method name, and per-camera metrics

```

```

00234     """
00235     params_all = combined.get("state", {}).get("parameters", {})
00236     if not isinstance(params_all, dict) or len(params_all) == 0:
00237         return {"health": 0.0, "method": "lifcal", "perCam": {}}
00238
00239     per_cam = {}
00240     scores = []
00241
00242     for cam, p in params_all.items():
00243         rep = (p or {}).get("reprojection", {}) or {}
00244
00245         x_std = rep.get("x_std", None)
00246         y_std = rep.get("y_std", None)
00247         x_mae = rep.get("x_mae", None)
00248         y_mae = rep.get("y_mae", None)
00249
00250         # Skip camera if any field is missing
00251         if any(v is None for v in [x_std, y_std, x_mae, y_mae]):
00252             per_cam[cam] = {"health": 0.0, "reason": "missing reprojection fields"}
00253             continue
00254
00255         std_mag = math.sqrt(float(x_std) ** 2 + float(y_std) ** 2)
00256         mae_mag = math.sqrt(float(x_mae) ** 2 + float(y_mae) ** 2)
00257
00258         E_std = clamp((std_mag - S_BEST) / (S_WORST - S_BEST), 0.0, 1.0)
00259         E_mae = clamp((mae_mag - R_BEST) / (R_WORST - R_BEST), 0.0, 1.0)
00260
00261         health = 100.0 * (1.0 - (0.6 * (E_std ** 2) + 0.4 * (E_mae ** 2)))
00262         health = clamp(health, 0.0, 100.0)
00263
00264         per_cam[cam] = {
00265             "health": float(health),
00266             "std_mag": float(std_mag),
00267             "mae_mag": float(mae_mag),
00268             "E_std": float(E_std),
00269             "E_mae": float(E_mae),
00270         }
00271         scores.append(float(health))
00272
00273     if len(scores) == 0:
00274         return {"health": 0.0, "method": "lifcal", "perCam": per_cam}
00275
00276     # Global health: average across all cameras
00277     global_health = sum(scores) / len(scores)
00278
00279     return {"health": float(global_health), "method": "lifcal", "perCam": per_cam}
00280
00281
00282 def compute_lifcal_health_from_combined_path(combined_json_path: str) -> Dict[str, Any]:
00283     """
00284     @brief Load combined.json from file and compute LiFCal health.
00285
00286     @param combined_json_path Path to the combined.json file
00287     @return Dictionary with global health score, method name, and per-camera metrics
00288     """
00289     with open(combined_json_path, "r", encoding="utf-8") as f:
00290         combined = json.load(f)
00291     return compute_lifcal_health_from_combined_dict(combined)

```

6.7 calibrationLFC/ImageSet.py File Reference

Classes

- class [ImageSet.ImageSet](#)

Namespaces

- namespace [ImageSet](#)

6.8 ImageSet.py

[Go to the documentation of this file.](#)

```
00001 from typing import List
00002
00003 class ImageSet:
00004     """
00005     @brief Represents a set of calibration images organized by pose and camera.
00006     """
00007
00008     def __init__(self, baseDir: str, numPoses: int, cameraIds: List[str]):
00009         """
00010         @brief Constructor for ImageSet.
00011
00012         @param baseDir Base directory containing calibration images
00013         @param numPoses Total number of poses in the calibration set
00014         @param cameraIds List of camera identifiers
00015         """
00016         self.baseDir = baseDir
00017         self.numPoses = numPoses
00018         self.cameraIds = cameraIds
00019
00020     def getImagePath(self, poseIndex: int, cameraId: str) -> str:
00021         """
00022         @brief Construct the file path for a specific pose and camera.
00023
00024         @param poseIndex Zero-based pose index
00025         @param cameraId Camera identifier string
00026         @return String path in format: baseDir/pose_XXX_cameraId.png
00027         """
00028         poseStr = f"pose_{poseIndex:03d}"
00029         return f"{self.baseDir}/{poseStr}_{cameraId}.png"
00030
00031         """
00032         Beispiel:
00033         baseDir = "/data/images"
00034         poseIndex = 5 → poseStr = "pose_005"
00035         cameraId = "center"
00036
00037         Ergebnis: "/data/images/pose_005_center.png"
00038         """
```

6.9 calibrationLFC/InitialCalibration.py File Reference

Classes

- class [InitialCalibration.InitialCalibration](#)

Namespaces

- namespace [InitialCalibration](#)

6.10 InitialCalibration.py

[Go to the documentation of this file.](#)

```
00001 import cv2
00002 import numpy as np
00003 from CalibrationState import CalibrationState
00004 from scipy.optimize import least_squares
00005
00006
00007 class InitialCalibration:
00008     """
00009     @brief Multi-camera calibration using chessboard patterns and bundle adjustment.
00010     """
00011
00012     def __init__(self, chessboardSize=(8,8), squareSize=2/9):
00013         """
```

```

00014         @brief Constructor for InitialCalibration.
00015
00016         @param chessboardSize Tuple (cols, rows) defining chessboard dimensions (default: (8,8))
00017         @param squareSize Physical size of one chessboard square in meters (default: 2/9)
00018         """
00019         self.chessboardSize = chessboardSize
00020         self.squareSize = squareSize
00021         self.criteria = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001)
00022
00023     def createObjectPoints(self):
00024         """
00025         @brief Generate 3D coordinates of chessboard corners.
00026
00027         @return Numpy array of shape (N, 3) with 3D corner coordinates
00028         """
00029         cols, rows = self.chessboardSize
00030         objp = np.zeros((cols * rows, 3), np.float32)
00031         objp[:, :2] = np.mgrid[0:cols, 0:rows].T.reshape(-1, 2)
00032         objp *= self.squareSize
00033         return objp
00034
00035     def detectCorners(self, imagePath):
00036         """
00037         @brief Detect and refine chessboard corner positions in image.
00038
00039         @param imagePath Path to the image file
00040         @return Tuple (success, corners, imageSize) where success is bool, corners are refined 2D
00041         points, and imageSize is (width, height)
00042         """
00043         img = cv2.imread(imagePath)
00044         if img is None:
00045             return False, None, None
00046         gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
00047         found, corners = cv2.findChessboardCorners(gray, self.chessboardSize)
00048         if not found:
00049             return False, None, None
00050
00051         refined = cv2.cornerSubPix(gray, corners, (11, 11), (-1, -1), self.criteria)
00052         return True, refined, gray.shape[::-1]
00053
00054     def calibrateIntrinsics(self, imageSet):
00055         """
00056         @brief Calibrate intrinsic parameters independently for each camera.
00057
00058         @param imageSet ImageSet object containing calibration images
00059         @return CalibrationState with populated intrinsics for all cameras
00060         """
00061         state = CalibrationState()
00062         objp = self.createObjectPoints()
00063
00064         objpoints = {cam: [] for cam in imageSet.cameraIds}
00065         imgpoints = {cam: [] for cam in imageSet.cameraIds}
00066         imgsize = {}
00067
00068         # Collect corner points from all poses for each camera
00069         for pose in range(imageSet.numPoses):
00070             for cam in imageSet.cameraIds:
00071                 path = imageSet.getImagePath(pose, cam)
00072                 ok, corners, size = self.detectCorners(path)
00073                 if ok:
00074                     objpoints[cam].append(objp)
00075                     imgpoints[cam].append(corners)
00076                     imgsize[cam] = size
00077
00078         # Calibrate each camera separately
00079         for cam in imageSet.cameraIds:
00080             if len(objpoints[cam]) < 5:
00081                 print(f"[WARN] Not enough samples for camera {cam}")
00082                 continue
00083
00084             err, K, dist, rvecs, tvecs = cv2.calibrateCamera(
00085                 objpoints[cam], imgpoints[cam], imgsize[cam], None, None)
00086
00087             state.setIntrinsics(cam, K, dist, err)
00088
00089         return state
00090
00091     def calibrateExtrinsics(self, imageSet, state, refCamId):
00092         """
00093         @brief Compute pairwise stereo calibration between reference and each other camera.
00094
00095         @param imageSet ImageSet object containing calibration images
00096         @param state CalibrationState with intrinsics already computed
00097         @param refCamId Camera ID to use as reference coordinate system
00098         @return Updated CalibrationState with extrinsics for all cameras
00099         """

```

```

00100         objectPoints = self.createObjectPoints()
00101         K_ref = state.intrinsics[refCamId]["cameraMatrix"]
00102         dist_ref = state.intrinsics[refCamId]["distortionCoeffs"]
00103
00104         for cam in imageSet.cameraIds:
00105             if cam == refCamId:
00106                 state.setExtrinsics(cam, np.eye(3), np.zeros((3,1)), 0.0)
00107                 continue
00108
00109             if cam not in state.intrinsics:
00110                 continue
00111
00112             K_cam = state.intrinsics[cam]["cameraMatrix"]
00113             dist_cam = state.intrinsics[cam]["distortionCoeffs"]
00114
00115             objList = []
00116             imgRef = []
00117             imgCam = []
00118             imageSize = None
00119
00120             for pose in range(imageSet.numPoses):
00121                 pRef = imageSet.getImagePath(pose, refCamId)
00122                 pCam = imageSet.getImagePath(pose, cam)
00123
00124                 okR, cornersR, sizeR = self.detectCorners(pRef)
00125                 okC, cornersC, sizeC = self.detectCorners(pCam)
00126
00127                 if okR and okC:
00128                     objList.append(objectPoints)
00129                     imgRef.append(cornersR)
00130                     imgCam.append(cornersC)
00131                     imageSize = sizeR
00132
00133             if len(objList) < 3:
00134                 continue
00135
00136             flags = cv2.CALIB_FIX_INTRINSIC
00137
00138             stereo_rms, KR0, d0, KC1, d1, R, T, E, F = cv2.stereoCalibrate(
00139                 objList,
00140                 imgRef,
00141                 imgCam,
00142                 K_ref,
00143                 dist_ref,
00144                 K_cam,
00145                 dist_cam,
00146                 imageSize,
00147                 criteria=self.criteria,
00148                 flags=flags
00149             )
00150
00151             state.setExtrinsics(cam, R, T, stereo_rms)
00152
00153         return state
00154
00155     def bundleAdjustExtrinsics(self, imageSet, state, refCamId):
00156         """
00157         @brief Global bundle adjustment optimizing extrinsics and board poses.
00158
00159         @param imageSet ImageSet object containing calibration images
00160         @param state CalibrationState with initial intrinsics and extrinsics
00161         @param refCamId Camera ID to use as reference coordinate system
00162         @return Updated CalibrationState with refined extrinsics after bundle adjustment
00163         """
00164         objp = self.createObjectPoints()
00165         cameraIds = list(imageSet.cameraIds)
00166         if refCamId not in cameraIds:
00167             raise ValueError("refCamId not in imageSet.cameraIds")
00168
00169         ref_idx = cameraIds.index(refCamId)
00170
00171         # Initialize board poses via PnP-RANSAC in reference camera
00172         K_ref = state.intrinsics[refCamId]["cameraMatrix"]
00173         dist_ref = state.intrinsics[refCamId]["distortionCoeffs"]
00174
00175         pose_rvecs = {}
00176         pose_tvecs = {}
00177         valid_poses = []
00178
00179         for pose in range(imageSet.numPoses):
00180             img_path = imageSet.getImagePath(pose, refCamId)
00181             ok, corners, size = self.detectCorners(img_path)
00182             if not ok:
00183                 continue
00184
00185             # Use PnP-RANSAC for robust board pose estimation
00186             ok_pnp, rvec, tvec, inliers = cv2.solvePnP(Ransac(

```

```

00187         objp,
00188         corners,
00189         K_ref,
00190         dist_ref,
00191         reprojectionError=2.0,
00192         confidence=0.99,
00193         flags=cv2.SOLVEPNP_ITERATIVE
00194     )
00195
00196     if not ok_pnp or inliers is None or len(inliers) < 10:
00197         # Skip pose if too few reliable points
00198         continue
00199
00200     pose_rvecs[pose] = rvec.astype(np.float64).reshape(3)
00201     pose_tvecs[pose] = tvec.astype(np.float64).reshape(3)
00202     valid_poses.append(pose)
00203
00204     if len(valid_poses) < 3:
00205         print("[BA] Not enough valid poses for bundle adjustment.")
00206         return state
00207
00208     # Initialize extrinsics from state
00209     # Only optimize non-reference cameras
00210     optim_cams = []
00211     cam_param_index = {}
00212
00213     for ci, cam in enumerate(cameraIds):
00214         if cam == refCamId:
00215             continue
00216         if cam not in state.extrinsics:
00217             continue
00218
00219         optim_cams.append(ci)
00220         cam_param_index[ci] = len(cam_param_index)
00221
00222     num_optim_cams = len(optim_cams)
00223     if num_optim_cams == 0:
00224         print("[BA] No extrinsic data to optimize, skipping BA.")
00225         return state
00226
00227     # Collect measurements (all cameras, all valid poses)
00228     # Each measurement: (cam_idx, pose, 2D points)
00229     measurements = []
00230
00231     for ci, cam in enumerate(cameraIds):
00232         for pose in valid_poses:
00233             img_path = imageSet.getImagePath(pose, cam)
00234             ok, corners, size = self.detectCorners(img_path)
00235             if not ok:
00236                 continue
00237             pts2d = corners.reshape(-1, 2).astype(np.float64)
00238             measurements.append((ci, pose, pts2d))
00239
00240     if len(measurements) == 0:
00241         print("[BA] No overlapping detections found, skipping BA.")
00242         return state
00243
00244     # Initialize parameter vector
00245     # Layout: [cams (without ref): (r_cam(3), t_cam(3))... , poses: (r_p(3), t_p(3))...]
00246
00247     param_cam = []
00248     for ci in optim_cams:
00249         cam = cameraIds[ci]
00250         extr = state.extrinsics[cam]
00251         R = extr["rotationMatrix"]
00252         T = extr["translationVector"].reshape(3)
00253         rvec_cam, _ = cv2.Rodrigues(R)
00254         param_cam.append(rvec_cam.reshape(3))
00255         param_cam.append(T)
00256     param_cam = np.concatenate(param_cam).astype(np.float64)
00257
00258     param_pose = []
00259     for pose in valid_poses:
00260         param_pose.append(pose_rvecs[pose])
00261         param_pose.append(pose_tvecs[pose])
00262     param_pose = np.concatenate(param_pose).astype(np.float64)
00263
00264     x0 = np.concatenate([param_cam, param_pose])
00265
00266     # Helper functions for parameter decoding
00267     def decode_params(params):
00268         # Cameras
00269         cam_rvecs = {}
00270         cam_tvecs = {}
00271         idx = 0
00272         for ci in optim_cams:
00273             r = params[idx:idx+3]; idx += 3

```

```

00274         t = params[idx:idx+3]; idx += 3
00275         cam_rvecs[ci] = r
00276         cam_tvecs[ci] = t
00277
00278     # Poses
00279     pose_r = {}
00280     pose_t = {}
00281     for pose in valid_poses:
00282         r = params[idx:idx+3]; idx += 3
00283         t = params[idx:idx+3]; idx += 3
00284         pose_r[pose] = r
00285         pose_t[pose] = t
00286
00287     return cam_rvecs, cam_tvecs, pose_r, pose_t
00288
00289 # Residuals function
00290 def residuals(params):
00291     cam_rvecs, cam_tvecs, pose_r, pose_t = decode_params(params)
00292
00293     # Precompute rotation matrices
00294     R_cam = {}
00295     for ci, r in cam_rvecs.items():
00296         R_cam[ci], _ = cv2.Rodrigues(r.astype(np.float64))
00297
00298     R_pose = {}
00299     for pose, r in pose_r.items():
00300         R_pose[pose], _ = cv2.Rodrigues(r.astype(np.float64))
00301
00302     residual_list = []
00303
00304     for ci, pose, pts2d in measurements:
00305         cam_id = cameraIds[ci]
00306         K = state.intrinsics[cam_id]["cameraMatrix"]
00307         dist = state.intrinsics[cam_id]["distortionCoeffs"]
00308
00309         # Board pose in reference frame
00310         r_p = pose_r[pose]
00311         t_p = pose_t[pose]
00312         R_p = R_pose[pose]
00313
00314         if ci == ref_idx:
00315             # Reference camera: use board pose directly
00316             r_total = r_p
00317             t_total = t_p
00318         else:
00319             if ci not in cam_rvecs:
00320                 # For non-optimized cameras, use static extrinsics
00321                 extr = state.extrinsics[cam_id]
00322                 R_c = extr["rotationMatrix"]
00323                 t_c = extr["translationVector"].reshape(3)
00324             else:
00325                 R_c = R_cam[ci]
00326                 t_c = cam_tvecs[ci]
00327
00328             # Composition:  $X_{cam} = R_c * (R_p * X + t_p) + t_c$ 
00329             R_total = R_c @ R_p
00330             t_total = R_c @ t_p + t_c
00331             r_total, _ = cv2.Rodrigues(R_total.astype(np.float64))
00332             r_total = r_total.reshape(3)
00333
00334             # Projection
00335             proj, _ = cv2.projectPoints(
00336                 objp,
00337                 r_total.astype(np.float64),
00338                 t_total.astype(np.float64),
00339                 K,
00340                 dist
00341             )
00342             proj = proj.reshape(-1, 2)
00343
00344             res = (proj - pts2d).reshape(-1)
00345             residual_list.append(res)
00346
00347     if not residual_list:
00348         return np.zeros(0, dtype=np.float64)
00349
00350     return np.concatenate(residual_list)
00351
00352 # Optimization (robust, soft_l1)
00353 result = least_squares(
00354     residuals,
00355     x0,
00356     verbose=2,
00357     method="trf",
00358     loss="soft_l1",
00359     f_scale=1.0
00360 )

```

```

00361
00362     print(
00363         f"[BA] Done. Final cost: {result.cost:.4f}, "
00364         f"RMS per coord: {np.sqrt(2*result.cost/len(result.fun)):.4f} px"
00365     )
00366
00367     # Write results back to state.extrinsics
00368     cam_rvecs_opt, cam_tvecs_opt, pose_r_opt, pose_t_opt = decode_params(result.x)
00369
00370     # Reference camera remains identity
00371     state.setExtrinsics(
00372         refCamId,
00373         np.eye(3, dtype=np.float64),
00374         np.zeros((3, 1), dtype=np.float64),
00375         0.0
00376     )
00377
00378     # Store optimized cameras
00379     for ci in optim_cams:
00380         cam_id = cameraIds[ci]
00381         r = cam_rvecs_opt[ci]
00382         t = cam_tvecs_opt[ci]
00383         R, _ = cv2.Rodrigues(r.astype(np.float64))
00384         T = t.reshape(3, 1).astype(np.float64)
00385
00386         # Approximate RMS for this camera from global residuals
00387         # (simple average across all measurements)
00388         cam_residuals = []
00389         for m_ci, pose, pts2d in measurements:
00390             if m_ci != ci:
00391                 continue
00392             # Reproject with final parameters
00393             R_p, _ = cv2.Rodrigues(pose_r_opt[pose].astype(np.float64))
00394             t_p = pose_t_opt[pose].astype(np.float64)
00395
00396             R_total = R @ R_p
00397             t_total = R @ t_p + t.flatten()
00398             r_total, _ = cv2.Rodrigues(R_total)
00399             K = state.intrinsics[cam_id]["cameraMatrix"]
00400             dist = state.intrinsics[cam_id]["distortionCoeffs"]
00401
00402             proj, _ = cv2.projectPoints(
00403                 objp, r_total, t_total, K, dist
00404             )
00405             proj = proj.reshape(-1, 2)
00406             res = (proj - pts2d).reshape(-1)
00407             cam_residuals.append(res)
00408
00409         if cam_residuals:
00410             cam_residuals = np.concatenate(cam_residuals)
00411             stereo_rms = float(np.sqrt(np.mean(cam_residuals**2)))
00412         else:
00413             stereo_rms = state.extrinsics[cam_id].get("stereoRms", 0.0)
00414
00415         state.setExtrinsics(cam_id, R, T, stereo_rms)
00416
00417     return state
00418
00419 def run(self, imageSet, bundleAdjust=True):
00420     """
00421     @brief Main entry point for calibration pipeline.
00422
00423     @param imageSet ImageSet object containing calibration images
00424     @param bundleAdjust Whether to apply bundle adjustment (default: True)
00425     @return CalibrationState with complete intrinsic and extrinsic calibration
00426     """
00427     print("Running intrinsic calibration...")
00428     state = self.calibrateIntrinsics(imageSet)
00429
00430     print("\nRunning pairwise extrinsic calibration...")
00431     state = self.calibrateExtrinsics(imageSet, state, refCamId=imageSet.cameraIds[0])
00432
00433     if bundleAdjust:
00434         print("\nRunning global bundle adjustment on extrinsics...")
00435         state = self.bundleAdjustExtrinsics(imageSet, state, refCamId=imageSet.cameraIds[0])
00436
00437     return state

```

6.11 calibrationLFC/ResultLogger.py File Reference

Classes

- class [ResultLogger.ResultLogger](#)

Namespaces

- namespace [ResultLogger](#)

6.12 ResultLogger.py

[Go to the documentation of this file.](#)

```

00001 import os
00002 import json
00003 import logging
00004 import numpy as np
00005 from datetime import datetime
00006
00007
00008 class ResultLogger:
00009     """
00010     @brief Manages calibration result logging and baseline versioning.
00011     """
00012
00013     def __init__(self, baseDir="/data/baseline"):
00014         """
00015         @brief Constructor for ResultLogger.
00016
00017         @param baseDir Directory for storing calibration results and baseline (default:
00018         "/data/baseline")
00019         """
00019         self.baseDir = baseDir
00020         os.makedirs(baseDir, exist_ok=True)
00021
00022         self.logger = logging.getLogger("calibration")
00023         self.logger.setLevel(logging.INFO)
00024
00025         if not self.logger.handlers:
00026             log_path = os.path.join(baseDir, "calibration.log")
00027             file_handler = logging.FileHandler(log_path, encoding="utf-8")
00028             file_handler.setFormatter(logging.Formatter("%(asctime)s [% (levelname)s] %(message)s"))
00029             self.logger.addHandler(file_handler)
00030
00031     def to_serializable(self, obj):
00032         """
00033         @brief Convert numpy arrays and types to JSON-serializable format.
00034
00035         @param obj Object to convert (numpy array, dict, list, or primitive type)
00036         @return JSON-serializable version of the object
00037         """
00038         if isinstance(obj, np.ndarray):
00039             return obj.tolist()
00040         if isinstance(obj, (np.float32, np.float64)):
00041             return float(obj)
00042         if isinstance(obj, (np.integer, np.int32, np.int64)):
00043             return int(obj)
00044         if isinstance(obj, dict):
00045             return {k: self.to_serializable(v) for k, v in obj.items()}
00046         if isinstance(obj, (list, tuple)):
00047             return [self.to_serializable(v) for v in obj]
00048         return obj
00049
00050     def read_baseline_health(self, baseline_path: str) -> float:
00051         """
00052         @brief Extract health score from baseline.json.
00053
00054         @param baseline_path Path to the baseline.json file
00055         @return Health score as float, or -1.0 if not found or cannot be determined
00056         """
00057         if not os.path.isfile(baseline_path):
00058             return -1.0
00059
00060         try:
00061             with open(baseline_path, "r", encoding="utf-8") as f:
00062                 base = json.load(f)
00063         except Exception:
00064             return -1.0
00065
00066         # Try direct health value from metadata
00067         try:
00068             h = float(base.get("meta", {}).get("health", -1.0))
00069             if np.isfinite(h) and h >= 0:
00070                 return h
00071         except Exception:
00072             pass

```

```

00073
00074     # Fallback: compute based on calibration type
00075     try:
00076         from HealthMonitor import (
00077             compute_initial_health_from_state,
00078             compute_lifcal_health_from_combined_dict,
00079         )
00080     except Exception:
00081         return -1.0
00082
00083     run_type = (base.get("meta", {}) or {}).get("runType", "")
00084     state = base.get("state", {})
00085
00086     try:
00087         if run_type == "initial":
00088             rep = compute_initial_health_from_state(state)
00089             return float(rep.get("health", -1.0))
00090         else:
00091             # Treat as lifcal/combined
00092             rep = compute_lifcal_health_from_combined_dict(base)
00093             return float(rep.get("health", -1.0))
00094     except Exception:
00095         return -1.0
00096
00097 def update_baseline(self, candidate_obj: dict, candidate_score: float, method: str) -> None:
00098     """
00099     @brief Update baseline.json with new calibration if health improved.
00100
00101     @param candidate_obj Dictionary containing calibration data
00102     @param candidate_score Health score of the candidate calibration
00103     @param method Calibration method identifier ("initial" or "lifcal")
00104     """
00105     baseline_path = os.path.join(self.baseDir, "baseline.json")
00106     old_score = self.read_baseline_health(baseline_path)
00107
00108     if (not os.path.isfile(baseline_path)) or (candidate_score > old_score):
00109         try:
00110             with open(baseline_path, "w", encoding="utf-8") as f:
00111                 json.dump(candidate_obj, f, indent=2)
00112                 if old_score < 0:
00113                     self.logger.info(f"Baseline created: baseline.json (new={candidate_score:.2f},
00114 method={method})")
00115             else:
00116                 self.logger.info(
00117                     f"Baseline updated: baseline.json (old={old_score:.2f},
00118 new={candidate_score:.2f}, method={method})"
00119                 )
00120             except Exception as e:
00121                 self.logger.error(f"Failed to update baseline.json: {e}")
00122         else:
00123             self.logger.info(
00124                 f"Baseline kept: baseline.json (old={old_score:.2f} >= new={candidate_score:.2f},
00125 method={method})"
00126             )
00127
00128 def logInitialCalibration(self, calibrationState, meta=None):
00129     """
00130     @brief Store initial calibration result with health score.
00131
00132     @param calibrationState CalibrationState object with intrinsic and extrinsic parameters
00133     @param meta Optional dictionary with metadata about the calibration run
00134     @return Health score as float
00135     """
00136     if meta is None:
00137         meta = {}
00138
00139     # Extract calibration state
00140     try:
00141         state_dict = calibrationState.__getState__()
00142     except Exception as e:
00143         self.logger.error(f"Could not extract state dict from CalibrationState: {e}")
00144         return 0.0
00145
00146     safe_state = self.to_serializable(state_dict)
00147     safe_meta = self.to_serializable(meta)
00148
00149     # Compute health score from metrics
00150     try:
00151         from HealthMonitor import compute_initial_health_from_state
00152         report = compute_initial_health_from_state(state_dict)
00153         score = float(report.get("health", 0.0))
00154     except Exception as e:
00155         self.logger.error(f"Initial HealthScore computation failed: {e}", exc_info=True)
00156         score = 0.0
00157
00158     # Create timestamped filename
00159     timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")

```



```

00157         imagesetnumber = meta.get("imageDir", "unknown")
00158     if isinstance(imagesetnumber, str) and "imageset" in imagesetnumber:
00159         imagesetnumber = imagesetnumber.split("imageset")[-1]
00160
00161     json_name = f"calibration_initial_imageset{imagesetnumber}_{timestamp}.json"
00162     json_path = os.path.join(self.baseDir, json_name)
00163
00164     # Prepare output structure
00165     out = {
00166         "meta": safe_meta,
00167         "state": safe_state
00168     }
00169
00170     # Attach health score to metadata
00171     out.setdefault("meta", {})
00172     out["meta"]["health"] = score
00173     out["meta"]["method"] = "initial"
00174     out["meta"]["runType"] = out["meta"].get("runType", "initial")
00175
00176     # Save to file
00177     try:
00178         with open(json_path, "w", encoding="utf-8") as f:
00179             json.dump(out, f, indent=2)
00180     except Exception as e:
00181         self.logger.error(f"Failed to write JSON result file '{json_path}': {e}")
00182         return 0.0
00183
00184     intr = state_dict.get("intrinsics", {})
00185     extr = state_dict.get("extrinsics", {})
00186     self.logger.info(
00187         f"Stored initial calibration to '{json_name}' "
00188         f"(intrinsics for {len(intr)} cams, extrinsics for {len(extr)} cams, bundleAdjustf:
00189 {meta.get('bundleAdjust', True)})."
00190     )
00191     self.logger.info(f"HealthScore = {score:.2f} (method=initial)")
00192
00193     # Update baseline (creates baseline on first run)
00194     self.update_baseline(out, score, method="initial")
00195
00196     return score
00197
00198 def logRecalibration(self, combined_json_path: str, meta=None):
00199     """
00200     @brief Store LiFCal recalibration result with health score.
00201
00202     @param combined_json_path Path to the combined.json file from LiFCal
00203     @param meta Optional dictionary with metadata about the recalibration run
00204     @return Health score as float
00205     """
00206     if meta is None:
00207         meta = {}
00208
00209     # Import locally to avoid errors when not needed
00210     from HealthMonitor import compute_lifcal_health_from_combined_path
00211
00212     # Read combined.json
00213     try:
00214         with open(combined_json_path, "r", encoding="utf-8") as f:
00215             combined = json.load(f)
00216     except Exception as e:
00217         self.logger.error(f"Could not read combined.json '{combined_json_path}': {e}")
00218         return 0.0
00219
00220     # Compute health score
00221     try:
00222         report = compute_lifcal_health_from_combined_path(combined_json_path)
00223         score = float(report.get("health", 0.0))
00224     except Exception as e:
00225         self.logger.error(f"LiFCal HealthScore computation failed: {e}", exc_info=True)
00226         score = 0.0
00227
00228     # Create timestamped copy
00229     timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
00230     json_name = f"calibration_lifcal_{timestamp}.json"
00231     out_path = os.path.join(self.baseDir, json_name)
00232
00233     combined.setdefault("meta", {})
00234     combined["meta"].update(self.to_serializable(meta))
00235     combined["meta"]["health"] = score
00236     combined["meta"]["method"] = "lifcal"
00237     combined["meta"]["runType"] = combined["meta"].get("runType", "recalib")
00238
00239     # Save to file
00240     try:
00241         with open(out_path, "w", encoding="utf-8") as f:
00242             json.dump(combined, f, indent=2)
00243     except Exception as e:

```

```

00243         self.logger.error(f"Failed to write LiFCal result '{out_path}': {e}")
00244         return 0.0
00245
00246     num_cams = len(combined.get("state", {}).get("parameters", {}))
00247     self.logger.info(f"Stored lifcal recalibration to '{json_name}' (cams={num_cams}).")
00248     self.logger.info(f"HealthScore = {score:.2f} (method=lifcal)")
00249
00250     # Update baseline only if better
00251     self.update_baseline(combined, score, method="lifcal")
00252
00253     return score

```

6.13 calibrationLFC/results/evaluate_calibration.py File Reference

Namespaces

- namespace [evaluate_calibration](#)

Functions

- [evaluate_calibration.rot_angle_deg](#) (R)
- [evaluate_calibration.as_vec3](#) (v)
- [evaluate_calibration.compute_metrics_for_file](#) (Path calib_path, Path gt_path)
- [evaluate_calibration.print_per_file_table](#) (results_by_name)
- [evaluate_calibration.print_scenario_table](#) (results_by_name)
- [evaluate_calibration.main](#) ()

Variables

- [evaluate_calibration.THIS_DIR](#) = Path(__file__).resolve().parent
- [evaluate_calibration.CALIB_LFC_DIR](#) = THIS_DIR.parent
- [evaluate_calibration.PROJECT_ROOT](#) = CALIB_LFC_DIR.parent
- str [evaluate_calibration.RESULTS_DIR](#) = [CALIB_LFC_DIR](#) / "results"
- str [evaluate_calibration.GT_DIR](#) = [PROJECT_ROOT](#) / "TestEnvironment" / "params"
- list [evaluate_calibration.CAM_IDS](#)
- dict [evaluate_calibration.CALIB_FILES](#)
- dict [evaluate_calibration.GT_FILES](#)
- dict [evaluate_calibration.SCENARIOS](#)

6.14 evaluate_calibration.py

[Go to the documentation of this file.](#)

```

00001 import json
00002 import numpy as np
00003 from pathlib import Path
00004
00005 # Paths
00006 THIS_DIR = Path(__file__).resolve().parent
00007 CALIB_LFC_DIR = THIS_DIR.parent # calibrationLFC/
00008 PROJECT_ROOT = CALIB_LFC_DIR.parent # CameraCalibration/
00009 RESULTS_DIR = CALIB_LFC_DIR / "results"
00010
00011 # Groundtruth base directory
00012 GT_DIR = PROJECT_ROOT / "TestEnvironment" / "params"
00013
00014 # Camera order
00015 CAM_IDS = [
00016     "Center",
00017     "Up1", "Up2", "Up3",

```

```

00018     "Down1", "Down2", "Down3",
00019     "Left1", "Left2", "Left3",
00020     "Right1", "Right2", "Right3",
00021 ]
00022
00023 # Calibration JSONs
00024 CALIB_FILES = {
00025     "imageset1": RESULTS_DIR / "calibration_initial_imageset1_20251215_173701.json",
00026     "imageset2": RESULTS_DIR / "calibration_initial_imageset2_20251215_181903.json",
00027     "imageset3": RESULTS_DIR / "calibration_initial_imageset3_20251215_185041.json",
00028     "imageset4": RESULTS_DIR / "calibration_initial_imageset4_20251215_165644.json",
00029     "imageset5": RESULTS_DIR / "calibration_initial_imageset5_20251217_011656.json",
00030     "imageset6": RESULTS_DIR / "calibration_initial_imageset6_20251215_201527.json",
00031     "imageset7": RESULTS_DIR / "calibration_initial_imageset7_20251215_203849.json",
00032 }
00033
00034 # Groundtruth per calibration
00035 GT_FILES = {
00036     # Rig0 for imageset1-4
00037     "imageset1": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00038     "imageset2": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00039     "imageset3": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00040     "imageset4": GT_DIR / "groundtruth_extrinsics_Rig0.json",
00041
00042     # Different rigs/sets:
00043     "imageset5": GT_DIR / "groundtruth_extrinsics_Rig1_set5.json",
00044     "imageset6": GT_DIR / "groundtruth_extrinsics_Rig2_set6.json",
00045     "imageset7": GT_DIR / "groundtruth_extrinsics_Rig3_set7.json",
00046 }
00047
00048 # Scenario grouping for thesis table
00049 SCENARIOS = {
00050     "A_Robustheit": ["imageset1", "imageset2", "imageset3", "imageset4"],
00051     "B_Generalisation": ["imageset5", "imageset6", "imageset7"],
00052 }
00053
00054 # Helper functions
00055 def rot_angle_deg(R):
00056     """Extract rotation angle (degrees) from 3x3 rotation matrix."""
00057     tr = np.trace(R)
00058     c = (tr - 1.0) / 2.0
00059     c = np.clip(c, -1.0, 1.0)
00060     return float(np.degrees(np.arccos(c)))
00061
00062 def as_vec3(v):
00063     """Convert any translation vector to robust 3-element vector."""
00064     arr = np.array(v, dtype=float).reshape(-1)
00065     if arr.size != 3:
00066         raise ValueError(f"Expected 3 entries for translation, got {arr.size}")
00067     return arr
00068
00069
00070 def compute_metrics_for_file(calib_path: Path, gt_path: Path):
00071     """
00072     Loads a calibration JSON and groundtruth JSON and computes:
00073     - numPoses
00074     - mean T (m)
00075     - mean R (°)
00076     - mean reprojection error (px, from intrinsics)
00077
00078     Definitions as in your extrinsic plot:
00079     R_cam = angle(R_est * R_gt^T)
00080     T_cam = || T_est - T_gt ||_2
00081     """
00082     with open(gt_path, "r", encoding="utf-8") as f:
00083         gt = json.load(f)
00084
00085     with open(calib_path, "r", encoding="utf-8") as f:
00086         calib = json.load(f)
00087
00088     meta = calib.get("meta", {})
00089     extr_est = calib["state"]["extrinsics"]
00090     intr_est = calib["state"]["intrinsics"]
00091
00092     bundle_adjust = bool(meta.get("bundleAdjust", False))
00093     num_poses = int(meta.get("numPoses", -1))
00094
00095     rot_err = []
00096     trans_err = []
00097     reproj_err = []
00098
00099     for cam in CAM_IDS:
00100         if cam not in extr_est or cam not in gt:
00101             print(f"[WARN] Camera '{cam}' not in both files for {calib_path.name}, skipping this
cam.")
00102             continue
00103

```

```

00104         # Extrinsics
00105         R_est = np.array(extr_est[cam]["rotationMatrix"], dtype=float)
00106         R_gt = np.array(gt[cam]["rotationMatrix"], dtype=float)
00107
00108         T_est = as_vec3(extr_est[cam]["translationVector"])
00109         T_gt = as_vec3(gt[cam]["translationVector"])
00110
00111         # Rotation error as in plot
00112         R_err = R_est @ R_gt.T
00113         ang = rot_angle_deg(R_err)
00114
00115         # Translation error (absolute, m)
00116         d_abs = float(np.linalg.norm(T_est - T_gt))
00117
00118         rot_err.append(ang)
00119         trans_err.append(d_abs)
00120
00121         # Intrinsic reprojection error
00122         if cam in intr_est:
00123             reproj_err.append(float(intr_est[cam]["reprojectionError"]))
00124
00125         # Mean values
00126         mean_dT = float(np.mean(trans_err)) if trans_err else float("nan")
00127         mean_dR = float(np.mean(rot_err)) if rot_err else float("nan")
00128         mean_reproj = float(np.mean(reproj_err)) if reproj_err else float("nan")
00129
00130         return {
00131             "bundleAdjust": bundle_adjust,
00132             "numPoses": num_poses,
00133             "mean_dT": mean_dT,
00134             "mean_dR": mean_dR,
00135             "mean_reproj": mean_reproj,
00136         }
00137
00138     # Table output (console)
00139     def print_per_file_table(results_by_name):
00140         """
00141         Prints one line per calibration JSON.
00142         """
00143         header = (
00144             f"{'Name':<12} "
00145             f"{'#Poses':>6} "
00146             f"{'BA':<4} "
00147             f"{'mean dT [m]':>12} "
00148             f"{'mean dR [deg]':>13} "
00149             f"{'mean reproj [px]':>16}"
00150         )
00151         print(header)
00152         print("-" * len(header))
00153
00154         for name, res in sorted(results_by_name.items()):
00155             ba_flag = "yes" if res["bundleAdjust"] else "no"
00156             print(
00157                 f"{name:<12} "
00158                 f"{res['numPoses']:6d} "
00159                 f"{ba_flag:<4} "
00160                 f"{res['mean_dT']:12.5f} "
00161                 f"{res['mean_dR']:12.5f} "
00162                 f"{res['mean_reproj']:16.5f}"
00163             )
00164
00165     def print_scenario_table(results_by_name):
00166         """
00167         Aggregates over scenarios (Robustness / Generalization)
00168         and prints a compact table.
00169         """
00170         print("\n\nScenario Summary:\n")
00171
00172         header = (
00173             f"{'Scenario':<22} "
00174             f"{'#Runs':>6} "
00175             f"{'Avg#Poses':>9} "
00176             f"{'mean dT [m]':>12} "
00177             f"{'mean dR [deg]':>13} "
00178             f"{'mean reproj [px]':>16}"
00179         )
00180         print(header)
00181         print("-" * len(header))
00182
00183         for scen_name, calib_names in SCENARIOS.items():
00184             dT_vals = []
00185             dR_vals = []
00186             reproj_vals = []
00187             pose_counts = []
00188
00189             for cname in calib_names:
00190                 res = results_by_name.get(cname)

```

```

00191         if res is None:
00192             continue
00193
00194         dT_vals.append(res["mean_dT"])
00195         dR_vals.append(res["mean_dR"])
00196         reproj_vals.append(res["mean_reproj"])
00197         pose_counts.append(res["numPoses"])
00198
00199     if not dT_vals:
00200         continue
00201
00202     mean_dT = float(np.mean(dT_vals))
00203     mean_dR = float(np.mean(dR_vals))
00204     mean_reproj = float(np.mean(reproj_vals))
00205     mean_poses = float(np.mean(pose_counts))
00206     num_runs = len(dT_vals)
00207
00208     print(
00209         f"{scen_name:<22} "
00210         f"{num_runs:6d} "
00211         f"{mean_poses:9.1f} "
00212         f"{mean_dT:12.5f} "
00213         f"{mean_dR:13.5f} "
00214         f"{mean_reproj:16.5f}"
00215     )
00216
00217 def main():
00218     results_by_name = {}
00219
00220     for name, calib_path in CALIB_FILES.items():
00221         if not calib_path.exists():
00222             print(f"[WARN] File not found: {calib_path}")
00223             continue
00224
00225         gt_path = GT_FILES.get(name)
00226         if gt_path is None:
00227             print(f"[WARN] No GT file mapped for {name}, skipping.")
00228             continue
00229         if not gt_path.exists():
00230             print(f"[WARN] GT file not found for {name}: {gt_path}")
00231             continue
00232
00233         metrics = compute_metrics_for_file(calib_path, gt_path)
00234         results_by_name[name] = metrics
00235
00236     if not results_by_name:
00237         print("[ERROR] No valid results computed. Check file paths.")
00238         return
00239
00240     print("Detailed results per calibration:\n")
00241     print_per_file_table(results_by_name)
00242
00243     print_scenario_table(results_by_name)
00244
00245
00246 if __name__ == "__main__":
00247     main()

```

6.15 calibrationLFC/run_initial_calibration.py File Reference

Namespaces

- namespace [run_initial_calibration](#)

Functions

- [run_initial_calibration.parse_args](#) ()
- [run_initial_calibration.main](#) (args)

Variables

- str [run_initial_calibration.DEFAULT_BASE_DIR](#) = "/data/calibrationLFC/sets/imageset5"
- int [run_initial_calibration.DEFAULT_NUM_POSES](#) = 28
- list [run_initial_calibration.CAM_IDS](#)
- [run_initial_calibration.repoRoot](#) = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))

6.16 run_initial_calibration.py

[Go to the documentation of this file.](#)

```

00001 """
00002 Test runner for initial calibration.
00003
00004 Usage:
00005     cd calibrationLFC
00006     python3 run_initial_calibration.py --base_dir /path/to/imageset --num_poses 28 --bundle-adjust
00007     python3 run_initial_calibration.py --base_dir /path/to/imageset --num_poses 28 --no-bundle-adjust
00008
00009 Performs validation checks (file existence, corner detection on pose 0)
00010 then runs the complete initial calibration pipeline. Outputs summary statistics
00011 for intrinsics, extrinsics, and health score.
00012 """
00013
00014 import argparse
00015 import os
00016 import sys
00017
00018 from ImageSet import ImageSet
00019 from Controller import Controller
00020
00021
00022 # Default configuration parameters
00023 DEFAULT_BASE_DIR = "/data/calibrationLFC/sets/imageset5"
00024 DEFAULT_NUM_POSES = 28
00025 CAM_IDS = [
00026     "Center",
00027     "Up1", "Up2", "Up3",
00028     "Down1", "Down2", "Down3",
00029     "Left1", "Left2", "Left3",
00030     "Right1", "Right2", "Right3",
00031 ]
00032
00033
00034 def parse_args():
00035     """
00036     @brief Parse command line arguments for calibration configuration.
00037
00038     @return Parsed arguments with base_dir, num_poses, and bundle_adjust settings
00039     """
00040     parser = argparse.ArgumentParser(description="Run initial camera calibration")
00041     parser.add_argument("--base_dir", default=DEFAULT_BASE_DIR, help="Path to imageset directory")
00042     parser.add_argument("--num_poses", type=int, default=DEFAULT_NUM_POSES, help="Number of poses in
the imageset")
00043     parser.add_argument(
00044         "--bundle-adjust",
00045         dest="bundle_adjust",
00046         action="store_true",
00047         help="Enable bundle adjustment (default)",
00048     )
00049     parser.add_argument(
00050         "--no-bundle-adjust",
00051         dest="bundle_adjust",
00052         action="store_false",
00053         help="Disable bundle adjustment",
00054     )
00055     parser.set_defaults(bundle_adjust=True)
00056     return parser.parse_args()
00057
00058
00059 def main(args):
00060     """
00061     @brief Run complete initial calibration pipeline with validation and summary output.
00062
00063     @param args
00064     """
00065     print("Creating ImageSet...")
00066     imageSet = ImageSet(args.base_dir, args.num_poses, CAM_IDS)
00067
00068     print("Instantiating Controller and InitialCalibration...")
00069     controller = Controller(bundleAdjust=args.bundle_adjust)
00070     logger = controller.resultLogger
00071
00072     print("\nQuick checks (pose 0): file exists / corners detected")
00073     for cam in CAM_IDS:
00074         path = imageSet.getImagePath(0, cam)
00075         exists = os.path.exists(path)
00076         cornersFound = None
00077
00078         try:
00079             cornersFound, __, __ = controller.initialCalibration.detectCorners(path)
00080         except Exception as e:
00081             cornersFound = f"error: {e.__class__.__name__}"

```

```

00082         print(f" {cam}: {path} - exists={exists} - corners={cornersFound}")
00083
00084     print("\nRunning initial calibration (this may take time depending on data)...")
00085     try:
00086         calibrationState = controller.runInitialCalibration(imageSet)
00087     except Exception as e:
00088         logger.logger.error(f"Calibration failed with exception: {e}", exc_info=True)
00089         print("Calibration failed, see calibration.log for details.")
00090         raise
00091
00092     print("\nCalibration finished. Attempting to print summary...")
00093     try:
00094         stateDict = calibrationState.__getState__()
00095     except Exception:
00096         stateDict = None
00097
00098     if stateDict is None:
00099         print("Could not extract a state dictionary.")
00100         return
00101
00102     intr = stateDict.get("intrinsics", {})
00103     extr = stateDict.get("extrinsics", {})
00104
00105     # Print calibration results per camera
00106     print("\nCalibration Results by Camera")
00107     print("=" * 50)
00108
00109     for cam in CAM_IDS:
00110         print(f"\n--- {cam} ---")
00111
00112         # Intrinsic parameters
00113         if cam in intr:
00114             data = intr[cam]
00115             K = data["cameraMatrix"]
00116             dist = data["distortionCoeffs"].ravel()
00117             err = data["reprojectionError"]
00118
00119             print("Intrinsics:")
00120             print(" K =")
00121             print(f" {K[0][0]:.3f} {K[0][1]:.3f} {K[0][2]:.3f}")
00122             print(f" {K[1][0]:.3f} {K[1][1]:.3f} {K[1][2]:.3f}")
00123             print(f" {K[2][0]:.3f} {K[2][1]:.3f} {K[2][2]:.3f}")
00124             print(" distCoeffs =", dist)
00125             print(f" reprojectionError = {err:.6f}")
00126         else:
00127             print("Intrinsics: (no data)")
00128
00129         # Extrinsic parameters
00130         if cam in extr:
00131             data = extr[cam]
00132             R = data["rotationMatrix"]
00133             T = data["translationVector"].ravel()
00134             rms = data["stereoRms"]
00135
00136             print("Extrinsics:")
00137             print(" R =")
00138             print(f" {R[0][0]:.5f} {R[0][1]:.5f} {R[0][2]:.5f}")
00139             print(f" {R[1][0]:.5f} {R[1][1]:.5f} {R[1][2]:.5f}")
00140             print(f" {R[2][0]:.5f} {R[2][1]:.5f} {R[2][2]:.5f}")
00141             print(" T =", T)
00142             print(f" stereoRMS = {rms:.6f}")
00143         else:
00144             print(" Extrinsics: (no data)")
00145
00146         # Print health score
00147         print("\nOverall Health Score")
00148         print("=" * 50)
00149         score = controller.selfHealthCheck(calibrationState)
00150         print(f"Global HealthScore = {score:.2f} / 100")
00151
00152 if __name__ == "__main__":
00153     # Set up Python path to find calibrationLFC modules
00154     repoRoot = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
00155     if repoRoot not in sys.path:
00156         sys.path.insert(0, repoRoot)
00157     main(parse_args())

```

6.17 calibrationLFC/run_periodic_lifcal.py File Reference

Namespaces

- namespace `run_periodic_lifcal`

Functions

- `run_periodic_lifcal.setup_logger()`
- `bool run_periodic_lifcal.acquire_lock(logger)`
- `run_periodic_lifcal.release_lock(logger)`
- `int run_periodic_lifcal.run_cmd(logger, cmd, cwd=None)`
- `int run_periodic_lifcal.count_images_in_dir(str d, exts=(".png", ".jpg", ".jpeg"))`
- `bool run_periodic_lifcal.needs_imageset_creation(logger)`
- `str run_periodic_lifcal.find_latest_run_dir(str out_root)`
- `str run_periodic_lifcal.combined_json_path(str run_dir)`
- `run_periodic_lifcal.try_log_lifcal_with_resultlogger(logger, str combined_path)`
- `run_periodic_lifcal.run_cycle(logger)`
- `run_periodic_lifcal.main()`

Variables

- `int run_periodic_lifcal.INTERVAL_SECONDS = 5 * 60`
- `str run_periodic_lifcal.RUN_IMAGESET_CREATION = "/data/external/LiFCal/LiFCal_Data/run_imageset_↵
creation.py"`
- `str run_periodic_lifcal.RUN_LIFCAL_CALIB = "/data/external/LiFCal/LiFCal_Data/run_LiFCal_calibration.py"`
- `str run_periodic_lifcal.LIFCAL_OUT_ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/Calibration↵
ByCamera"`
- `str run_periodic_lifcal.BASELINE_DIR = "/data/baseline"`
- `run_periodic_lifcal.LOG_PATH = os.path.join(BASELINE_DIR, "calibration.log")`
- `str run_periodic_lifcal.LOCK_PATH = "/tmp/lifcal_periodic.lock"`
- `str run_periodic_lifcal.ROOT_IMAGESET = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_↵
Imageset"`
- `run_periodic_lifcal.FOCUS_ROOT = os.path.join(ROOT_IMAGESET, "focus")`
- `run_periodic_lifcal.DEPTH_ROOT = os.path.join(ROOT_IMAGESET, "depth")`
- `list run_periodic_lifcal.CAMERA_IDS`
- `int run_periodic_lifcal.MIN_POSES_PER_CAM = 1`

6.18 run_periodic_lifcal.py

[Go to the documentation of this file.](#)

```
00001 import os
00002 import sys
00003 import time
00004 import json
00005 import logging
00006 import subprocess
00007 from datetime import datetime
00008
00009 """
00010 @brief Periodic LiFCal scheduler that prepares imagesets, runs calibration, and logs health.
00011
00012 """
00013
00014 INTERVAL_SECONDS = 5 * 60 # 5 Minutes interval
00015
00016 RUN_IMAGESET_CREATION = "/data/external/LiFCal/LiFCal_Data/run_imageset_creation.py"
00017 RUN_LIFCAL_CALIB = "/data/external/LiFCal/LiFCal_Data/run_LiFCal_calibration.py"
00018
00019 LIFCAL_OUT_ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/CalibrationByCamera"
00020 BASELINE_DIR = "/data/baseline"
00021 LOG_PATH = os.path.join(BASELINE_DIR, "calibration.log")
00022
00023 LOCK_PATH = "/tmp/lifcal_periodic.lock"
00024
00025 # Prepared images
00026 ROOT_IMAGESET = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
00027 FOCUS_ROOT = os.path.join(ROOT_IMAGESET, "focus") # focus/<CamId>/*.png
```



```

00028 DEPTH_ROOT      = os.path.join(ROOT_IMAGESET, "depth")    # depth/<CamId>/*_depth.png
00029
00030 CAMERA_IDS = [
00031     "Center", "Up1", "Up2", "Up3",
00032     "Down1", "Down2", "Down3",
00033     "Left1", "Left2", "Left3",
00034     "Right1", "Right2", "Right3"
00035 ]
00036
00037 # Minimum number of poses required per camera
00038 MIN_POSES_PER_CAM = 1
00039
00040
00041 def setup_logger():
00042     """
00043     @brief Create logger that writes to file and stdout.
00044
00045     @return Logger instance configured for calibration logging
00046     """
00047     os.makedirs(BASELINE_DIR, exist_ok=True)
00048     logger = logging.getLogger("lifcal_periodic")
00049     logger.setLevel(logging.INFO)
00050
00051     if not logger.handlers:
00052         fh = logging.FileHandler(LOG_PATH, encoding="utf-8")
00053         fh.setFormatter(logging.Formatter("%(asctime)s [%(levelname)s] %(message)s"))
00054         logger.addHandler(fh)
00055
00056         sh = logging.StreamHandler(sys.stdout)
00057         sh.setFormatter(logging.Formatter("%(asctime)s [%(levelname)s] %(message)s"))
00058         logger.addHandler(sh)
00059
00060     return logger
00061
00062
00063 def acquire_lock(logger) -> bool:
00064     """
00065     @brief Create lock file to prevent overlapping scheduler cycles.
00066
00067     @param logger Logger instance for status messages
00068     @return True if lock was acquired, False otherwise
00069     """
00070     if os.path.exists(LOCK_PATH):
00071         logger.warning(f"Lock exists, skipping this cycle: {LOCK_PATH}")
00072         return False
00073     try:
00074         with open(LOCK_PATH, "w") as f:
00075             f.write(str(os.getpid()))
00076         return True
00077     except Exception as e:
00078         logger.error(f"Could not create lock: {e}")
00079         return False
00080
00081
00082 def release_lock(logger):
00083     """
00084     @brief Remove lock file after a cycle completes.
00085
00086     @param logger Logger instance for error messages
00087     """
00088     try:
00089         if os.path.exists(LOCK_PATH):
00090             os.remove(LOCK_PATH)
00091     except Exception as e:
00092         logger.error(f"Could not remove lock: {e}")
00093
00094
00095 def run_cmd(logger, cmd, cwd=None) -> int:
00096     """
00097     @brief Run a subprocess, log combined stdout/stderr, and return exit code.
00098
00099     @param logger Logger instance for command output
00100     @param cmd Command as list of strings
00101     @param cwd Working directory for the command (default: None)
00102     @return Exit code of the subprocess
00103     """
00104     logger.info(f"RUN: {' '.join(cmd)}")
00105     try:
00106         p = subprocess.run(
00107             cmd,
00108             cwd=cwd,
00109             stdout=subprocess.PIPE,
00110             stderr=subprocess.STDOUT,
00111             text=True
00112         )
00113         out = p.stdout or ""
00114         if len(out) > 5000:

```

```

00115         logger.info(out[:5000] + "\n... (truncated) ...")
00116     else:
00117         logger.info(out)
00118     return int(p.returncode)
00119 except Exception as e:
00120     logger.error(f"Command failed: {e}", exc_info=True)
00121     return 1
00122
00123
00124 def count_images_in_dir(d: str, exts=(".png", ".jpg", ".jpeg")) -> int:
00125     """
00126     @brief Count images in directory matching expected extensions.
00127
00128     @param d Directory path to scan
00129     @param exts Tuple of file extensions to match (default: (".png", ".jpg", ".jpeg"))
00130     @return Number of matching image files
00131     """
00132     if not os.path.isdir(d):
00133         return 0
00134     cnt = 0
00135     for fn in os.listdir(d):
00136         if fn.lower().endswith(exts):
00137             cnt += 1
00138     return cnt
00139
00140
00141 def needs_imageset_creation(logger) -> bool:
00142     """
00143     @brief Check if imageset exists and has enough images per camera; trigger creation if not.
00144
00145     @param logger Logger instance for status messages
00146     @return True if imageset needs to be created, False if it already exists
00147     """
00148     if not os.path.isdir(ROOT_IMAGESET):
00149         logger.info("Imageset missing: ROOT_IMAGESET not found.")
00150         return True
00151     if not os.path.isdir(FOCUS_ROOT) or not os.path.isdir(DEPTH_ROOT):
00152         logger.info("Imageset missing: focus/depth folders not found.")
00153         return True
00154
00155     for cam in CAMERA_IDS:
00156         fdir = os.path.join(FOCUS_ROOT, cam)
00157         ddir = os.path.join(DEPTH_ROOT, cam)
00158
00159         fcnt = count_images_in_dir(fdir, exts=(".png", ".jpg", ".jpeg"))
00160         dcnt = count_images_in_dir(ddir, exts=(".png", ".jpg", ".jpeg"))
00161
00162         if fcnt < MIN_POSES_PER_CAM:
00163             logger.info(f"Imageset not ready: focus/{cam} has {fcnt} images (<{MIN_POSES_PER_CAM}).")
00164             return True
00165         if dcnt < MIN_POSES_PER_CAM:
00166             logger.info(f"Imageset not ready: depth/{cam} has {dcnt} images (<{MIN_POSES_PER_CAM}).")
00167             return True
00168     logger.info("Imageset looks ready. Skipping run_imageset_creation.py.")
00169     return False
00170
00171
00172 def find_latest_run_dir(out_root: str) -> str:
00173     """
00174     @brief Return latest run_* directory inside output root, or empty string if none.
00175
00176     @param out_root Root directory containing run_* folders
00177     @return Path to the latest run directory, or empty string
00178     """
00179     if not os.path.isdir(out_root):
00180         return ""
00181     runs = []
00182     for name in os.listdir(out_root):
00183         p = os.path.join(out_root, name)
00184         if os.path.isdir(p) and name.startswith("run_"):
00185             runs.append(p)
00186     if not runs:
00187         return ""
00188     runs.sort(key=lambda p: os.path.getmtime(p))
00189     return runs[-1]
00190
00191
00192 def combined_json_path(run_dir: str) -> str:
00193     """
00194     @brief Return path to combined.json inside run directory if present.
00195
00196     @param run_dir Path to run directory
00197     @return Path to combined.json file, or empty string if not found
00198     """
00199     if not run_dir:
00200         return ""
00201     p = os.path.join(run_dir, "combined.json")

```

```

00202     return p if os.path.isfile(p) else ""
00203
00204
00205 def try_log_lifcal_with_resultlogger(logger, combined_path: str):
00206     """
00207     @brief Use ResultLogger to record LiFCal health score and update baseline.
00208
00209     @param logger Logger instance for status messages
00210     @param combined_path Path to combined.json from LiFCal
00211     @return True if logging succeeded, False otherwise
00212     """
00213     try:
00214         repo_dir = "/data/calibrationLFC"
00215         if repo_dir not in sys.path:
00216             sys.path.insert(0, repo_dir)
00217
00218         from ResultLogger import ResultLogger
00219         rl = ResultLogger(baseDir=BASELINE_DIR)
00220
00221         meta = {
00222             "runType": "recalib",
00223             "source": "periodic_scheduler",
00224             "timestamp": datetime.now().isoformat(timespec="seconds"),
00225         }
00226         score = rl.logRecalibration(combined_path, meta=meta)
00227         logger.info(f"LiFCal HealthScore logged via ResultLogger: {score:.2f}")
00228         return True
00229     except Exception as e:
00230         logger.warning(f"Could not use ResultLogger for LiFCal logging: {e}")
00231         return False
00232
00233
00234 def run_cycle(logger):
00235     """
00236     @brief Execute one scheduler cycle: prepare imageset, run LiFCal, log health.
00237
00238     @param logger Logger instance for cycle logging
00239     """
00240     # 1) Only prepare imageset if needed
00241     if needs_imageset_creation(logger):
00242         if not os.path.isfile(RUN_IMAGESET_CREATION):
00243             logger.error(f"Missing: {RUN_IMAGESET_CREATION}")
00244             return
00245
00246         rc = run_cmd(logger, [sys.executable, RUN_IMAGESET_CREATION])
00247         if rc != 0:
00248             logger.error(f"run_imageset_creation failed (rc={rc}). Skipping LiFCal.")
00249             return
00250
00251     # 2) LiFCal calibration
00252     if not os.path.isfile(RUN_LIFCAL_CALIB):
00253         logger.error(f"Missing: {RUN_LIFCAL_CALIB}")
00254         return
00255
00256     rc = run_cmd(logger, [sys.executable, RUN_LIFCAL_CALIB])
00257     if rc != 0:
00258         logger.error(f"run_LiFCal_calibration failed (rc={rc}).")
00259         return
00260
00261     # 3) locate combined.json (latest run)
00262     latest = find_latest_run_dir(LIFCAL_OUT_ROOT)
00263     cpath = combined_json_path(latest)
00264
00265     if not cpath:
00266         logger.warning("No combined.json found after LiFCal run.")
00267         return
00268
00269     logger.info(f"Latest combined.json: {cpath}")
00270
00271     # 4) log health + baseline update using your ResultLogger
00272     ok = try_log_lifcal_with_resultlogger(logger, cpath)
00273     if not ok:
00274         logger.info("LiFCal finished, but no baseline/health logging was executed.")
00275
00276
00277 def main():
00278     """
00279     @brief Periodic loop that runs LiFCal recalibration on a schedule.
00280     """
00281     logger = setup_logger()
00282     logger.info("Starting periodic LiFCal scheduler.")
00283     logger.info(f"Interval: {INTERVAL_SECONDS}s")
00284     logger.info(f"Lock: {LOCK_PATH}")
00285
00286     while True:
00287         start = time.time()
00288

```

```

00289         if acquire_lock(logger):
00290             try:
00291                 logger.info("=== CYCLE START ===")
00292                 run_cycle(logger)
00293                 logger.info("=== CYCLE END ===")
00294             finally:
00295                 release_lock(logger)
00296
00297         elapsed = time.time() - start
00298         sleep_s = max(1.0, INTERVAL_SECONDS - elapsed)
00299         logger.info(f"Sleeping {sleep_s:.1f}s\n")
00300         time.sleep(sleep_s)
00301
00302
00303 if __name__ == "__main__":
00304     main()

```

6.19 external/LiFCal/LiFCal_Data/depthmap.py File Reference

Namespaces

- namespace [depthmap](#)

Functions

- [depthmap.build_matchers\(\)](#)
- bool [depthmap.imwrite_u16](#) (str path, np.ndarray img_u16)
- np.ndarray [depthmap.disparity](#) (np.ndarray left_gray, np.ndarray right_gray)
- np.ndarray [depthmap.to_u16_from_disp](#) (np.ndarray disp)
- np.ndarray [depthmap.fuse_disparities_median](#) (list disp_list)
- Dict[str, np.ndarray] [depthmap.generate_depthmaps_for_pose](#) (Dict[str, str] cam_to_focus_path)

Variables

- [depthmap._LEFT_MATCHER](#)
- [depthmap._RIGHT_MATCHER](#)
- [depthmap._WLS](#)

6.20 depthmap.py

[Go to the documentation of this file.](#)

```

00001 import os
00002 import cv2
00003 import numpy as np
00004 from typing import Optional, Dict
00005
00006 """
00007 @brief Depth map generation for LiFCal calibration using stereo matching and fusion.
00008
00009 """
00010
00011 def build_matchers():
00012     """
00013     @brief Build stereo matchers with SGBM and optional Weighted Least Squares (WLS) filtering.
00014
00015     @return Tuple (left_matcher, right_matcher, wls_filter)
00016     """
00017     left_matcher = cv2.StereoSGBM_create(
00018         minDisparity=0,
00019         numDisparities=128, # must be divisible by 16
00020         blockSize=7,
00021         P1=8 * 7 * 7,

```

```

00022         P2=32 * 7 * 7,
00023         uniquenessRatio=5,
00024         speckleWindowSize=100,
00025         speckleRange=2,
00026         disp12MaxDiff=1,
00027         preFilterCap=31,
00028         mode=cv2.STEREO_SGBM_MODE_SGBM_3WAY
00029     )
00030
00031     right_matcher = None
00032     wls = None
00033
00034     try:
00035         if hasattr(cv2, "ximgproc"):
00036             right_matcher = cv2.ximgproc.createRightMatcher(left_matcher)
00037             wls = cv2.ximgproc.createDisparityWLSFilter(left_matcher)
00038             wls.setLambda(6000)
00039             wls.setSigmaColor(1.0)
00040         except Exception:
00041             right_matcher = None
00042             wls = None
00043
00044     return left_matcher, right_matcher, wls
00045
00046
00047 _LEFT_MATCHER, _RIGHT_MATCHER, _WLS = build_matchers()
00048
00049
00050 def imwrite_ul6(path: str, img_ul6: np.ndarray) -> bool:
00051     """
00052     @brief Write uint16 png, create dirs automatically.
00053
00054     @param path Output file path
00055     @param img_ul6 Image array in uint16 format
00056     @return True if write succeeded, False otherwise
00057     """
00058     os.makedirs(os.path.dirname(path), exist_ok=True)
00059     if img_ul6.dtype != np.uint16:
00060         img_ul6 = img_ul6.astype(np.uint16)
00061     return bool(cv2.imwrite(path, img_ul6))
00062
00063
00064 def disparity(left_gray: np.ndarray, right_gray: np.ndarray) -> np.ndarray:
00065     """
00066     @brief Compute stereo disparity map with optional WLS filtering.
00067
00068     @param left_gray Left grayscale image
00069     @param right_gray Right grayscale image
00070     @return Float32 disparity map with NaNs for invalid pixels (<=0)
00071     """
00072     dl = _LEFT_MATCHER.compute(left_gray, right_gray) # int16 scaled by 16
00073
00074     if _RIGHT_MATCHER is not None and _WLS is not None:
00075         dr = _RIGHT_MATCHER.compute(right_gray, left_gray)
00076         d = _WLS.filter(dl, left_gray, None, dr)
00077     else:
00078         d = dl
00079
00080     d = d.astype(np.float32) / 16.0
00081     d[d <= 0] = np.nan
00082     return d
00083
00084
00085 def to_ul6_from_disp(disparity: np.ndarray) -> np.ndarray:
00086     """
00087     @brief Convert disparity to uint16 depth map to be compatible with LiFCal.
00088
00089     @param disp Float32 disparity map
00090     @return Uint16 depth map
00091     """
00092     disp = np.nan_to_num(disp, nan=0.0, posinf=0.0, neginf=0.0)
00093
00094     mx = float(np.percentile(disp, 99))
00095     if not np.isfinite(mx) or mx <= 1e-6:
00096         return np.zeros(disp.shape, dtype=np.uint16)
00097
00098     disp = np.clip(disp, 0.0, mx)
00099     depth16 = (disp / mx * 65535.0).astype(np.uint16)
00100     return depth16
00101
00102
00103 def fuse_disparities_median(disp_list: list) -> np.ndarray:
00104     """
00105     @brief Fuse multiple disparity maps robustly using median.
00106
00107     @param disp_list List of float32 disparity maps to fuse
00108     @return Float32 fused disparity map with 0 where no valid data exists

```

```

00109     """
00110     stack = np.stack(disparity_list, axis=0) # (K,H,W)
00111
00112     # Locations where at least one value is valid
00113     valid_any = np.any(~np.isnan(stack), axis=0)
00114
00115     # Initialize result with zeros
00116     disp_final = np.zeros(stack.shape[1:], dtype=np.float32)
00117
00118     # Compute median only where we have valid data
00119     if np.any(valid_any):
00120         disp_final[valid_any] = np.nanmedian(stack[:, valid_any], axis=0)
00121
00122     # Remove non-positive disparities
00123     disp_final[disp_final <= 0] = 0.0
00124     return disp_final
00125
00126
00127 def generate_depthmaps_for_pose(cam_to_focus_path: Dict[str, str]) -> Dict[str, np.ndarray]:
00128     """
00129     @brief Generate depth maps for all cameras in a multi-camera pose.
00130
00131     @param cam_to_focus_path Dictionary mapping camera IDs to image file paths
00132     @return Dictionary mapping camera IDs to uint16 depth maps
00133     """
00134
00135     # 1) Load images
00136     imgs = {}
00137     for cam, path in cam_to_focus_path.items():
00138         im = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
00139         if im is None:
00140             continue
00141         imgs[cam] = im
00142
00143     if len(imgs) < 2:
00144         return {}
00145
00146     # helper
00147     def opposite_name(name: str) -> Optional[str]:
00148         if name.startswith("Left"):
00149             return name.replace("Left", "Right", 1)
00150         if name.startswith("Right"):
00151             return name.replace("Right", "Left", 1)
00152         if name.startswith("Up"):
00153             return name.replace("Up", "Down", 1)
00154         if name.startswith("Down"):
00155             return name.replace("Down", "Up", 1)
00156         return None
00157
00158     depthmaps = {}
00159
00160     for cam, img in imgs.items():
00161         disparity_list = []
00162
00163         if cam == "Center":
00164             # Center uses all other views
00165             for other_cam, other_img in imgs.items():
00166                 if other_cam == "Center":
00167                     continue
00168                 d = disparity(img, other_img)
00169                 # Keep only if there is any valid disparity
00170                 if np.any(~np.isnan(d)):
00171                     disparity_list.append(d)
00172
00173         else:
00174             # Opposite direction if available
00175             opp = opposite_name(cam)
00176             if opp is not None and opp in imgs:
00177                 if cam.startswith("Right") or cam.startswith("Down"):
00178                     d = disparity(imgs[opp], img)
00179                 else:
00180                     d = disparity(img, imgs[opp])
00181
00182                 if np.any(~np.isnan(d)):
00183                     disparity_list.append(d)
00184
00185             # Add Center view
00186             if "Center" in imgs:
00187                 d = disparity(img, imgs["Center"])
00188                 if np.any(~np.isnan(d)):
00189                     disparity_list.append(d)
00190
00191         if not disparity_list:
00192             # No valid disparity found; skip
00193             continue
00194
00195     # Robust fusion

```

```

00196         disp_final = fuse_disparities_median(disp_list)
00197
00198         # Convert to uint16 depth map
00199         depth16 = to_ul6_from_disp(disp_final)
00200         depthmaps[cam] = depth16
00201
00202     return depthmaps

```

6.21 external/LiFCal/LiFCal_Data/lifcal_to_json.py File Reference

Namespaces

- namespace [lifcal_to_json](#)

Functions

- dict [lifcal_to_json.parse_protocol](#) (str protocol_path)
- Dict[str, Any] [lifcal_to_json.parse_extrinsics_xml](#) (str xml_path)
- str [lifcal_to_json.export_lifcal_parameters_json_in_place](#) (str result_dir, str cam_id, str image_dir, str run_name="recalib", float pixel_size_mm_fallback=0.0055, str protocol_name="calibrationProtocol.txt", str extrinsic_name="extrinsicOrientations.xml", str out_name="parameters.json", Optional[float] pixel_size_mm=None, Optional[float] fallback_cx=None, Optional[float] fallback_cy=None)
- str [lifcal_to_json.build_combined_parameters_json](#) (str run_root, list camera_ids, str out_name="combined.json")

6.22 lifcal_to_json.py

[Go to the documentation of this file.](#)

```

00001 import os
00002 import re
00003 import json
00004 import xml.etree.ElementTree as ET
00005 from typing import Dict, Any, Optional
00006
00007 import numpy as np
00008 import cv2
00009
00010 """
00011 @brief code to parse LiFCal calibration protocol and extrinsic XML files, and export parameters.json
00012 for each camera, as well as a combined.json for all cameras. This is used to convert LiFCal's output
00013 into a structured JSON format that can be easily consumed for further analysis or processing.
00014
00015 def parse_protocol(protocol_path: str) -> dict:
00016     """
00017     @brief Parse LiFCal calibration protocol text file.
00018
00019     @return Dictionary containing parsed calibration parameters and reprojection errors
00020     """
00021     txt = open(protocol_path, "r", encoding="utf-8", errors="ignore").read()
00022
00023     def find_float(pattern: str, default=None) -> Optional[float]:
00024         m = re.search(pattern, txt, re.MULTILINE)
00025         if not m:
00026             return default
00027         try:
00028             return float(m.group(1))
00029         except Exception:
00030             return default
00031
00032     pixel_size = find_float(r"Pixel Size:\s*([0-9.+-eE+])\s*mm", None)
00033
00034     fL = find_float(r"^\s*fL\s*:\s*([0-9.+-eE+])\s*$", None)
00035     bL0 = find_float(r"^\s*bL0\s*:\s*([0-9.+-eE+])\s*$", None)
00036     B = find_float(r"^\s*B\s*:\s*([0-9.+-eE+])\s*$", None)
00037     cx = find_float(r"^\s*cX\s*:\s*([0-9.+-eE+])\s*$", None)

```

```

00038     cy = find_float(r"^\\s*cy\\s*:\\s*([0-9.+-eE]+)\\s*$", None)
00039
00040     a0 = find_float(r"^\\s*a0\\s*:\\s*([0-9.+-eE]+)\\s*$", 0.0)
00041     a1 = find_float(r"^\\s*a1\\s*:\\s*([0-9.+-eE]+)\\s*$", 0.0)
00042     b0 = find_float(r"^\\s*b0\\s*:\\s*([0-9.+-eE]+)\\s*$", 0.0)
00043     b1 = find_float(r"^\\s*b1\\s*:\\s*([0-9.+-eE]+)\\s*$", 0.0)
00044
00045     x_std = find_float(r"std\\.\\s*Dev\\.\\s*x:\\s*([0-9.+-eE]+)", None)
00046     y_std = find_float(r"std\\.\\s*Dev\\.\\s*y:\\s*([0-9.+-eE]+)", None)
00047     x_mae = find_float(r"mae\\s*x:\\s*([0-9.+-eE]+)", None)
00048     y_mae = find_float(r"mae\\s*y:\\s*([0-9.+-eE]+)", None)
00049
00050     return {
00051         "pixelSizeMm": pixel_size,
00052         "fL_mm": fL,
00053         "bL0_mm": bL0,
00054         "B_mm": B,
00055         "cx_px": cx,
00056         "cy_px": cy,
00057         "a0": a0,
00058         "a1": a1,
00059         "b0": b0,
00060         "b1": b1,
00061         "reprojection": {
00062             "x_std": x_std,
00063             "y_std": y_std,
00064             "x_mae": x_mae,
00065             "y_mae": y_mae
00066         }
00067     }
00068
00069
00070 def parse_extrinsics_xml(xml_path: str) -> Dict[str, Any]:
00071     """
00072     @brief Parse LiFCal extrinsic orientations XML file.
00073
00074     @param xml_path Path to extrinsicOrientations.xml file
00075     @return Dictionary mapping pose keys to rotation and translation data
00076     """
00077     root = ET.parse(xml_path).getroot()
00078     out: Dict[str, Any] = {}
00079
00080     for frame in root.findall("Frame"):
00081         fid = int(frame.attrib.get("id"))
00082         pose_key = "pose%03d" % fid
00083
00084         rot = frame.find("Rotation")
00085         tra = frame.find("Translation")
00086
00087         rvec = [0.0, 0.0, 0.0]
00088         tvec = [0.0, 0.0, 0.0]
00089
00090         if rot is not None:
00091             for c in rot.findall("Coeff"):
00092                 i = int(c.attrib["i"])
00093                 rvec[i] = float(c.text.strip())
00094
00095         if tra is not None:
00096             for c in tra.findall("Coeff"):
00097                 i = int(c.attrib["i"])
00098                 tvec[i] = float(c.text.strip())
00099
00100         rvec_np = np.array(rvec, dtype=np.float64).reshape(3, 1)
00101         R, _ = cv2.Rodrigues(rvec_np)
00102
00103         out[pose_key] = {
00104             "rotationMatrix": R.astype(float).tolist(),
00105             "rotationVector": [[float(rvec[0])], [float(rvec[1])], [float(rvec[2])]],
00106             "translationVector": [[float(tvec[0])], [float(tvec[1])], [float(tvec[2])]],
00107         }
00108
00109     return out
00110
00111
00112 def export_lifcal_parameters_json_in_place(
00113     result_dir: str,
00114     cam_id: str,
00115     image_dir: str,
00116     run_type: str = "recalib",
00117     pixel_size_mm_fallback: float = 0.0055,
00118     protocol_name: str = "calibrationProtocol.txt",
00119     extr_name: str = "extrinsicOrientations.xml",
00120     out_name: str = "parameters.json",
00121     # Aliases (kept for runner compatibility)
00122     pixel_size_mm: Optional[float] = None,
00123     fallback_cx: Optional[float] = None, # accepted for compatibility, not used
00124     fallback_cy: Optional[float] = None, # accepted for compatibility, not used

```



```

00125 ) -> str:
00126     """
00127     @brief Write LiFCal parameters.json for a single camera.
00128
00129     @param result_dir Directory containing LiFCal output files
00130     @param cam_id Camera identifier
00131     @param image_dir Directory containing input images
00132     @param run_type Type of calibration run (default: "recalib")
00133     @param pixel_size_mm_fallback Fallback pixel size in mm (default: 0.0055)
00134     @param protocol_name Protocol filename (default: "calibrationProtocol.txt")
00135     @param extr_name Extrinsic filename (default: "extrinsicOrientations.xml")
00136     @param out_name Output JSON filename (default: "parameters.json")
00137     @param pixel_size_mm Optional pixel size override
00138     @param fallback_cx Optional fallback cx (compatibility only, not used)
00139     @param fallback_cy Optional fallback cy (compatibility only, not used)
00140     @return Path to written parameters.json file
00141     """
00142     if pixel_size_mm is not None:
00143         pixel_size_mm_fallback = float(pixel_size_mm)
00144
00145     protocol_path = os.path.join(result_dir, protocol_name)
00146     extr_path = os.path.join(result_dir, extr_name)
00147
00148     if not os.path.isfile(protocol_path):
00149         raise FileNotFoundError(protocol_path)
00150     if not os.path.isfile(extr_path):
00151         raise FileNotFoundError(extr_path)
00152
00153     protocol = parse_protocol(protocol_path)
00154     extr_by_pose = parse_extrinsics_xml(extr_path)
00155
00156     # Ensure pixel size is available
00157     px = protocol.get("pixelSizeMm", None)
00158     if px is None or (not np.isfinite(px)) or px <= 0:
00159         protocol["pixelSizeMm"] = float(pixel_size_mm_fallback)
00160
00161     # Compact reprojection error (average MAE)
00162     rep = protocol.get("reprojection", {})
00163     x_mae = rep.get("x_mae", None)
00164     y_mae = rep.get("y_mae", None)
00165     avg_mae = None
00166     if x_mae is not None and y_mae is not None and np.isfinite(x_mae) and np.isfinite(y_mae):
00167         avg_mae = float((x_mae + y_mae) / 2.0)
00168
00169     data = {
00170         "meta": {
00171             "runType": run_type,
00172             "numPoses": int(len(extr_by_pose)),
00173             "imageDir": image_dir,
00174             "cameraIds": [cam_id],
00175             "bundleAdjust": True,
00176         },
00177         "state": {
00178             "parameters": {
00179                 cam_id: {
00180                     "pixelSizeMm": protocol.get("pixelSizeMm"),
00181                     "fL_mm": protocol.get("fL_mm"),
00182                     "bL0_mm": protocol.get("bL0_mm"),
00183                     "B_mm": protocol.get("B_mm"),
00184                     "cx_px": protocol.get("cx_px"),
00185                     "cy_px": protocol.get("cy_px"),
00186                     "a0": protocol.get("a0"),
00187                     "a1": protocol.get("a1"),
00188                     "b0": protocol.get("b0"),
00189                     "b1": protocol.get("b1"),
00190                     "reprojection": rep,
00191                     "reprojectionAvgMae": avg_mae,
00192                 }
00193             },
00194             "extrinsicsByPose": {cam_id: extr_by_pose},
00195             "timeStamp": None
00196         }
00197     }
00198
00199     out_path = os.path.join(result_dir, out_name)
00200     with open(out_path, "w", encoding="utf-8") as f:
00201         json.dump(data, f, indent=2)
00202
00203     return out_path
00204
00205
00206 def build_combined_parameters_json(run_root: str, camera_ids: list, out_name: str = "combined.json")
-> str:
00207     """
00208     @brief Combine per-camera parameters.json files into a single combined.json.
00209
00210     @param run_root Root directory containing per-camera subdirectories

```

```

00211     @param camera_ids List of camera identifiers
00212     @param out_name Output filename (default: "combined.json")
00213     @return Path to written combined.json file
00214     """
00215     combined = {
00216         "meta": {
00217             "runType": "combined",
00218             "cameraIds": camera_ids,
00219         },
00220         "state": {
00221             "parameters": {},
00222             "extrinsicsByPose": {},
00223             "timeStamp": None
00224         }
00225     }
00226
00227     found_any = False
00228
00229     for cam in camera_ids:
00230         cam_dir = os.path.join(run_root, cam)
00231         jpath = os.path.join(cam_dir, "parameters.json")
00232
00233         if not os.path.isfile(jpath):
00234             continue
00235
00236         with open(jpath, "r", encoding="utf-8") as f:
00237             data = json.load(f)
00238
00239         st = data.get("state", {})
00240         params = st.get("parameters", {})
00241         extr_pose = st.get("extrinsicsByPose", {})
00242
00243         if cam in params:
00244             combined["state"]["parameters"][cam] = params[cam]
00245             found_any = True
00246
00247         if cam in extr_pose:
00248             combined["state"]["extrinsicsByPose"][cam] = extr_pose[cam]
00249             found_any = True
00250
00251     if not found_any:
00252         raise RuntimeError("No parameters.json found to combine.")
00253
00254     out_path = os.path.join(run_root, out_name)
00255     with open(out_path, "w", encoding="utf-8") as f:
00256         json.dump(combined, f, indent=2)
00257
00258     return out_path

```

6.23 external/LiFCal/LiFCal_Data/run_imageset_creation.py File Reference

Namespaces

- namespace [run_imageset_creation](#)

Functions

- [run_imageset_creation.ensure_camera_subfolders](#) (str base_dir)
- [run_imageset_creation.group_focus_images_by_pose](#) ()
- [run_imageset_creation.copy_focus_into_camera_folders](#) (str pose, dict cam_to_path)
- [run_imageset_creation.cleanup_focus_root](#) ()
- [run_imageset_creation.main](#) ()

Variables

- str [run_imageset_creation.ROOT](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
- [run_imageset_creation.FOCUS_DIR](#) = os.path.join([ROOT](#), "focus")
- [run_imageset_creation.DEPTH_DIR](#) = os.path.join([ROOT](#), "depth")
- bool [run_imageset_creation.CLEAN_FOCUS_ROOT_AFTER](#) = True
- list [run_imageset_creation.CAMERA_IDS](#)
- [run_imageset_creation.PATTERN](#)

6.24 run_imageset_creation.py

[Go to the documentation of this file.](#)

```

00001 import os
00002 import re
00003 import shutil
00004 from collections import defaultdict
00005 import depthmap
00006
00007 """
00008 @brief Organizes focus images by camera and generates depth maps for LiFCal recalibration.
00009
00010 """
00011
00012 ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
00013 FOCUS_DIR = os.path.join(ROOT, "focus")
00014 DEPTH_DIR = os.path.join(ROOT, "depth")
00015
00016 # If True: original files in focus root directory are deleted after sorting
00017 CLEAN_FOCUS_ROOT_AFTER = True
00018
00019 CAMERA_IDS = [
00020     "Center", "Up1", "Up2", "Up3",
00021     "Down1", "Down2", "Down3",
00022     "Left1", "Left2", "Left3",
00023     "Right1", "Right2", "Right3"
00024 ]
00025
00026 # Expected filenames: pose000_Center.png ... pose000_Down3.png
00027 PATTERN = re.compile(
00028     r"^(pose\d+)_([Center|Up|Down|Left|Right][123])\.([png|jpg|jpeg])$",
00029     re.IGNORECASE
00030 )
00031
00032 def ensure_camera_subfolders(base_dir: str):
00033     """
00034     @brief Create subdirectory for each camera ID under base_dir.
00035
00036     @param base_dir Base directory path
00037     """
00038     os.makedirs(base_dir, exist_ok=True)
00039     for cid in CAMERA_IDS:
00040         os.makedirs(os.path.join(base_dir, cid), exist_ok=True)
00041
00042 def group_focus_images_by_pose():
00043     """
00044     @brief Group focus images by pose ID from flat directory structure.
00045
00046     @return Dictionary mapping pose IDs to camera-path dictionaries
00047     """
00048     pose_to_cam = defaultdict(dict)
00049     for fn in sorted(os.listdir(FOCUS_DIR)):
00050         m = PATTERN.match(fn)
00051         if not m:
00052             continue
00053         pose, cam, ext = m.group(1), m.group(2), m.group(3)
00054         pose_to_cam[pose][cam] = os.path.join(FOCUS_DIR, fn)
00055     return pose_to_cam
00056
00057 def copy_focus_into_camera_folders(pose: str, cam_to_path: dict):
00058     """
00059     @brief Copy focus images into camera-specific subdirectories.
00060
00061     @param pose Pose identifier string (e.g., "pose000")
00062     @param cam_to_path Dictionary mapping camera IDs to source image paths
00063     """
00064     for cam, src in cam_to_path.items():
00065         ext = os.path.splitext(src)[1].lower()
00066         dst = os.path.join(FOCUS_DIR, cam, f"{pose}_{cam}{ext}")
00067         if os.path.abspath(src) != os.path.abspath(dst):
00068             shutil.copy2(src, dst)
00069
00070 def cleanup_focus_root():
00071     """
00072     @brief Delete only loose files directly in focus/ (not in subdirectories).
00073     """
00074     for fn in os.listdir(FOCUS_DIR):
00075         p = os.path.join(FOCUS_DIR, fn)
00076         if os.path.isfile(p) and PATTERN.match(fn):
00077             os.remove(p)
00078
00079 def main():
00080     """
00081     @brief Main pipeline: organize focus images and generate depth maps.
00082     """

```

```

00083     if not os.path.isdir(FOCUS_DIR):
00084         raise SystemExit(f"focus folder not found: {FOCUS_DIR}")
00085
00086     ensure_camera_subfolders(FOCUS_DIR)
00087     ensure_camera_subfolders(DEPTH_DIR)
00088
00089     pose_to_cam = group_focus_images_by_pose()
00090     if not pose_to_cam:
00091         raise SystemExit("No focus images found. Expected: pose000_Center.png etc.")
00092
00093     poses = sorted(pose_to_cam.keys())
00094     print("Found poses:", len(poses))
00095
00096     for i, pose in enumerate(poses, start=1):
00097         cam_to_focus_path = pose_to_cam[pose]
00098
00099         # Sort focus images into camera folders
00100         copy_focus_into_camera_folders(pose, cam_to_focus_path)
00101
00102         # Generate depth maps for this pose
00103         depthmaps = depthmap.generate_depthmaps_for_pose(cam_to_focus_path)
00104
00105         # Save depth maps: depth/<CamId>/<pose>_<CamId>_depth.png
00106         for cam, depth_ul6 in depthmaps.items():
00107             out_path = os.path.join(DEPTH_DIR, cam, f"{pose}_{cam}_depth.png")
00108             ok = depthmap.imwrite_ul6(out_path, depth_ul6)
00109             if not ok:
00110                 print("FAILED write:", out_path)
00111
00112         print(f"[{i}/{len(poses)}] {pose}: views={len(cam_to_focus_path)} depthmaps={len(depthmaps)}")
00113
00114     if CLEAN_FOCUS_ROOT_AFTER:
00115         cleanup_focus_root()
00116
00117     print("DONE.")
00118     print("Focus by camera:", FOCUS_DIR)
00119     print("Depth by camera:", DEPTH_DIR)
00120
00121 if __name__ == "__main__":
00122     main()

```

6.25 external/LiFCal/LiFCal_Data/run_LiFCal_calibration.py File Reference

Namespaces

- namespace [run_LiFCal_calibration](#)

Functions

- str [run_LiFCal_calibration.read_text](#) (str path)
- None [run_LiFCal_calibration.write_text](#) (str path, str s)
- str [run_LiFCal_calibration.replace_yaml_key_line](#) (str text, str key, str new_val)
- str [run_LiFCal_calibration.patch_settings_yaml](#) (str template_text, str focus_dir, str depth_dir, str mla_xml)
- int [run_LiFCal_calibration.run_lifcal_recalib](#) (str settings_path, str fixed_params_path, str store_dir)
- str [run_LiFCal_calibration.find_results_folder](#) (str store_dir)
- None [run_LiFCal_calibration.copy_essentials](#) (str result_dir, str out_cam_dir)
- [run_LiFCal_calibration.main](#) ()

Variables

- str [run_LiFCal_calibration.LIFCAL_BIN](#) = "/data/external/LiFCal/build/bin/LiFCal"
- str [run_LiFCal_calibration.SETTINGS](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/Settings.yaml"
- str [run_LiFCal_calibration.FIXED_PARAMS](#) = "/data/external/LiFCal/LiFCal_Data/Recalibration/Fixed_Parameters.txt"

- str `run_LiFCal_calibration.MLA_CALIB_XML` = `"/data/external/LiFCal/LiFCal_Data/Recalibration/MLA_Calibration.xml"`
- str `run_LiFCal_calibration.ROOT_IMAGESET` = `"/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"`
- `run_LiFCal_calibration.FOCUS_ROOT` = `os.path.join(ROOT_IMAGESET, "focus")`
- `run_LiFCal_calibration.DEPTH_ROOT` = `os.path.join(ROOT_IMAGESET, "depth")`
- str `run_LiFCal_calibration.OUT_ROOT` = `"/data/external/LiFCal/LiFCal_Data/Recalibration/CalibrationByCamera"`
- list `run_LiFCal_calibration.CAMERA_IDS`

6.26 run_LiFCal_calibration.py

[Go to the documentation of this file.](#)

```
00001 import os
00002 import re
00003 import shutil
00004 import subprocess
00005 from datetime import datetime
00006 import lifcal_to_json
00007
00008 """
00009 @brief Runs LiFCal recalibration for all cameras and exports results to JSON.
00010 """
00011
00012 # Configuration paths
00013 LIFCAL_BIN = "/data/external/LiFCal/build/bin/LiFCal"
00014
00015 SETTINGS = "/data/external/LiFCal/LiFCal_Data/Recalibration/Settings.yaml"
00016 FIXED_PARAMS = "/data/external/LiFCal/LiFCal_Data/Recalibration/Fixed_Parameters.txt"
00017 MLA_CALIB_XML = "/data/external/LiFCal/LiFCal_Data/Recalibration/MLA_Calibration.xml"
00018
00019 ROOT_IMAGESET = "/data/external/LiFCal/LiFCal_Data/Recalibration/LiFCal_Imageset"
00020 FOCUS_ROOT = os.path.join(ROOT_IMAGESET, "focus") # focus/<CamId>/
00021 DEPTH_ROOT = os.path.join(ROOT_IMAGESET, "depth") # depth/<CamId>/
00022
00023 OUT_ROOT = "/data/external/LiFCal/LiFCal_Data/Recalibration/CalibrationByCamera"
00024
00025 CAMERA_IDS = [
00026     "Center", "Up1", "Up2", "Up3",
00027     "Down1", "Down2", "Down3",
00028     "Left1", "Left2", "Left3",
00029     "Right1", "Right2", "Right3"
00030 ]
00031
00032
00033 def read_text(path: str) -> str:
00034     """
00035     @brief Read text file with UTF-8 encoding.
00036
00037     @param path Path to the text file
00038     @return File content as string
00039     """
00040     with open(path, "r", encoding="utf-8", errors="replace") as f:
00041         return f.read()
00042
00043
00044 def write_text(path: str, s: str) -> None:
00045     """
00046     @brief Write text to file, creating directories if needed.
00047
00048     @param path Output file path
00049     @param s Text content to write
00050     """
00051     os.makedirs(os.path.dirname(path), exist_ok=True)
00052     with open(path, "w", encoding="utf-8") as f:
00053         f.write(s)
00054
00055
00056 def replace_yaml_key_line(text: str, key: str, new_val: str) -> str:
00057     """
00058     @brief Replace YAML key-value lines.
00059
00060     @param text YAML text content
00061     @param key YAML key to find and replace
00062     @param new_val New value to set (will be quoted)
00063     @return Modified YAML text
00064     """
```

```

00065     pattern = re.compile(rf"^{(re.escape(key))}\s*:\s*(.*)$", re.MULTILINE)
00066     return pattern.sub(rf'\1{new_val}"', text)
00067
00068
00069 def patch_settings_yaml(template_text: str, focus_dir: str, depth_dir: str, mla_xml: str) -> str:
00070     """
00071     @brief Update YAML settings with camera-specific paths.
00072
00073     @param template_text Original YAML template content
00074     @param focus_dir Path to focus images directory
00075     @param depth_dir Path to depth maps directory
00076     @param mla_xml Path to MLA calibration XML file
00077     @return Modified YAML text with updated paths
00078     """
00079     out = template_text
00080     out = replace_yaml_key_line(out, "Path.totalFocusImages", focus_dir)
00081     out = replace_yaml_key_line(out, "Path.virtualDepthData", depth_dir)
00082     out = replace_yaml_key_line(out, "Path.microLensCalibration", mla_xml)
00083     return out
00084
00085
00086 def run_lifcal_recalib(settings_path: str, fixed_params_path: str, store_dir: str) -> int:
00087     """
00088     @brief Run LiFCal recalibration and return exit code.
00089
00090     @param settings_path Path to Settings.yaml file
00091     @param fixed_params_path Path to fixed parameters file
00092     @param store_dir Directory where LiFCal will store results
00093     @return Exit code from LiFCal process
00094     """
00095     os.makedirs(store_dir, exist_ok=True)
00096
00097     cmd = [LIFCAL_BIN, "recalib", settings_path, fixed_params_path]
00098
00099     proc = subprocess.Popen(
00100         cmd,
00101         stdin=subprocess.PIPE,
00102         stdout=subprocess.PIPE,
00103         stderr=subprocess.STDOUT,
00104         text=True,
00105         cwd=os.path.dirname(LIFCAL_BIN),
00106     )
00107
00108     out, _ = proc.communicate(input=f"y\n{store_dir}\n")
00109     write_text(os.path.join(store_dir, "lifcal_stdout.log"), out)
00110
00111     return proc.returncode
00112
00113
00114 def find_results_folder(store_dir: str) -> str:
00115     """
00116     @brief Find LiFCal results folder.
00117
00118     @param store_dir LiFCal output directory
00119     @return Path to results folder (typically Calibration_Results... subdirectory)
00120     """
00121     if not os.path.isdir(store_dir):
00122         raise RuntimeError(f"store_dir does not exist: {store_dir}")
00123
00124     subdirs = []
00125     for name in os.listdir(store_dir):
00126         p = os.path.join(store_dir, name)
00127         if os.path.isdir(p):
00128             subdirs.append(p)
00129
00130     if not subdirs:
00131         return store_dir
00132
00133     subdirs.sort(key=lambda p: os.path.getmtime(p))
00134     return subdirs[-1]
00135
00136
00137 def copy_essentials(result_dir: str, out_cam_dir: str) -> None:
00138     """
00139     @brief Copy calibrationProtocol.txt and extrinsicOrientations.xml to output directory.
00140
00141     @param result_dir Source directory containing LiFCal results
00142     @param out_cam_dir Destination directory for essential files
00143     """
00144     proto = os.path.join(result_dir, "calibrationProtocol.txt")
00145     extr = os.path.join(result_dir, "extrinsicOrientations.xml")
00146
00147     missing = []
00148     if not os.path.isfile(proto):
00149         missing.append("calibrationProtocol.txt")
00150     if not os.path.isfile(extr):
00151         missing.append("extrinsicOrientations.xml")

```

```

00152     if missing:
00153         raise RuntimeError(f"Missing files in {result_dir}: {missing}")
00154
00155     shutil.copy2(proto, os.path.join(out_cam_dir, "calibrationProtocol.txt"))
00156     shutil.copy2(extr, os.path.join(out_cam_dir, "extrinsicOrientations.xml"))
00157
00158
00159 def main():
00160     """
00161     @brief Main pipeline: run LiFCal for each camera, export JSON, and log health.
00162     """
00163     if not os.path.isfile(LIFCAL_BIN):
00164         raise SystemExit(f"LiFCal binary not found: {LIFCAL_BIN}")
00165     if not os.path.isfile(SETTINGS):
00166         raise SystemExit(f"Settings template not found: {SETTINGS}")
00167     if not os.path.isfile(FIXED_PARAMS):
00168         raise SystemExit(f"Fixed parameters file not found: {FIXED_PARAMS}")
00169     if not os.path.isfile(MLA_CALIB_XML):
00170         raise SystemExit(f"MLA_Calibration.xml not found: {MLA_CALIB_XML}")
00171
00172     templ = read_text(SETTINGS)
00173
00174     timestamp = datetime.now().strftime("%Y_%m_%d_%H%M%S")
00175     session_out = os.path.join(OUT_ROOT, f"run_{timestamp}")
00176     os.makedirs(session_out, exist_ok=True)
00177
00178     print("Output:", session_out)
00179
00180     for cam in CAMERA_IDS:
00181         focus_dir = os.path.join(FOCUS_ROOT, cam)
00182         depth_dir = os.path.join(DEPTH_ROOT, cam)
00183
00184         if not os.path.isdir(focus_dir):
00185             print(f"[SKIP] {cam}: focus dir missing: {focus_dir}")
00186             continue
00187         if not os.path.isdir(depth_dir):
00188             print(f"[SKIP] {cam}: depth dir missing: {depth_dir}")
00189             continue
00190
00191         cam_out = os.path.join(session_out, cam)
00192         os.makedirs(cam_out, exist_ok=True)
00193
00194         settings_cam = os.path.join(cam_out, "Settings.yaml")
00195         patched = patch_settings_yaml(templ, focus_dir, depth_dir, MLA_CALIB_XML)
00196         write_text(settings_cam, patched)
00197
00198         store_dir = os.path.join(cam_out, "lifcal_store")
00199         rc = run_lifcal_recalib(settings_cam, FIXED_PARAMS, store_dir)
00200
00201         if rc != 0:
00202             print(f"[FAIL] {cam}: LiFCal returncode={rc} (siehe {store_dir}/lifcal_stdout.log)")
00203             continue
00204
00205         result_dir = find_results_folder(store_dir)
00206
00207         try:
00208             copy_essentials(result_dir, cam_out) # ???iert protocol + xml nach cam_out
00209         except Exception as e:
00210             print(f"[FAIL] {cam}: {e}")
00211             print(f"check log: {store_dir}/lifcal_stdout.log")
00212             continue
00213
00214         # store dir wegräumen
00215         shutil.rmtree(store_dir, ignore_errors=True)
00216
00217         # ? JSON EXPORT: result_dir muss cam_out sein, weil DA liegen jetzt die 2 Dateien
00218         try:
00219             json_path = lifcal_to_json.export_lifcal_parameters_json_in_place(
00220                 result_dir=cam_out,
00221                 cam_id=cam,
00222                 image_dir="LiFCal_Imageset",
00223                 run_type="recalib",
00224                 pixel_size_mm=0.0055,
00225                 fallback_cx=640.0, # compatibility only
00226                 fallback_cy=360.0, # compatibility only
00227                 out_name="parameters.json"
00228             )
00229         except Exception as e:
00230             print(f"[WARN] {cam}: JSON export failed: {e}")
00231
00232         # Build combined JSON from all cameras
00233         try:
00234             combined_path = lifcal_to_json.build_combined_parameters_json(
00235                 run_root=session_out,
00236                 camera_ids=CAMERA_IDS,
00237                 out_name="combined.json"
00238             )

```

```
00239     print("[OK] COMBINED:", combined_path)
00240 except Exception as e:
00241     print("[WARN] combined.json not created:", e)
00242
00243 # Log health score and update baseline
00244 try:
00245     import sys
00246     CALIB_ROOT = "/data/calibrationLFC"
00247     if CALIB_ROOT not in sys.path:
00248         sys.path.insert(0, CALIB_ROOT)
00249
00250     from ResultLogger import ResultLogger
00251     logger = ResultLogger(baseDir="/data/baseline")
00252     score = logger.logRecalibration(
00253         combined_json_path=combined_path,
00254         meta={
00255             "runType": "recalib",
00256             "imageDir": "LiFCal_Imageset",
00257             "bundleAdjust": True
00258         }
00259     )
00260     print(f"[OK] HealthScore (lifcal) = {score:.2f}")
00261 except Exception as e:
00262     print("[WARN] LiFCal logging/health failed:", e)
00263
00264
00265 if __name__ == "__main__":
00266     main()
```